Integrating temporal dynamics with transcriptomics and metabolomics to improve abiotic stress in Brassicas

A Dissertation

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Dedication

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Chapter 1: Temporal subfunctionalization in paralogous gene expression provide novel resources for crop improvement

1.1 Gene duplication in the plant lineage

Gene duplication is a key evolutionary mechanism that generates genetic material for functional divergence and can facilitate species diversification[1–3]. Duplicate genes arise through multiple mechanisms, including tandem duplication (adjacent, collinear copies often generated through unequal crossing over), transposon-mediated duplication (gene copies mobilized and inserted into new genomic locations), segmental duplication (duplication of a chromosomal block containing multiple syntenic genes), and importantly, Whole Genome Duplications (WGD)[1]. WGD, or polyploidy, represents a pervasive large-scale mechanism particularly prevalent in plant lineages[4,5] but is also found to a lesser degree in fishes, amphibians, reptiles and insects[5]. Nearly all extant angiosperm genomes show signatures of at least one historical WGD event[1], and multiple WGD have been shown in grasses and angiosperms[5]. However, most duplicated genes are ultimately lost and as such WGD is often considered to be an evolutionary 'dead end'[5]. This of course raises a clear paradox which has resulted in decades of debate on why and how such duplication events are still so predominant in plant lineages[6]. Notably, genes that are retained following WGD are not randomly distributed across the genome but are frequently enriched for regulatory and stress responsive functions[1]. This review will focus on conservation of duplicate copies with particular emphasis on how regulatory and temporal divergence among paralogs can be leveraged to improve breeding strategies.

Immediately following duplication, gene copies are redundant and experience relaxed purifying selection, as selective constraints are initially shared. In the absence of selection favoring retention, one duplicate typically accumulates mutations through genetic drift. Over evolutionary time, this degeneration can lead to pseudogenization which is characterized by deleterious mutations such as premature stop codons, frameshifts, splice-site disruptions, or loss of regulatory elements[1,7]. For example, following the most recent WGD in Arabidopsis estimates show that approximately 70% of duplicate genes were lost through fractionation[8]. At the whole genome scale, continual loss of redundant duplicates following WGD is referred to as fractionation, a pervasive process that shapes the long-term structure of polyploid genomes. Comparative genomic analyses indicate that the majority of WGD-derived duplicates are lost within tens of millions of years, although rates and patterns vary across lineages[7,8]. Duplicates arising from smaller scales such as tandem, segmental, or transposon-mediated duplication events are lost through gene-level processes, such as pseudofunctionalization and deletion[9]. Local recombination rates, dosage sensitivity, tissue specificity, and chromatin context, to name a few examples, all influence the evolutionary fate of duplicated genes[8–10].

Retention of duplicates, although the minority outcome, reflects genes with a selective advantage. One mechanism for retention is through neofunctionalization, in which one paralog acquires a novel biochemical or regulatory function, while the other maintains the ancestral role. Alternatively, subfunctionalization occurs when ancestral functions become partitioned between duplicates, such that both copies are required to recapitulate the original function[11]. In this context, 'function' can refer to a protein or biochemical function (ex: coding sequences), or to regulatory functions (ex: cis-regulatory sequences). Although divergence in both regulatory and coding functions contribute to the fate of duplicate copies, coding sequences are often more constrained[7] while regulatory divergence appears to be more widespread and rapid in plants[5]. Consequently, regulatory divergence is thought to play a prominent role in the long-term preservation of plant paralogs[8]. In this review, we focus on examples of regulatory divergence among polyploid plant species, with an emphasis on crops, and discuss the implications of expression variation on abiotic stress tolerance.

1.2 Genome expansion as a source of regulatory variation

WGDs generate novel opportunities for genes to diverge in regulatory function, providing a more flexible genetic framework in which conserved functions can be partitioned across distinct regulatory contexts. Although the majority of duplicated genes are lost through fractionation, a surprising 65-85% of annotated plant genes have at least one additional copy[1], suggesting that, during each round of WGD, there are a nonconsequential amount of genes that are retained. Comparative analyses across diverse plant lineages demonstrate that these multi-copy genes are disproportionately enriched in regulatory gene families[1,5,12], including Transcription Factors (TFs), most notable of which include WRKY, NAC, bZIP, and MYB families[13]. Additional hormone and signaling regulators (auxin; ABA), calcium-dependent signaling components (CBL; CIPKs), circadian clock genes, and other stress-responsive families such as heat shock factors (HSFs) and glutathione S-transferases (GSTs), represent some of the most expanded gene families in plant genomes, suggesting that WGD favors developmental, signaling, and stress related pathways[1,5,14,15].

In some cases, these regulatory proteins function within interconnected regulatory networks and their relative abundance influences the activity of numerous interacting partners[10]. For example, many regulatory proteins form multimeric complexes and/or participate in tightly coordinated signaling cascades, where changes in gene dosage can alter network output[10]. Following whole genome duplication, all components of such networks are duplicated simultaneously to preserve stoichiometric balance among interacting proteins. However, subsequent loss of a single duplicate copy can disrupt stoichiometry, potentially destabilizing these regulatory interactions[10]. As a result, selection is likely to favor retention of duplicated regulatory genes that maintain balanced gene expression and signaling activity[12]. This principle, more commonly referred to as the ‘dosage balance hypothesis’[10,16], is one explanation behind preferential retention of regulatory gene families following WGD.

While dosage balance may constrain divergence of duplicate copies following duplication, long-term retention of paralogs is frequently associated with subsequent regulatory divergence[8,17]. This allows duplicate gene copies to partition their expression profiles across spatial, developmental, tissue-specific, environmental, or temporal contexts[9,18–20]. Accumulating evidence indicates that a substantial proportion of this divergence arises from changes in cis-regulatory regions, including promoter evolution and gain or loss of transcription factor binding sites, although trans-acting and chromatin-level mechanisms may also contribute[9,18]. By reducing functional redundancy, these regulatory changes facilitate subfunctionalization and stabilize persistence of duplicated genes over longer evolutionary periods[11,17].

1.3 Circadian and diel control in retained paralog expression patterns

Because plant physiology and gene expression are tightly coordinated with daily environmental cycles, divergence in temporal regulation provides one mechanism through which redundant copies may be retained. This provides a means to balance resource allocation among metabolism, growth, and environmental responsiveness, which is made possible through diel and circadian regulation[21,22]. Comparative analyses across *Archaeplastida* demonstrate that diel gene expression is evolutionarily conserved but increases in regulatory complexity in land plants, consistent with expansion of multiple gene families[23]. Diel (24h) regulation arises through exogenous cues such as light and temperature, which directly entrains daily rhythms in gene expression and physiology[24]. In parallel, circadian regulation is driven by an endogenous clock consisting of interconnected transcriptional-translational feedback loops, allowing expression patterns to be maintained with reinforcement of external cues[25]. As a result, rhythmic gene expression patterns are tightly aligned with the surrounding environment. This confers a fitness advantage by enabling organisms to anticipate, and potentially prepare for, reoccurring changes in the environment, such as dawn, dusk, and daily temperature fluctuations[21,22].

Time-of-day dependent regulation is evident in core physiological processes that underlie plant growth and productivity[26]. Photosynthesis, for example, is a well described process that exhibits strong circadian and diel regulation in stomatal opening, non-photochemical quenching (NPQ), and light-adapted photosystem II efficiency (Fv'/Fm')[27]. Stomatal conductance, or the rate at which CO_2 enters the leaf, exhibits robust circadian rhythms indicating a clear role in the clock in optimizing carbon acquisition[21]. Plants with disrupted clocks exhibit reduced stomatal conductance followed by a decrease in photosynthetic efficiency, ultimately leading to reduced plant growth[21]. Interestingly, circadian control of stomatal behavior requires the guard cell specific clock, and clock regulation of stomatal aperture drives carbon assimilation in Arabidopsis[28]. This is consistent with a growing body of research that indicates circadian regulation is cell and tissue-specific, with distinct clocks controlling localized outputs[29]. While this spatial partitioning further complicates our understanding of how individual clocks coordinate a plant-wide response, it provides additional opportunities for paralog diversification across both space and time.

The majority of staple crop species have experienced multiple rounds of WGD[1,2] followed by varying degrees of genome fractionation. This raises important questions about how paralog retention has modified clock control of physiology, metabolism, and stress responses, particularly in polyploid crops. Clock components comprise a substantial portion of preferentially retained gene families, suggesting that flexibility in temporal regulation provides a benefit[14]. A clear example of this is the *PSEUDO-RESPONSE REGULATOR* (*PRR)* family. *PRRs* play a central role in modulating the pace of the clock and restrict both plant growth and environmental responsiveness (light; temperature) to specific times of day[30]. *PRR* copy number varies substantially across plant lineages with green algae (*Ostreococcus tauri*) possessing a relatively simple clock that contains a single *PRR* (*PRR1/TOC1*) while liverwort (*Marchantia polymorpha)* possesses two copies (*PRR1; PRR7*). In eudicots, Arabidopsis has five temporally distinct *PRR* copies (*PRR1/3/5/7/9*). Other polyploids including Brassica's have retained additional paralogs of the five Arabidopsis orthologs, with Chinese cabbage containing a total of 11 *PRR* copies[31]. Across eudicots, *PRR* paralogs are expressed in a conserved, sequential order across the day, with *PRR9* peaking in the early morning followed by *PRR7*, *PRR5*, and *PRR3* later throughout the day[30]. The distinct separation in phasing demonstrates an adaptive role in limiting functional redundancy of paralogs while preserving the overall function. Notably, despite extensive genome fractionation following the Brassica-specific whole-genome triplication, nearly all *PRR* copies were retained[32]. Together, these examples support a generalized model in which paralog retention within the clock is facilitated by temporal divergence over functional redundancy. This widespread retention highlights the importance of genetic flexibility in maintaining and diversifying circadian control on extensive regulatory networks.

1.4 Pervasiveness of paralog partitioning in abiotic stress response

Growth and stress responses are often at odds with one another, as activation of stress-response pathways frequently imposes tradeoffs that delay growth and development[33]. The circadian clock gates abiotic stress responses in plants, restricting sensitivity to environmental cues such as cold, drought, heat, and osmotic stress to specific times of day[20,34–36]. Within the clock, paralogs are frequently retained when they occupy distinct temporal phases across the diel cycle, consistent with partitioning of regulatory roles by time of day[26]; it is unclear if this principle extends to abiotic stress pathways, however. Enrichment of duplicated genes within stress-associated pathways would suggest that paralog partitioning throughout the day represents a general strategy for balancing growth and stress responsiveness to optimize productivity.

Comparative genomic analyses across diverse crop species support this possibility and show that stress pathways are disproportionately enriched for duplicated genes following whole-genome duplication[1,37]. This is observed across grasses such as rice and wheat[38], oilseeds[39], and vegetable crops[40]. In Brassicales, for example, variation within the *METHYLTHIOALKYLMALATE SYNTHASE* (*MAM*) locus, arising from WGD followed by tandem duplication and rearrangement, has generated multiple paralogous enzymes that underlie natural variation in glucosinolate composition, illustrating how duplication within stress-related metabolic pathways contributes to functional diversification[41].

Studies in bread wheat and sorghum provide complementary evidence that both gene duplication and temporal regulation are central to plant stress responses. For example, in hexaploid bread wheat genes belonging to expanded gene families are disproportionately represented among abiotic stress responsive loci and individual paralogs within these families often differ in whether, and how strongly, they respond to stress[38], highlighting a clear separation in which paralog was responsive. This separation was further distinguished by time of day where cold responsive genes across the genome were gated by the clock[38]. In sorghum, transcriptomic analyses in heat and cold tolerant and intolerant accessions demonstrate that stress and hormone related genes exhibit pronounced context-dependent regulation between accessions, with transcription factors and signaling genes showing particularly dynamic behavior throughout the day[42]. Together, these studies indicate that while regulatory plasticity is a pervasive feature of plant stress responses, the extent to which temporal regulation structures paralog-specific stress responses across crops isn't fully understood.

1.5 Expansion of temporal regulation in polyploid crops

Foundational work in Arabidopsis established that circadian and diel regulation structure a substantial fraction of the plant transcriptome[43]. Early genome wide time course studies demonstrated that ~30% of transcripts cycle under constant conditions[44], and under entrained light–dark conditions this proportion can exceed 50–70%[19,20]. Subsequent work reinforced that rhythmic transcription is not restricted to clock components, but extends broadly to genes involved in metabolism, hormone signaling, photosynthesis, growth, and stress responses[26,43].

In polyploid crops such as Brassicas, genome duplication has dramatically expanded gene copy number within these temporally structured networks. Transcriptomic analyses in *Brassica rapa* revealed that a large fraction of expressed genes exhibit time-of-day dependent circadian regulation, exceeding levels observed in *Arabidopsis*[19]. Notably, many retained paralogs display divergence in rhythmic behavior, including shifts in peak expression phase, changes in amplitude, or context-dependent rhythmicity.

Temporal regulation also structures abiotic stress responses in crops. Studies in wheat and Brassicas demonstrate that stress-responsive genes are frequently gated by time-of-day and that paralogs within expanded gene families often differ in both magnitude and timing of response[19,20,38]. In polyploid contexts, individual homoeologs or paralogs may be differentially activated (gated) under stress depending on the time of day, suggesting that temporal divergence contributes to functional specialization within stress pathways. In *B. rapa*, enrichment analyses revealed that multi-copy genes were significantly overrepresented where only a single paralog copy was drought responsive, rather than both[19]. These patterns are more complex when examined across morphotypes, as the specific paralog that responds to stress frequently differs among accessions[20]. In *Brassica napus*, and allotetraploid, prolonged water limitation uncovered significant homoeolog expression divergence, with time-of-day specific shifts in sulfoquinovose and glucosinolate biosynthesis genes which occurred in distinct copies rather than uniformly across duplicates. In *Brassica rapa*, analysis of thirteen diverse accessions following cold acclimation further demonstrated that the identity of the stress-responsive paralog frequently differed among accessions. These findings indicate that the interaction between genome duplication and time-of-day regulation may be particularly relevant for understanding adaptive stress responses. This expansion of the "temporal transcriptome" suggests that whole genome duplication does not merely increase gene dosage but increases the dimensionality of time-of-day regulation itself.

1.6 Leveraging time of day dynamics in polyploids as a strategy for identifying candidate breeding targets

Among the mechanisms that facilitate long term retention, regulatory divergence, which can be mediated through changes in cis-regulatory elements, provides a mechanistic framework for understanding how duplicate genes partition function over time. Examples from core circadian regulators illustrate how retention of multiple copies can coincide with distinct phasing that minimize redundancy while preserving overall network integrity[19,32]. Similar patterns are observed in abiotic stress pathways, where duplicated regulatory genes frequently display context-dependent and time-of-day specific responses[20,38,42]. Together, these observations suggest that while genome expansion generates additional genetic material and dosage balance may constrain early divergence, differences in temporal regulation represent one pathway through which paralogs persist and diversify.

While temporal divergence among paralogs is frequently observed, the cis-regulatory mechanisms underlying phase shifts and rhythmic differences are rarely resolved. Direct links between promoter sequence variation, transcription factor binding dynamics, chromatin accessibility, and paralog-specific phasing[9,19,20,24] are still rarely examined in an integrated framework. Moreover, most time-of-day studies have been conducted in single reference genotypes, limiting our understanding of how temporal regulation varies across morphotypes or stress. Connections between paralog specific timing, downstream metabolic outputs, and whole-plant physiological performance remain largely correlative rather than mechanistically defined. Incorporating temporal dynamics into functional genomics and breeding strategies may improve identification of adaptive regulatory variants by revealing which paralog, and at what time of day, contributes most strongly to stress tolerance or growth optimization. In polyploid crops, where duplicated regulatory genes are abundant, resolving time-of-day specific expression patterns can reveal how paralogs partition regulatory roles across the diel cycle, improving our ability to identify candidate genes underlying stress tolerance.

Chapter 2: Paralog diversification underlies diversity in B. rapa time of day responses to cold acclimation

This chapter is adapted from:

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A.R. designed the study, performed analyses, and wrote the manuscript. Co-authors contributed to experimental design, data collection, and manuscript revisions.

2.1 Introduction

Extreme temperature events, including cold and freezing snaps, significantly hinder crop productivity and limit our ability to meet increasing food demand[45–47]. To overcome these climatic pressures, we need crops that can more effectively acclimate to cold. Capturing genomic variation is a critical first step in elucidating regulatory drivers (transcription factors; cis-regulatory elements) underlying complex traits such as cold tolerance for identifying breeding targets. Duplication events, fractionation/loss, and genomic rearrangements are key evolutionary consequences that contribute to genomic diversity[3,48–50] and desirable crop traits[51–54]. However, much of this diversity is masked when using a single reference genome. Pangenomes, which incorporate genomes from multiple individuals within a species, represent the entire breadth of genes that are present among genetically similar populations, and allow us to capture both core (conserved among genomes) and accessory (variable retention among genomes) genes. Across diverse lineages, pangenome analyses have revealed that presence absence variation (PAV) and copy number variation (CNV) of accessory genes is often associated with abiotic stresses including freezing tolerance[53,55–57]. PAV is also prevalent across crop species. Maize and peanut, for example, exhibit substantial PAV among accessions, with 33% and 44% of all detected genes classified as accessory, respectively[58,59]. Substantial proportions of accessory genes have been reported in other species, independent of the number of genomes included, such as *Solanum lycopersicum L.* (38.4%[60]), *Brassica oleracea* (18.7%[54]), and *Brassica rapa* (17.8%[48]).

Accessory genes are often associated with pathways that are of interest to breeders[54,61], including defense and signaling[48,54], flowering time[48], fruit quality/productivity[62], and specialized metabolism pathways[48,54]. For example, glucosinolate biosynthetic genes, which encode sulfur-rich specialized metabolites with well established roles in biotic and abiotic stress tolerance[63,64], show extensive copy-number variation across Brassicaceae[41,54,63,65] and contribute to differences in glucosinolate content[66,67], defense responses[64], and sulfur allocation[68]. Likewise, CNV and PAV in *FLOWERING LOCUS C (FLC)* is associated with differences in vernalization requirement and flowering time among Brassica morphotypes[54,69–71]. These examples demonstrate how structural variation in the genome imparts key adaptations to stress related pathways to facilitate adaptations to challenging climatic regimes[54,72].

Brassica rapa (*B. rapa*) is a genetically diverse mesopolyploid that diverged from Arabidopsis approximately 20 MYA[73]. Following this divergence, *B. rapa* underwent an additional whole-genome triplication and subsequent genome fractionation[74]. Its unique evolutionary history has generated substantial CNV and PAV which underpins its remarkable morphological diversity[48]. *B. rapa* encompasses numerous specialty crops (chinese cabbage, pak choi, oilseed, turnip and leafy vegetable varieties) that vary dramatically in their leafy-head architecture, storage-root formation and oilseed forms[40,75,76]. The morphological diversity in *B. rapa* is also due to the unique domestication histories of morphologically distinct cultivars (herein referred to as ‘morphotypes’). Historical and population genomic analyses support multiple domestication trajectories, but a model describing an early Eurasian-East Asia origin is generally accepted[76–78]. Yellow sarsons, oilseeds grown for both vegetable oil and fodder, originated from southern Asia and were eventually brought to drier climates in northern India. Chinese cabbages and other east asian leafy vegetables including pak choi and pak choi-like (herein referred to as choy sum) were domesticated alongside the Yangtze river[77,79] which is known for warm and moist climates, while winter oilseed (*Brassica rapa ssp. oleifera*) are generally considered to be adapted to cooler climates from higher latitudes such as Finland[80]. The domestication of *B. rapa* occurred across temperate, arid, and semi-arid environments, supporting the idea that while *B. rapa* crops were being domesticated we were artificially selecting for different abiotic stress tolerances.

A common challenge for temperate crops including Brassicas is exposure to low, non-freezing temperatures, often followed by subsequent freezing events, which are mitigated through a process known as cold acclimation[81]. The genomic and physiological diversity of *B. rapa*, in addition to its close evolutionary relationship with both economically important crops such as Canola, and the model plant Arabidopsis, positions it as an attractive model for identifying key regulatory genes underlying cold acclimation and freezing tolerance. Physiological studies directed at classifying cold hardy varieties have identified both cold sensitive (Chinese cabbage[82]) and cold hardy lines, some of which diverge within the same morphotype (Turnip rape[83]). Cold acclimation is a complex process that primes plants for future cold and freezing events. The acclimation process broadly involves the coordination of hormonal and transcriptional cascades that remodel membrane lipids to maintain structural fluidity[84], increase antioxidant pools (glutathione; anthocyanins)[85,86], and increase intracellular osmolyte content (ex: proline; sugars; carbohydrates), to regulate water homeostasis[81,85,87]. After exposure to sub-zero temperatures, ice forms outside the cell, reducing extracellular water potential and effectively dehydrating the cell as water flows out due to osmosis[84,88]. However, even a small increase in solutes within the cell can reduce intracellular water potential thereby reducing the amount of water lost[88]. The degree of tolerance, particularly under freezing, can vary depending on the initial membrane lipid content[88], osmolyte pools[89] and variation in gene regulation[90].

Although the direct mechanisms by which plants initially perceive cold temperatures remains unclear, early cold exposure rigidifies membranes and triggers a transient influx of calcium into the cytoplasm that initiates downstream signaling pathways. After calcium accumulates in the cytoplasm it initiates post translational modification pathways (sumoylation; ubiquitination; phosphorylation) that involve cold responsive transcriptional regulators including MYB, bZIP, NAC, and WRKY transcription factor (TF) families[91]. bHLH factors such as ICE1 (Inducer of CBF Expression-1) and CAMTA (CALMODULIN-BINDING TRANSCRIPTION ACTIVATOR) promote induction of CBFs/DREBs (C-repeat binding factors/Dehydration-responsive element binding; AP2/ERF family) alongside additional cold-responsive TF families[91]. Together, these TFs activate large sets of cold-responsive (*COR*) genes, where the ICE–CBF–COR module is broadly conserved, but the identity and contribution of cold-inducible paralogs varies by species and genotype. For example, in Swedish and Italian Arabidopsis accessions with different freezing tolerances, expression levels of *CBF1*, *CBF2*, and *CBF3* are all similarly induced within minutes of exposure to cold, but expression levels of their target genes (*CORs*) vary[90]. This was found to be due at least in part to a copy of *CBF2* that was truncated in freeze sensitive plants but was fully intact in the tolerant line[90]. Comparative transcriptomic studies in winter rapeseed have shown that key cold-responsive TFs, including *ICE1* paralogs[92] and numerous *WRKY* TFs[93], are differentially induced among freezing-tolerant versus freezing-sensitive *B. rapa* lines[94]. Transgenic studies in wheat show *TaCBF14* and *TaCBF15* homologs improve freezing tolerance by upregulating expression of downstream *COR* genes[95], while other *CBF/DREB* homologs contribute to cold tolerance in rice[96–98] and canola[99] by upregulating expression of additional *COR* or photosynthesis genes[98–100]. Collectively, these examples emphasize how paralog diversification shapes accession specific tolerances to cold which can be leveraged for engineering tolerance into crops. However, few studies examine multiple morphotypes under similar experimental conditions, impeding our ability to link cold and freezing phenotypes across such a genomic and geographically diverse landscape.

Because acclimation responses can be metabolically costly, the magnitude and timing of acclimation are strongly time-of-day dependent and are regulated in part by the circadian clock, particularly through clock control of the CBF regulon[101,102]. The circadian clock is an endogenous ~24h oscillator composed of interlocked transcriptional-translational feedback loops which, in plants, are primarily entrained by light and temperature[103]. Unlike diel regulation, which is restricted to 24 hour day/light cycles, circadian regulation of the transcriptome allows organisms to anticipate external fluctuations such as changes in day length or temperature. This anticipation enables temporal allocation of resources and responsiveness to the times of day when they are most beneficial, a phenomenon referred to as circadian gating[38]. Cold response is one example of this where cold induction of the CBF pathway is regulated by the clock through the morning regulators, CIRCADIAN CLOCK ASSOCIATED 1 (CCA1) and LATE ENLONGATED HYPOCOTYL (LHY), which restrict the activity of the CBFs and other downstream CORs (*i.e.* COR15A; COR47; COR78[102]) to specific times of day. This is true for other cold responsive TFs in Arabidopsis (*i.e.* RAV1; ZAT12[104]) as well as other plant species including Brassicas. For example, allelic variation at *GIGANTEA* (*GI*), a key regulator of the clock and photoperiodic signalling[105], is associated with differences in circadian period in *B. rapa*, and these differences correlate with variation in cold and freezing tolerance[106]. In Chinese cabbage, alleles of *EARLY FLOWERING 3* (*ELF3*), a well known clock component best described for integrating light and temperature signals into the clock[107], alter thermal responsiveness of circadian rhythms[108]. In *B. rapa*, circadian clock genes have been preferentially retained in multiple copies, and genome-wide analyses showed extensive divergence in temporal expression among retained paralogs[19]. Consistent with other work[19], this divergence suggests that paralogs have the potential to generate accession specific differences in abiotic stress tolerance. However, our understanding of how time-of-day regulation and paralog-level diversification jointly contribute to stress responses remains largely unexplored.

To identify regulatory genes underlying beneficial cold acclimation responses across genetically and morphologically diverse *B. rapa*, we combined time-of-day-resolved physiology assays under freezing stress with a five-day cold acclimation RNA-seq time course across 13 *B. rapa* accessions representing six distinct morphotypes. Following the freeze stress, accessions exhibited distinct physiological responses that did not segregate by morphotype as might be suggested by their shared domestication histories. Differential expression analysis of temporally resolved transcriptomes revealed thousands of genes with altered diel expression patterns under cold acclimation; however, the number of cold-responsive genes did not correlate with relative freezing tolerance, indicating extensive transcriptome reprogramming across all accessions. To explore functional differences within these varying transcriptomes, we grouped cold-responsive genes by the time of day at which they reached maximal expression ('phase') and assessed enrichment of biological pathways across these groups. Although photosynthesis, sulfur assimilation, and redox homeostasis were enriched across most accessions, the timing of the response differed. This divergence was observed not only between morphotypes but also among accessions within the same morphotype, again indicating accession-specific alterations to essential biological pathways under cold acclimation.

Surprisingly, we also identified divergence at the paralog level, with different accessions preferentially exploiting distinct copies of the same Arabidopsis ortholog under cold acclimation. In several cases, paralogs were partitioned into different phase groups, revealing time-of-day specific specialization of cold response. To explore the transcriptional regulation underlying this paralog level divergence, we built Gene Regulatory networks (GRNs) and found markedly increased similarity among networks when we used the Arabidopsis ortholog over the *B. rapa* paralog, emphasizing the conservation of functional responses to cold despite divergence in the paralog that was utilized. Together, these observations indicate that cold acclimation responses are shaped by both temporal regulation and paralog-level diversification rather than by conserved pathway regulation. To further resolve key regulators underlying these differences, we pulled Arabidopsis network edges that were shared among accessions with more similar transcriptional networks and found broad representation in hormone signaling, circadian clock regulation, and sulfur metabolism. When we explore differences between the more freeze tolerant and intolerant accessions, we found both previously characterized cold regulators (*VRN1*; *DREB1A/2B/2C*) and TFs not previously uncharacterized as cold regulators (*MYB28*; *MYB34*; *MYB122*; *RRTF1*). Overall, this work demonstrates how temporal and genomic diversity intersect to shape cold acclimation responses in a polyploid crop and highlights the importance of incorporating time-of-day information with pangenome approaches to modify breeding strategies aimed at improving cold tolerance.

2.2 Methods

2.2.1 Plant growth and experimental design

Seeds for 14 *B. rapa* lines (Supplemental Table 1, Accession Information tab) were germinated on wet whatman paper for three days before sowing in 3.5" by 3.5" by 3.5" (height; width; depth) rectangular pots filled with moistened Sungrow Professional Growing Mix soil. Plants were grown in 14h/10h light/dark photoperiods with 200-250 m^2s^{-1} light intensity, 21°C/16°C temperature cycles, and 60-70% relative humidity. All plants were fertilized with 30mL of 0.5X Hoaglands liquid fertilizer 14 days after planting. Ten replicates per accession were cold acclimated after three weeks of growth in control conditions (Figure1A), with an additional five replicates per line for electrolyte leakage assays. Cold acclimation conditions for the first day consisted of a six hour ramp down from 21°C day to a 4°C night. Temperatures increased to 10°C at ZT0 (Day 2; ZT: Zeightgeber or hours after lights on) and were ramped back down to 4°C for eight hours at ZT13, followed by a freeze 'snap' at -4°C for two hours. Temperatures were brought back up to -2°C at ZT0 for two hours (Day 3) then slowly warmed to 4°C and maintained at 4°C for the remainder of the day. Nighttime temperatures were slowly ramped up from 4°C to 16°C (end of Day 3) over the course of 10 hours. Temperatures were then slowly ramped from 16°C (nighttime control conditions) to 21°C (daytime control conditions) over the course of eight hours (Day 4). These conditions were maintained for the remainder of the experiment.

2.2.2 Physiology assays following freezing stress

To track alterations to physiology throughout the experiment we measured several aspects of plant physiology before (Day 1), during (‘Acclimation’ or Day 2; ‘Freeze’ or Day 3; and ‘Post-Freeze’ or Day 4) and after (Recovery on Days 5, 9 and 10) cold acclimation (Figure1A). These metrics include light adapted photosystem II (PSII) efficiency (Fv’/Fm’), leaf damage, electrolyte leakage, and above ground (fresh) biomass. PSII efficiency was measured on ten biological replicates using a handheld fluorometer (‘FluorPen’) at various timepoints throughout the experiment (Figure1A). Leaf tissue from five biological replicates were collected for electrolyte leakage at ZT0 the day following the freeze (Day 4). To avoid cell rupture that might occur by attempting to submerge a whole leaf in a small test tube, we cut one half-inch by half-inch section on the left and right side of the midvein for each leaf using a sharp razor blade. Sections were fully submerged in 4mL of DI water in glass test tubes and shaken for three hours before measuring the initial conductivity. Leaf sections without DI water were then frozen at -80℃ overnight to rupture the remaining intact cells before measuring the final conductivity as described above. Three conductance readings per leaf section were taken and then averaged for each section. The average leakage was calculated as between the two leaf sections and used for statistical analyses and plotting. To assess both freezing damage and recovery capacity, we calculated the percent change in post-freeze and recovery measurements relative to the pre-acclimation levels. Leaf damage was assessed on ten biological replicates by giving each plant a ‘damage score’ which is a whole number rating that describes the percentage of damage observed, where ‘10’ is no damage to any leaves, ‘1’ is a fully dead plant, and ‘5’ is approximately 50% of the plant exhibiting damage. Above ground biomass was collected on five biological replicates on the tenth and last day of the experiment. Data was processed and visualized in our RMarkdown which is available on our Github (https://github.com/greenhamlab/).

2.2.3 Nucleic Acid extractions

High molecular weight DNA was extracted from young tissue using the protocol of Doyle and Doyle[109] with minor modifications. Flash-frozen young leaves were ground to a fine powder in a frozen mortar with liquid nitrogen followed by very gentle extraction in 2% CTAB buffer (that included proteinase K, PVP-40 and beta-mercaptoethanol) for 30min to 1h at 50 °C. After centrifugation, the supernatant was gently extracted twice with 24:1 Chloroform:Isoamyl alcohol. The upper phase was transferred to a new tube and 1/10th volume 3 M Sodium acetate was added, gently mixed, and DNA precipitated with iso-propanol. DNA precipitate was collected by centrifugation, washed with 70% ethanol, air dried for 5-10 minutes and dissolved thoroughly in elution buffer at room temperature followed by RNAse treatment. DNA purity was measured with Nanodrop; DNA concentration was measured with Qubit HS kit (Invitrogen), and DNA size was validated by Femto Pulse System (Agilent).

2.2.4 Genome assembly

The seven genome assemblies were constructed with identical pipelines and identified only minor differences in coverage and support (Table1). Initial assemblies were constructed with PacBio HiFi and Dovetail OmniC Hi-C data and assembled using HiFiAsm+HIC[110]. The initial assembly was polished with RACON[111]. Contigs for both haplotypes were oriented, ordered, and joined with JUICER[112]. Contigs containing significant telomeric sequence were considered properly oriented in the assembly. All initial assemblies were manually screened to identify misjoins using Hi-C data. Any misjoins present were corrected. Where necessary, adjacent alternative haplotypes were identified on the joined contig set and collapsed using the longest common substring between the two haplotypes. Homozygous SNPs and INDELs were identified with 2x150, 400bp insert Illumina reads mapped to each reference with GATK and corrected in the release sequence[113]. Chromosomes were numbered and oriented using version 1.0 Brassica rapa var. FPsc (Turnip mustard) obtained from Phytozome (https://phytozome-next.jgi.doe.gov). The resulting sequence was screened for retained vector and/or contaminants.

2.2.5 Genome annotation

Transcript assemblies were built using PERTRAN (Lovell et al. 2018), which conducts genome-guided transcriptome short read assembly via GSNAP (Wu, 2010) and builds splice alignment graphs after alignment validation, realignment and correction. To obtain putative full length transcripts, PacBio Iso-Seq CCSs were corrected and collapsed by genome guided correction pipeline, which aligns CCS reads to genome with GMAP (Wu, 2010) with intron correction for small indels in splice junctions if any and clusters alignments when all introns are the same or 95% overlap for single exon. Subsequently transcript assemblies were constructed using PASA (Haas, 2003) from RNA-seq transcript assemblies. Loci were determined by transcript assembly alignments and/or EXONERATE alignments of proteins from arabi (Arabidopsis thaliana), soybean, poplar, cotton, rice, sorghum, peach, citrus, aquilegia, tomato, grape, *Eutrema salsugineum*, *Nymphaea colorata*, *Amborella trichopoda*, and Swiss-Prot proteomes to repeat-soft-masked assemblies using RepeatMasker (Smit, 2013-2015) with up to 2K BP extension on both ends unless extending into another locus on the same strand. Gene models were predicted by homology-based predictors, FGENESH+ (Salamov, 2000), FGENESH_EST (similar to FGENESH+, but using EST to compute splice site and intron input instead of protein/translated ORF), EXONERATE (Slater, 2005), PASA assembly ORFs (in-house homology constrained ORF finder), and AUGUSTUS (Stanke, 2006) trained by the high confidence PASA assembly ORFs and with intron hints from short read alignments. The best scored predictions for each locus are selected using multiple positive factors including EST and protein support, and one negative factor: overlap with repeats. The selected gene predictions were improved by PASA. Improvement includes adding UTRs, splicing correction, and adding alternative transcripts. PASA-improved gene model proteins were subject to protein homology analysis to above mentioned proteomes to obtain Cscore and protein coverage. Cscore is a protein BLASTP score ratio to MBH (mutual best hit) BLASTP score and protein coverage is the highest percentage of protein aligned to the best of homologs. PASA-improved transcripts were selected based on Cscore, protein coverage, EST coverage, and their CDS overlapping with repeats. The transcripts were selected if their Cscore is larger than or equal to 0.5 and protein coverage larger than or equal to 0.5, or it has EST coverage, but their CDS overlapping with repeats is less than 20%. For gene models whose CDS overlaps with repeats for more than 20%, their Cscore must be at least 0.9 and homology coverage at least 70% to be selected. The selected gene models were subject to Pfam analysis and gene models whose protein is more than 30% in Pfam TE domains were removed and weak gene models. The genome is hardmasked with its high confidence gene models produced above (transcriptome fully supported, high homology supported and complete gene models). The masked genome was BLASTXed and EXONERATEed against non-self high confidence and non-redundant peptide proteomes from 7 genomes (MIZ_19 of B. juncea, and O_302V, L58, PCGlu, A03, VT123, and WO_83 of B. raps) in Brassica pan-genome annotations to make EXONERATE gene predictions. The predicted gene models were scored with BLASTP using homology proteomes. These models were compared to ones from the first round and better homology supported ones (not contraindicated by transcriptome evidence) were kept to improve gene models from the first round or added if not found therein. Incomplete gene models, low homology supported without fully transcriptome supported gene models and short single exon (< 300 BP CDS) without protein domain nor good expression gene models were manually filtered out.

2.2.6 GENESPACE analyses

To identify syntenic relationships among the *B. rapa* lines we used the R-package GENESPACE[114], scripts are provided on our GitHub. We also included Arabidopsis in a separate GENESPACE run for comparative analyses and to identify paralogs within the morphotypes. Following this we employed an in-house script to assign Brassica genes that were assigned to multiple Arabidopsis orthologs to a single ortholog. We repeated this framework for Arabidopsis genes that may have been assigned to multiple Brassica genes. Finally, to account for tandem duplications which are known to be relatively predominant in polyploids, we outline a multi-step process to ensure that all syntenic tables reflect one-to-one relationships between Brassica annotations and Arabidopsis.

2.2.7 SNP-calling

Paired-end RNA-seq reads were trimmed using Trimmomatic (v0.33) and aligned to the *Brassica rapa* R500 reference genome (Brapassp_trilocularisR500_795_v2.0.fa) using STAR (v2.5.3a) in two-pass mode with read group information embedded in the BAM headers. PCR duplicates were marked using GATK MarkDuplicates (v4.1.2), and spliced alignments were processed with GATK SplitNCigarReads to remove intronic regions. Reads were filtered to retain only high-confidence alignments with mapping quality ≥20 using samtools (v1.21). BAM files from biological replicates of each genotype were merged into a single BAM file using samtools.

Variants were jointly called across all genotypes using GATK HaplotypeCaller (v4.1.2) in cohort mode with soft-clipped bases ignored. Jointly called variants were filtered using GATK hard-filtering following GATK best practices for RNA-seq variant calling. SNPs were filtered based on standard variant quality annotations, including quality by depth (QD), Fisher strand bias (FS), mapping quality (MQ), strand odds ratio (SOR), mapping quality rank sum (MQRankSum), and read position rank sum (ReadPosRankSum). Variants failing any of the following criteria were excluded: QD < 2.0, FS > 60.0, MQ < 40.0, SOR > 3.0, MQRankSum < −12.5, or ReadPosRankSum < −8.0. Only biallelic SNPs passing all quality filters were retained for downstream analyses.

Filtered SNPs were converted to PLINK format and pruned for linkage disequilibrium using a sliding window approach (50 kb window size, step size of 5 kb, and r^2 threshold of 0.2). Principal component analysis (PCA) was performed on the LD-pruned SNP set using PLINK (v1.90b6), and the first two principal components were visualized in R (v4.4.0) using ggplot2 to assess genetic relationships by morphotype among the 13 samples.

2.2.8 Plant growth and experimental design for cold acclimated RNA sequencing

Plants from 14 *B. rapa* accessions were grown for three weeks in in (3.25' x 3.625') pots with a soil mixture of two parts Metro-Mix PX1 + one part Pro-Mix amended with 0.5 mL of Osmocote 18-6-12 fertilizer (Scotts, Marysville, OH) under 24/14°C thermal cycles and 14h /10h light/dark cycles. Plants were cold acclimated for three days starting with a drop in temperature at ZT14 (lights off) to 4°C, followed by an increase to daytime temperatures of 10°C at ZT0. Whole tissue was collected for RNA sequencing starting at ZT17 on the third night of acclimation from four individual plants (biological replicates) per timepoint. Tissue was collected every four hours over a 24h period for a total of six timepoints. RNA extraction was performed at the Cornell Institute of Biotechnology using the Direct-zol-96-RNA extraction kit by Zymo (Irvine, CA) and sent to the Joint Genome Institute for sequencing as part of project CSP504418 (See below).

2.2.9 RNA sequencing and processing of transcripts

RNA was submitted to the DOE Joint Genome Institute for library preparation and sequencing as part of a Community Science Program grant CSP504418. Paired-end RNA sequencing was performed on a Novaseq S4. Data are available through the NCBI GEO database. Reference genomes are available on Phytozome (https://phytozome-next.jgi.doe.gov/). Raw RNA sequencing reads were filtered and trimmed using BBDuk[115] and were mapped to every reference genome. Mappings with the highest score were used for downstream analyses (Table 1: DiPALM tab). Libraries were aligned using HISAT2 v2.2.1[116] and strand specific files were generated using deepTools v3.1[117]. Feature count tables were generated using featureCounts[118] from hits assigned to the reverse strand and were normalized to Counts Per Million (CPMs). Genes that did not have at least four samples greater than zero ($\log_2$ CPM) were removed. Missing replicates were imputed by averaging two other randomly selected replicates (Supplemental Table

1: Imputed Replicates tab). If two of the four samples were missing for a given time point we imputed by taking the median expression of the remaining replicates. A list of imputed samples is provided in Table 1 (ImputedReplicates tab). Accession O_302V was removed from the dataset due to missingness.

Before analysing the transcriptomes we first assess each one by examining the variation among samples in a Principal Component Analysis and through hierarchical clustering. While the majority of the samples clustered relatively well (*i.e.* within a timepoint and/or treatment) we noticed one of FT005's cold treated samples (Replicate 2; time point 6) appeared to be an outlier and was removed. We then imputed this sample as described above. To validate the RNA-sequencing we plot expression for a set of clock genes with known patterns and phases for each genotype.

2.2.10 Cold responsive transcriptomes

To identify genes with significantly different diel (24h) expression patterns after cold acclimation we used the time-series based R package DiPALM[19]. For the Weighted Gene Co-Expression Networks (WGCNA[119]) step, we estimated soft thresholding power and mean connectivity. We selected soft-thresholding powers for each genotype by identifying the highest exponents (powers) that had reached stability with the lowest mean connectivity (Supplemental Table 1: Soft Thresholding Power tab). Samples were collected from individual plants at each time point and assigned a replicate number. These replicates were then strung together to generate an expression profile for each gene. To account for inherent variation between these arbitrarily assigned plants we permuted DiPALM runs for each genotype over 100 iterations and kept genes that had significantly different patterns (kME) or median expression (kMed) under cold in 95 of the 100 iterations. We use kME and kMed genes as our list of significantly cold responsive (CR) genes for the remainder of the analyses (Supplemental Table 1: DiPALM tab).

2.2.11 Calculating Phase Change Groups (PCGs)

We took all genes with differential expression patterns (kMEs) and calculated a 'Phase Change' as the difference in max expression (phase) under control from the max expression (phase) under cold. This gave us the direction and magnitude of every phase change in four, eight, or 12h increments. Some differentially patterned genes had no change in phase but still exhibited differences in the overall pattern, so they were included in the analysis. We collectively refer to these groups as 'Phase Groups'.

Genes in each Phase Group are shifting in the same magnitude and direction, but they are not necessarily shifting at the same time of day relative to the control phase. To account for this, we further delineated the Phase Groups based on the time of day in control they were shifting from. This gave us 36 Phase Change Groups (PCGs) that encompass all 36 comparisons between the different magnitudes and directions (0; +/-4; +/-8; +/-12). The 12h changes are difficult to separate in terms of advancing or delaying, and as such should be interpreted accordingly. For our dataset we assigned advancing 12h PCGs first and then flipped the direction for the opposing group (*i.e.* ZT1 to ZT13 was assigned a +12 but ZT13 to ZT1 would be hard coded as a -12). To assess the biological functions enriched in our PCGs we ran Gene Ontology (GO) analyses using an in-house script on each PCG and accession separately. We only kept terms that were significantly overenriched (Adjusted-p <= 0.05).

2.2.12 Circadian Period by Leaf Movement

To determine the circadian period under control and cold acclimation conditions we used leaf-movement assays. We germinated seeds of each line on wet Whatman paper in petri dishes for two days and individual seedlings were transplanted into PVC pipe connectors filled with Jolly Gardener C/GP Germinating Mix soil. All plants were entrained for three days in 14h light/10h dark cycles with 22°C/16 °C temperature cycles. Control plants were released into free-running conditions at 22°C. To align the timing of leaf-movement with the RNA-seq dataset, cold-acclimated plants received an additional two days of entrainment under 14h light/10h dark, 10°C/6°C cycles followed by release into free-running conditions at 10°C. After 24h in free running conditions constant light and temperature, time-lapse imaging began. Images were captured every 20 minutes for one week using Canon Powershot 2000 cameras modified with the Canon Hack Development Kit (CHDK; https://chdk.fandom.com/wiki/CHDK_1.6_User_Manual). Image stacks were analyzed using Tracking Rhythms in Plants (TRiP)[120], a computational pipeline that generates motion vectors for each plant and fits them to a circadian model to estimate circadian period length, which we have modified and uploaded to GitHub (https://github.com/GreenhamLab/TRiP). To visualize period length across lines by condition, data was plotted in R using the package ggplot2. A paired student's t-test was conducted in R between cold acclimation and control conditions for each line using the RStatix package.

2.2.13 Gene Regulatory Networks

We built temporally informed Gene Regulatory Networks (GRNs) for control and cold data separately using the GENIE3 workflow for each accession[121]. Known TFs from the Arabidopsis Plant Transcription Factor Data Base v.5.0 (https://planttfdb.gao-lab.org/index.php?sp=Ath) and known clock genes with known regulatory functions were used to make predictions (Supplemental Table 1: GRN_TFs tab). To determine a cutoff for significant edge weights we ran 100 permutations of each GRN for each genotype and selected weights with a FDR cutoff of 0.05. To assess similarity among the networks we created a syntenic table anchored by the R500 genome (*Brassica rapa ssp. Trilocularis* v2.1) which allowed us to compare connections (also referred to as 'edges') across all accessions with different genomes. We then converted the TF and target annotations to the R500 annotation and calculated a jaccard index for each network comparison. Comparisons were visualized using the R package pheatmap[122]. This process was repeated with the Arabidopsis ortholog to assess network similarity at the ortholog level.

2.2.14 Consensus Networks

Consensus networks were formed to identify Transcription Factor (TF)-Target connections underlying cold acclimation responses. We selected network comparisons from our Jaccard similarity heatmaps to identify shared connections among the accessions. For each treatment we created binary presence/absence matrices, where each row represents a TF-target edge and accession-specific columns indicate whether that edge is present (1) or absent (0) in a given accession. We then defined focal sets of accessions (single accessions or morphotype-based groups) and compared them to all remaining lines, unless otherwise specified. We then identified group-specific connections by applying explicit inclusion/exclusion rules across columns: an edge was retained when it showed full consensus/support in the focal set (e.g., shared presence for "group-only" edges) but was absent in the comparator set. This edge filtering was done separately for the cold and control networks. Then, within each group, we directly compared the two sets of edges to find connections present only in cold ("Cold_NOT_Control") or only in control ("Control_NOT_Cold"). Results are summarized as (1) the number of condition-specific TF–target connections and (2) the number of unique TFs found only in cold-specific edges (Cold_NOT_Control) or only in control-specific edges (Control_NOT_Cold).

2.3 RESULTS

2.3.1 Relative freeze tolerances across accessions

To evaluate relative freeze tolerance among the accessions, we exposed plants to conditions that model a late spring frost and measured changes in physiology. Briefly, plants were grown under controlled conditions (21°C/16°C; 14h light/10h dark) for three weeks before acclimating them to chilling temperatures (10°C day; 4°C night) for 24 hours (Figure 1A; 1B). Because late spring frosts in temperate climates tend to happen late at night or just after dawn[123], we treated plants with a -4°C freezing stress at ZT17

(Zeitgeiber; hours after lights on) for two hours followed by an additional two hours at -2°C before slowly warming back up to 4°C for the remainder of the day (Figure 1A).

Throughout these experiments we captured several aspects of physiology including leaf damage (Figure 1C), Electrolyte leakage (Figure 1D), above ground biomas (Figure 1E) and light adapted photosystem II efficiency (Fv'/Fm'; Figure 1F). To capture diel variation in Fv'/Fm' we took measurements one hour after lights on (ZT1), at ZT5 (midday) and ZT9 (late afternoon) on control, acclimation and recovery days. We only measured at ZT9 during the Post-Freeze day to avoid damaging the more delicate leaf tissue. Leaf damage was assessed on the recovery day as the proportion of the plant that showed visible signs of tissue damage or death (Figure 1C, Supplemental Figure 2).

While we expected accessions within the same morphotype to be similar (Supplemental Figure 1), we were surprised at the amount of within morphotype variation we observed. For example, all three yellow sarsons had a moderate reduction in Fv'/Fm' at ZT9 following the freeze, but ACC28 showed less damage (Figure 1C) and less change in biomass (Figure 1E) compared to R500 and ACC50. Similar within morphotype variation across metrics was observed in other groups, including the two vegetable turnips (VT123 and FT005). While both turnips had relatively little reduction in Fv'/Fm' and had the same mean damage score (Supplemental Figure 4), FT005 had nearly 100% electrolyte leakage on the recovery day while VT123 had the lowest percent leakage of all the genotypes (Figure 1D). The three Chinese cabbage accessions (CC168, A03, and HN53) group best in terms of electrolyte leakage and Fv'/Fm' (Figure 1D; Figure 1E), although CC168 had a much larger reduction in above ground biomass compared to HN53 or A03 (Figure 1E). Across all the physiology metrics, PC185 (pak choi) and HN53 (chinese cabbage) consistently performed the best under freezing stress, while R500 (yellow sarson) and ZCT (pak choi) were most sensitive, closely followed by FT005 (vegetable turnip). To assess if differences in physiology might be explained by similar genetic backgrounds, we performed a genome-wide single-nucleotide polymorphism (SNP) analysis for each of the accessions and clustered them using PCA. Consistent with our initial expectations, accessions group by morphotype (Supplemental Figure 2) and not relative freeze tolerance. These assays demonstrate that these accessions span a broad range of freeze tolerance phenotypes, providing a clear framework for comparing cold acclimated transcriptomes.

2.3.2 Diel transcriptome re-wiring during cold acclimation

To capture variation in diel regulation associated with differential cold acclimation across a genetically and morphologically diverse mesopolyploid crop, we conducted an RNA-sequencing time course under cold acclimation conditions. Because many physiological processes underlying stress tolerance are under diel and circadian regulation, understanding the temporal dynamics underlying cold responsive pathways is critical for determining how acclimation is coordinated at the whole plant level. To explore these dynamics we exposed plants to three days of cold at 10°C/4°C and leaf tissue was collected starting at ZT17 on the end of the third day for sequencing (Supplemental Figure 3A). Similar to our physiology experiments we sampled every four hours over a 24h period to capture the full diel response to cold. To identify genes with differential cold-responses we ran the R-package DiPALM[19] on each accession and kept genes with differential expression patterns (kMEs) or differences in median expression (kMeds) across the time series. These were collapsed into a single list of cold responsive (CR) genes used in downstream analyses.

Strikingly, we found that each accession had thousands of genes that were responsive (Supplemental Figure 3B), indicating extensive transcriptome rewiring under cold. To group these genes by their time of day pattern we re-clustered every accessions' CR genes and plotted the average expression of each cluster (Supplemental Figure 3C), where each cluster represents a unique time of day response across the diel cycle. However, due to the sheer number of clusters that arose, we realized we needed a way to compare specific phase (timing of max expression) changes under cold among the accessions to see if there were certain phases that were more associated with a given morphotype. To accomplish this we next calculated a "phase change" for every pattern change gene (kME) as the difference between the time of max expression ("phase") in cold relative to control and examined their distributions (Figure 2). We left out PCGlu for this analysis as nearly all of its CR genes exhibited changes in median expression instead of a pattern change. Genes for each accession fell into all phase change groups (PCGs) ranging from +/-4h, +/-8h, +/-12h as well as bins with differential patterns but no change in phase (0).

Overall, genotypes tended to have relatively few genes with extreme phase changes under cold (+/- 12 h; Figure 2). Nearly all accessions had the most genes in the +4 group, with a few exceptions (A03; ACC50; R500). A03 was the only accession with the most genes in the '0' bin while most others had relatively equal numbers across the groups. PC185 had the highest numbers of significantly patterned genes among the accessions, particularly in the group exhibiting eight hour advances under cold (Figure 2). Two genotypes had strikingly similar patterns in both the overall numbers and distribution of Phase Groups (ACC28; L58) but are from two different morphotypes (yellow sarson and choy sum; respectively). Two of the three Chinese cabbage accessions (HN53; CC168) have slightly more genes that exhibit a delay in phase; however, L58 (Choy sum), ZCT (Pak choi), FT005 (Vegetable turnip), and to a lesser degree, WO83 (Winter oil), also have a larger proportion of delayed genes. Overall, there wasn't a clear pattern in phase change that could be attributed to a single morphotype; instead we found accession specific changes to the timing of peak expression of thousands of genes. Given the large portion of the *B. rapa* transcriptome under circadian control[19] we wondered whether the variation in phase changes across accessions was due to differences in the pace of the clock during cold acclimation.Another explanation behind the pronounced within-morphotype variation in phase changes is that accessions were experiencing advances or delays to the pace of the circadian clock during cold acclimation.

To determine whether accessions were adjusting the pace of their clocks under cold acclimation,we performed leaf movement to determine the circadian period of all accessions under control and cold acclimation conditions (Figure 3). We entrained plants in 14h:10h light:dark, 22ºC days: 16ºC nights for three days before cold acclimating. Following entrainment plants were released into free running conditions (constant light and temperature) set to mimic control (22°C) or acclimation (10°C) conditions. The endogenous period length under control and cold acclimation was estimated for each accession using TRiP[120] (See Methods:Leaf Movement). Under control conditions, most accessions exhibited endogenous circadian periods close to 24 h (Figure 3). However, several accessions displayed longer control periods, including VT123 (25.5h), PC185 (24.8h), A03 (24.6h), and L58 (25.3h), whereas others exhibited slightly shorter control periods, such as HN53 (23h) and PCGlu (23.5h). Each accession's ability to maintain accurate timing (~24h periods) across physiologically-relevant temperatures - a phenomenon known as temperature compensation[124] - was then assessed under cold acclimation conditions. Several accessions appear to be well compensated under cold, including PC185, L58, and all three chinese cabbages (HN53; CC168; A03; Figure 3). In contrast, several accessions exhibited significantly shortened periods, including R500 and ACC50 (~3.5h), ACC29 (~2h), FT005 (~2h), and ZCT (1h) relative to control periods. VT123 was significantly overcompensated by approximately 3h, resulting in a dramatically long period (26.7h) under cold. Similarly, PCGlu and WO83 were also delaying the pace of the clock by ~1h and ~2h, respectively. The changes in circadian period under cold conditions could underlie some of the phase changes we have observed in gene expression[125]. As a result, genes that are not directly cold responsive but are under circadian control may still exhibit a change in expression pattern under cold acclimation and be called ‘cold responsive’ by our pipeline. However, because the RNA-seq experiment was performed in diel conditions (with light and temperature cues) we are unable to dissect whether our observed phase changes are driven by circadian regulation or are directly cold responsive.

2.3.3 Enrichment of Biological Pathways are phase and genotype specific

Because the Phase Groups do not tell us what time of day the changes were occurring, and to determine whether specific pathways were under conserved time of day control, we further delineated the phase groups based on the time of day the peak expression was changing under cold, resulting in a total of 36 Phase Change Groups (PCGs; Supplemental Figure 1). We next asked if there were morphotype or accession level differences in enrichment of biological pathways under cold acclimation that differed by time of day. To accomplish this, we ran Gene Ontology analyses on each accession and PCG separately.

We first identified GO terms that were enriched among the accessions irrespective of PCGs to identify global cold acclimation responses. We found that all accessions except for R500 were enriched in sulfur metabolism, and all accessions except for L58 were enriched in redox homoeostasis terms. The majority of accessions (except for R500; L58, PCGlu, and WO83) were enriched in photosynthetic related processes (Supplemental Table 1; Sulfur, Photo, and Redox GO tabs). L58 (choy sum), which was not enriched in either sulfur assimilation or redox homeostasis, had most of its enriched terms in the ATP binding or protein kinase activity groups, while accession lacking enrichment in photosynthesis (A03; PCGlu; R500; WO83) had a large portion of their enriched terms falling into ribosome or ATP-binding processes (Supplemental Table 1; GOresults_ByPCG tab).

Within photosynthesis related terms, we found that nine of the thirteen accessions seemed to be evenly represented in terms involved in photosystem I (PSI), photosystem II (PSII), chlorophyll biosynthesis, electron transport chain and light sensing/harvesting terms (Supplemental Table 1; Photo_GO tab). When we separated the terms into their respective PCGs we found that the direction (advanced; delayed) and magnitude (4, 8 or

12 hours) of the cold response varied among these accessions (Supplemental Figure 4A). Although six of the nine accessions in this group displayed enrichment in PSII related terms, four showed enrichment within PCGs corresponding with large advances (+8; +12) or delays (-12) in this term. These accessions include relatively tolerant (PC185; HN53) and relatively intolerant lines (FT005; ZCT). In PC185, expression of these genes were advanced by 12h from late in the day (ZT13) to early morning (ZT1; ZT5) while HN53 and ZCT had enrichment in PCGs that were delayed from early morning (ZT1; ZT5) to the evening (ZT17; ZT21; Figure 4A). In contrast, FT005 was lacking in enrichment of PSII terms, yet still exhibited 12h delays (early morning to end of day) in terms associated with PSI, light-harvesting reactions, and PSII reaction center proteins (*psb* family). The PHOTOSYSTEM II REACTION CENTER PROTEIN (*psb*) family plays an important role in stabilizing PSII and regulating non-photochemical quenching, processes that have been shown to be sensitive to low temperatures[126,127]. This prompted us to further examine *psb*-associated terms in more detail.

Across accessions, we found that ACC28, FT005, HN53, PC185, VT123, and ZCT each exhibited enrichment of terms that contained at least one member of the *psb* family (Supplemental Figure 4B), ranging from two *psb* orthologs (*psbE/F*; FT005) to five (*psbA/B/C/E/F*; ZCT). Perhaps even more striking; however, was that the number of paralogs corresponding to a given *psb* ortholog differed, both within and between accessions. For example, while PC185 and HN53 both had three *psb* orthologs in a PCG, HN53 had one copy of each while PC185 had one, two, or three copies (Figure 4B). ZCT showed enrichment across several PSII subunits (A, B, C, and E), and FT005 lacked enrichment for *psbB* but retained enrichment for *psbE/F* (Figure 4B). Interestingly, control and cold expression patterns were quite consistent within an accession, but differed markedly between accessions. For example, in ZCT, control gene expression peaked in the early morning (ZT1–ZT5), declined throughout the day, and was lowest at night before returning to peak levels (Figure 4B). This pattern is consistent despite the number of paralogs present in the analysis. Under cold; however, ZCT exhibited a pronounced phase advance such that max expression occurred at the time of day corresponding to the trough (lowest) expression under control conditions. This was similar to HN53's cold patterns, where the cold phase coincides with the lowest point in control; however, PC185 cold expression is phased at early morning and is lowest at night (Figure 4A).

Given the extent of paralog-level temporal divergence observed for *psb* genes, we next asked whether similar patterns of temporal partitioning among paralogs were evident in other pathways enriched under cold acclimation, including sulfur assimilation and redox metabolism.

2.3.4 Enrichment of sulfur and redox metabolism under cold acclimation

Plants assimilate sulfur from sulphate (SO_4^{2-}) in a tightly regulated process centered on the synthesis of APS (ADENOSINE-5'-PHOSPHOSULFATE) which is considered to be a core branch point between primary and specialized metabolism[128]. APS is synthesized in the cytosol by ATP sulfurylases (ATPSs) which can be reduced by APS reductases (APRs) to donate sulfur to primary metabolites (methionine; cysteine; glutathione), or it can be phosphorylated by APS kinases (APKs) to produce 3'-PHOSPHOADENOSINE-5'-PHOSPHOSULFATE (PAPS). PAPS acts as a primary donor of sulfur to specialized metabolites such as glucosinolates and sulfoflavanoids[128]. Diel regulation of sulfur metabolism provides a mechanism to allocate sulfur between growth and stress responses throughout the day[129]; yet, our understanding of how this regulation is modified during cold acclimation is lacking.

Nearly all accessions were enriched in at least one sulfur-associated process, though enrichment was most frequently observed in assimilatory steps involving APS synthesis (ATPS), reduction (APR), and phosphorylation (APK; Supplemental Figure 5). Within these major branches of sulfur assimilation we identified ~80 sulfur assimilation associated genes assigned to PCGs across accessions, revealing accession specific differences in which branch of the pathway was enriched. Among these three aspects of sulfur assimilation we noticed markedly similar patterns of expression under control conditions across accessions. *ATPSs*, for example, peak in the early morning (ZT1) while the cold phase differs depending on the accession (Figure 5B; Supplemental Figure 5). APRs are slightly more variable across accessions with A03, ACC50 and HN53 showing control phases at the end of the night while PCGlu and WO83 were phased at early morning. Under cold A03, HN53, and ACC50 also have noticeably similar expression patterns while PCGlu and WO83 are more unique (Supplemental Figure 5). However, unlike other accessions cold expression appears to be anchored between ZT1 and ZT5 but there are subtle differences that are unique to other timepoints among ATPS orthologs, such as a reduction in expression at ZT1 and an increase in expression at ZT9, but the increase depends on the observed ortholog. Of the accessions that had multiple copies in the enriched sulfur assimilation pathway (A03 ACC28, CC168, VT123, WO83, and ZCT) they all displayed a control phase at ZT1 but exhibited differences in when their expression under cold peaked. Within this pathway, the cold phases ranged from ZT17 (CC168), ZT1 (A03; VT123; ZCT) to ZT5 (ACC28, WO83). While the number of paralogs that fell into enriched PCGs in this pathway are less pronounced than in the *psb* family discussed earlier, the partitioning of paralog expression patterns under cold is equally notable. This diversification in paralog responsiveness suggests that temporal subfunctionalization of paralogous copies may be fine-tuning sulfur assimilation dynamics during cold acclimation, leading to important differences in downstream pathways.

Because glutathione synthesis depends on the availability of reduced sulfur, fluctuations in sulfur assimilation have immediate consequences for cellular redox balance, particularly during stress. For example, HN53 had multiple copies of *GSH1* in the enriched glutathione biosynthesis term that were phased at ZT17 under control and declined sharply overnight, with expression starting to recover at ZT9 (Figure 6B). Although ZCT's control expression pattern of *GSH1* was similar to HN53's during the day, the nighttime dynamics were different in both cold and control conditions. This was most noticeable at ZT17 where HN53's *GSH1* copies were highest in control but cold expression was greatly reduced. In contrast, ZCT's *GSH1* ortholog displayed similar expression between conditions at ZT17, slightly lower in cold throughout the night, and then higher throughout the day (Figure 6B). A comparable trend was observed for CC168's *GSH2* ortholog, which exhibited similar expression patterns as HN53 in cold and control during the day but diverged at night, where ZT17 patterns resembled HN53 but diverged at ZT21 (Figure 6B). Similarly, FT005's single GLN1/1.4 copy, enriched in glutamate biosynthesis exhibited largely overlapping cold and control expression profiles, differing modestly in magnitude from HN53's GLN1.3 paralogs only at ZT21 under cold (Figure 6). Collectively, variation in expression patterns of genes involved in redox homeostasis appear to differ between accessions and were most evident at night, usually manifesting at a single time point rather than across the day. This is in contrast to the sulfur assimilation pathway which was enriched for genes that were often different both between and within accessions and predominantly during early morning (Figure 5; Supplemental Figure 5).

Paralog diversification was less pronounced in this pathway compared to photosynthesis and sulfur assimilation; however, HN53 still showed multiple paralogs assigned to the same GO term and PCG within glutathione (*GSH1*) and glutamate biosynthesis (*GLN1.3*; Figure 6). The two relatively tolerant accessions nevertheless exhibited distinct but complementary redox strategies: HN53 emphasized glutathione and glutamate biosynthetic capacity, whereas PC185 showed stronger representation of *GLUTATHIONE S-TRANSFERASE 3/8*, thioredoxins, and peroxiredoxins (Supplemental Table 1; Redox_GO tab), suggesting a slightly different redox strategy centered on peroxide detoxification rather than glutathione synthesis. Despite these pathway level differences, both tolerant accessions maintained elevated redox gene expression at similar times of day (ZT5; Figure 6B), whereas ZCT displayed a delay in phase from late morning (ZT5) to late day for peoxiredoxins and GADPH terms as well as a 12h shift in glutathione biosynthesis from late day to early morning (ZT13; Figure 6B).

Collectively, our GO enrichment analyses reveal that cold acclimation engages conserved photosynthetic, sulfur, and redox pathways across *B. rapa* accessions, but timing and direction differ depending on the accession and the pathway. We found the extent of variation among accessions striking, not only in which orthologs contributed to enriched terms, but also in the number of paralogous copies responding under cold. Differences in regulatory regions within the paralogous copies are one explanation behind the extensive temporal diversification we see in complex pathways such as photosynthesis, sulfur, and redox metabolism.

We next sought to determine whether this temporal divergence among paralogs was pervasive across the transcriptome or was mostly restricted to these pathways. To assess how often single and multi-copy genes were assigned to the same PCG among the accessions we converted the *B. rapa* annotations to their Arabidopsis ortholog and quantified the number of accessions in which a given ortholog was assigned to the same PCG. We identified 19 *Arabidopsis* orthologs that consistently fell into the same PCG across multiple accessions, including five accessions (eight orthologs), six accessions (three orthologs), seven accessions (seven orthologs), and eight accessions (one ortholog). Although no clear functional trend emerged among these orthologs, more than half (12 of 21) were assigned to the same phase change group (ZT1 to ZT5) in at least five accessions. Within this group, *SULFUR LIMITATION 1* (*SLIM1*), which modulates both sulfur assimilation and glucosinolate catabolism[130], was shared among PC185, two chinese cabbages (HN53; CC168), and two yellow sarsons (ACC50; R500). Notably, several of these accessions also exhibited enrichment in the broader sulfur assimilation pathways, suggesting that aspects of sulfur metabolism may undergo cold-induced temporal reprogramming at similar times of day across distinct morphotypes. *COR28*, a well-characterized cold-responsive target of *CBF* transcription factors[131], was one of the single-copy orthologs consistently assigned to the same PCG with a four hour advance (ZT13 to ZT9) under cold across multiple accessions (CC168, FT005, L58, and VT123). This prompted us to examine the expression pattern of *COR28* further to see if other accessions were cold responsive for this gene or if there were differential patterns that placed *COR28* in multiple PCGs among the accessions. While several accessions did not exhibit cold-responsive changes for this gene, the yellow sarsons were notable for a pronounced four to eight hour phase advance under cold, as well as a substantial increase in daytime expression relative to control conditions (Supplemental Figure 6). In contrast, among multi-copy genes, no more than three accessions shared enrichment of the same Arabidopsis ortholog within the same PCG, suggesting that duplicated genes have greater accession specific temporal divergence under cold conditions.

Collectively, these patterns point to accession specific maintenance of diel regulation of cold acclimation across core metabolic pathways. We next look for TFs that may be regulating these time-of-day dependent responses.

2.3.5 Diverse regulatory networks among *B. rapa* accessions under cold acclimation

To identify transcriptional regulators underlying accession-specific cold responses, and to assess conservation of regulatory connections among accessions, we constructed Gene Regulatory Networks (GRNs) for each of the 13 *B. rapa* accessions under control and cold conditions, resulting in 26 networks. Networks were inferred using known TFs and circadian regulators to compare network topology across tolerant and intolerant accessions. Additionally, by comparing similar connections among the accessions we identify conserved regulatory strategies during cold acclimation, which may represent robust targets for breeding cold tolerance into crops.

To identify shared regulatory connections across accessions we computed Jaccard similarity scores for all pairwise combinations. Overall, similarity among control networks was relatively low, indicating substantial divergence in regulatory networks among *B. rapa* accessions (Figure 7A). However, two control networks from the relatively intolerant yellow sarsons (R500; ACC50) were more similar compared to others, while the third yellow sarson (ACC28) was more similar to the cabbages, turnips, and choy sum (Figure 7A). Similarly, in chinese cabbage, two of the accessions (CC168:

HN53) had relatively similar control networks but A03 had a similar score with the cabbages and the other accessions (Supplemental Table 1, BrapaJaccardSimilarity tab). This was somewhat surprising given our earlier snp-based PCA (Supplemental Figure 2) which showed that accessions grouped by morphotype.

Comparison of cold GRNs revealed further reduction in similarity among the accessions, consistent with accession specific regulatory rewiring in response to cold. However, the initial similarity between the two yellow sarsons and the cabbage-turnip group was maintained, although the score itself was lower in cold (Figure 7B). R500 and ACC50's networks are noticeably the most similar to each other compared to any other accessions, closely followed by CC168 and HN53.

From our GO analyses we found multiple examples of paralog divergence in temporal regulation of photosynthesis and sulfur assimilation pathways, highlighting that different *B. rapa* accessions are utilizing distinct paralogs of the same Arabidopsis ortholog in response to cold. This could indicate that, although there is divergence in the *B. rapa* copy that is used, the same regulatory function is likely maintained. To test whether divergence on the paralog level was contributing to the low conservation in regulatory networks, we converted the *B.rapa* genes in the GRNs to the Arabidopsis orthologs and calculated new similarity scores for the Arabidopsis anchored GRNs. At the ortholog level, we found that all accessions were noticeably more similar, consistent with our hypothesis (Figure 7C; 7D). This increase in similarity suggesting a level of conservation that was largely obscured when paralogs were considered separately and implies that divergence in regulatory network structure is driven at least in part by paralog diversification.

While GRNs that appeared more conserved in the *B. rapa* based networks were maintained in the Arabidopsis comparisons (ACC50-R500; CC168-turnips), one of the tolerant accessions, PC185, still did not have strong similarity with any other accession in the cold networks, implying unique regulatory strategies as opposed to divergence in paralogous copy. Interestingly, the other tolerant accession (HN53), which originally clustered with CC168 in the control networks, exhibited increased similarity to the third cabbage (A03) under cold conditions (Figure 7D), suggesting more similar regulation among cabbages under cold despite differences in control networks.

2.3.6 Identifying shared network connections in *B. rapa*

Although Arabidopsis anchored GRN comparisons increased overall network similarity, tolerant and intolerant accessions did not cluster strongly, indicating that differences among the networks extend beyond divergence in Brassica paralog usage. To identify edges that are conserved within two morphotypes (chinese cabbages; yellow sarsons) we compared the edges between cold and control consensus networks by comparing the shared Arabidopsis-anchored GRN edges within morphotypes (Figure 7). Examination of these TFs found among all morphotypes revealed broad conservation of TFs associated with heat stress (heat shock proteins; HSFs), hormone signaling (ABA, auxin, brassinosteroids), and core circadian control, although the corresponding targets differed. When we examined TFs that were unique to cold or control networks in specific morphotypes, we find that the cabbages have a substantial amount of core circadian clock connections unique to these accessions (A03; HN53; CC168). These include multiple *CONSTANS* and *CONSTANS-LIKE* family members (*CONSTANS*; C*OL1/2/4/5/9*), *EARLY FLOWERING 3/4*, *GIGANTEA*, *PRR3/5/7/9*, and *LATE ENLONGATED HYPOCOTYL* (*LHY*). Notably, *LHY* also had unique connections in the cabbage and cold only networks and was targeting key sulfur and glucosinolate associated regulators, including *SULFUR DEFICIENCY INDUCED 2* (*SDI2*), *MYB28*, and *APS REDUCTASE 3* (*APR3*). *APR3* was also targeted by an evening oscillator of the clock (*TIMING OF CAB EXPRESSION 1*; *TOC1*), linking core clock function to sulfur assimilation and glucosinolate metabolism under cold. Additional TFs such as *PIF4* (*PHYTOCHROME INTERACTING 4*) was predicted to target another glucosinolate biosynthesis gene

(*BCAT3*). Canonical cold regulators, including *CBF1/2*, *CAMTA2/4/5/6*, and *DREB1A* also had unique connections in the cabbage and cold only networks, reinforcing the coordination between circadian, metabolic, and cold responsive pathways.

The most pronounced accession and morphotype divergence emerged when comparing connections that were found in the cold networks of our two tolerant accessions (PC185; HN53) but were absent from the intolerant lines (FT005; R500; ZCT). Notably, *REDUCED VERNALIZATION 1* (*VRN1*) and *FLOWERING LOCUS C* (*FLC*) were exclusive to the tolerant cold networks which is consistent with previous work showing that both are key regulators of development and tolerance under cold[71,132]. Additional well-characterized cold regulators including *ICE1*, *CBF2/4* and multiple *DREB* family members (*DREB1A/2B/2C*) also had unique connections represented in this group. Their tolerant-specific targets included oxidoreductases (FAD proteins), *LONG VEGETATIVE PHASE 1* (*LOV1*), which coordinates flowering time with cold responses[133], and *SUPERSENSITIVE TO DROUGHT AND ABA 1*, collectively indicating tighter coordination among stress signaling, developmental control, and hormone pathways under cold acclimation in the tolerant lines.

A defining feature of these tolerant only cold networks was the integration of sulfur, glucosinolate, and redox metabolism with circadian regulation under cold. A component unique to the tolerant-only cold networks included *RRTF1* (*REDOX RESPONSIVE TRANSCRIPTION FACTOR 1*), which has been previously shown to induce ROS levels to induce stress response pathways[134]. In the tolerant lines this TF is targeting the well known cold responsive TF *DREB2A* (*DEHYDRATION-RESPONSIVE ELEMENT BINDING PROTEIN 2A*) indicating a direct link between redox signaling and cold acclimation. To our knowledge, this is the first recognition of *RRTF1* and *MYB28/34/122* playing direct roles in the regulation of cold acclimation in Brassicas, and certainly highlights the importance of regulating sulfur allocation and redox homeostasis during cold acclimation.

MYB28, a key regulator of glucosinolate biosynthesis, had cabbage and cold specific targets that included members of the Cysteine-Rich Receptor-Like Kinases (CRKs) and *AUXIN RESPONSE FACTOR* (*ARF*) gene families, both of which have been implicated in osmotic stress regulation[135,136]. Interestingly, *MYB28* was also targeting additional glucosinolate biosynthesis genes including *2-ISOPROPYLMALATE SYNTHASE 2* (*IMS2*) and both *MAM1/2* orthologs. Similarly, indolic (Tryptophan derived) glucosinolate biosynthesis genes *MYB34* and *MYB122* shared targets such as *MAM1* and *IMS2*, while a third glucosinolate biosynthesis gene (*MYB51*) targeted glutaredoxins (*GRXs*) and a thioredoxin (*TRX*), linking glucosinolate biosynthesis to redox homeostasis. *ICE1,* a core regulator of cold acclimation[91,92], is uniquely targeting *BETA-GLUCOSIDASE 28*, a myrosinase involved in glucosinolate turnover, further connecting sulfur allocation with cold responsive signaling. Notably, multiple circadian regulators - including *PRR7* and *GI* - collectively targeted yet another glucosinolate biosynthesis gene; *MYB122*, which is also regulating *IMS2* and *MAM1/2*. Together, the cabbage - cold only and tolerant - cold only networks reveal that TFs within the circadian oscillator and core cold regulators are uniquely targeting glucosinolate biosynthesis pathways, suggesting that glucosinolate pools may be under circadian regulation during cold acclimation.

Together, these results demonstrate that cold acclimation regulatory networks in *B. rapa* are shaped by both conserved TFs and morphotype specific time of day targeting. These results exemplify unique strategies predominantly centered around circadian, hormonal, and sulfur related TFs that underlie accession specific regulation of cold acclimation. By integrating time-of-day with transcriptomics we captured unique differences in regulatory networks across diverse accessions that are likely to have been otherwise obscured by single time point analyses. This research provides new insight into how temporal and genomic diversity jointly structure stress responsive gene regulation in a genetically and physiologically diverse *B. rapa* panel.

2.4 Discussion

Improving crop tolerance requires a detailed understanding of how stress responses are temporally coordinated with growth and metabolism[26]. Because stress responses must be coordinated with growth[137], maintaining precise regulation of physiology and metabolism is critical for allocating limited resource pools to the appropriate pathways at the time of day they are needed most[21]. Consequently, analyses that incorporate time-of-day patterns of gene expression are uniquely positioned to uncover regulatory variation that would otherwise be masked in single time-point comparisons. To uncover temporal dynamics underlying cold acclimation responses we integrated time-of-day resolved expression patterns during cold acclimation across diverse accessions and found that cold acclimation in *B.rapa* is strongly structured by time of day. This is consistent with previous work in *B. rapa*[19,34], poplar[138], and Arabidopsis[139] that show that the largest variation in stress responses is attributed to time of day. To determine whether accessions with similar cold acclimation transcriptional profiles also exhibited comparable freezing tolerance we conducted a series of freeze stress experiments and found pronounced variation within and between morphotypes in light-adapted PSII efficiency (Fv′/Fm′), electrolyte leakage, and fresh weight. Interestingly, one of our relatively tolerant lines (PC185) had a slight increase in fresh weight and a negligible reduction in Fv'/Fm' following freeze stress (Figure 1E and F). One possible explanation for this is that, during acclimation to cold, transcriptional regulation of photosynthesis related genes to favor PSII repair pathways under cold, thereby sustaining PS function under cold stress. These differences point toward distinct strategies of cold acclimation among accessions and demonstrates that freeze tolerance in *B. rapa* is more accession specific than previously appreciated.

Given that a substantial proportion of plant transcriptomes is under circadian regulation[19,20,43], including stress-related pathways[26], we hypothesized that shifts in the timing, rather than differences in the overall magnitude of expression, would exemplify major sources of regulatory divergence among accessions. Under cold acclimation, we found that accessions were altering expression patterns of thousands of genes across the diel cycle, consistent with previous work showing that up to ~70% of expressed genes in *B. rapa* exhibit circadian rhythmic expression[19]. These findings underscore that cold-induced transcriptome reprogramming occurs within a highly time-structured transcriptome.

To assess how circadian regulation of cold acclimation responses might be contributing to time of day changes in gene expression observed under cold conditions we performed leaf movement and found differential temperature compensation across the lines leading to one to three hour differences in period between cold and control. These changes in circadian timing under cold certainly influences gene expression patterns; however, we are unable to identify which of our 'cold responsive' genes are directly responding to cold over which are shifting phase due to a shorter or longer period in the over/under compensated accessions. As such, some proportion of our 'cold responsive' genes may instead be more representative of clock reprogramming under cold. This has been demonstrated in Arabidopsis *REVEILLE 8* mutants which exhibited a four hour delay in phase compared to wild type, but when that phase was accounted for only 13 genes were differentially expressed[125]. Given the pervasive influence of the clock on gene expression, future studies should consider how stress signaling and circadian timing jointly shape transcriptional responses when interpreting 'stress responsive' expression patterns.

Re-clustering cold-responsive genes into discrete phase change groups revealed extensive diversity in diel expression patterns among accessions with reprogramming of photosynthesis, sulfur assimilation, and redox pathways during cold acclimation. Photosynthesis related GO terms were enriched in multiple accessions and exhibited striking paralog level divergence. In PC185, multiple *PHOTOSYSTEM II REACTION CENTER PROTEIN (psb)* paralogs were upregulated early in the day (ZT1–ZT5) under cold, whereas corresponding orthologs in HN53, ZCT and FT005 were downregulated at the same time (Figure 4B). Members of the psb family contribute to repair and maintenance of the PSII reaction center, particularly the D1 protein (Marder et al., 1987), which is critical for avoiding photoinhibition under stress[140]. HN53 and FT005 had similar control patterns (Figure 4B) although HN53 and ZCT were more similar in the cold. PC185's patterns seemed relatively unique with an early-day upregulation of *psb* expression. It is possible that, under cold acclimation, PC185 is upregulating at this time of day to enhance repair capacity and contribute to maintenance of PSII efficiency before compounding effects of increased light and to avoid detrimental effects of non-photochemical quenching. This could be one explanation behind the minimal reduction in growth under freezing stress and the maintenance of Fv'/Fm' under freezing stress in PC185 (Figure 1E and F). While time-of-day effects on photosynthesis have been previously reported (Dodd et al., 2005), our results demonstrate that both the magnitude and timing of these responses vary among closely related *B. rapa* accessions, and suggest that coordinated recruitment of multiple *psb* paralogs with early phased expression patterns during acclimation may contribute to enhanced freeze tolerance by ensuring psb proteins are available during the coldest part of the day.

Cold stress perturbs photosynthesis by slowing enzymatic reactions of the Calvin–Benson cycle while light harvesting and electron transport continue, resulting in over-reduction of the photosynthetic electron transport chain and increased ROS production[141], which elevates reliance on glutathione-dependent antioxidant systems[142]. Because glutathione biosynthesis depends on cysteine derived from sulfur assimilation, and sulfate reduction itself requires photosynthetically generated reducing power[143], temporal shifts in sulfur metabolism may reflect coordinated regulation of reductant supply and redox demand across the diel cycle.

Enrichment in sulfur assimilation and redox pathways displayed similarly dynamic, accession specific changes in gene expression under cold that, in general, maintained the same expression pattern between treatments but was shifting the timing of peak expression. We noticed that both tolerant accessions (PC185 and HN53) aligned peak expression of genes associated with PSI and PSII to the early morning period (ZT1–ZT5; Figure 4A). Early morning phasing was also evident in sulfur assimilation where PC185 was delaying the phase of *ATPSs* and *APKs* to ZT5 under cold while HN53 was advancing *APRs* to ZT1 (Figure 5). However, redox related genes did not follow suit. In this pathway PC185 maintained the same phase under cold (ZT5) while other accessions were shifting maximum gene expression to multiple timepoints throughout the diel cycle (Figure 5A), with HN53's *GSH1* paralogs a single exception. The coordinated early morning phasing of PC185's *psb* paralogs during cold may lessen the requirement for widespread redox gene activation, potentially by mitigating photoinhibition early in the photoperiod. However, further experiments aimed at quantifying ROS levels are needed to support this.

Together, these pathway-level differences reflect a broader pattern in which duplicated genes (paralogs) are partitioned into different PCGs relative to single copy genes. A subset of single copy orthologs (~20 Arabidopsis genes) maintained conserved phase assignments across five or more accessions. This convergence suggests that single copy responsive genes are more likely to be regulated in a similar way across accessions, whereas paralogs have diverged in phase under cold. Similar patterns of homoeolog expression divergence have been reported in *Brassica napus*, where paralog partitioning contributes to time-of-day specific stress responses[20]. Collectively, these comparisons suggest that paralog specialization is an important source of regulatory divergence during cold acclimation, whereas single copy genes may be more conserved in their temporal regulation compared to multi-copy paralogs, at least within our dataset. To determine how these accession specific temporal patterns are regulated, we next examined gene regulatory networks to identify upstream transcription factors and unique regulatory connections driving these differences. Gene regulatory network comparisons across all accessions revealed greater similarity in networks that were constructed using Arabidopsis orthologs compared to *B. rapa* annotations, indicating that diversification of regulatory strategies is driven largely by which paralogs make up the connections. Similar divergence in copy number has been reported for *FLOWERING LOCUS C* (*FLC*), where copy-number variation and presence–absence variation contribute to differences in vernalization requirement and flowering time among Brassica morphotypes[54,70]. Additionally, glucosinolate biosynthetic genes also exhibit extensive copy-number variation[41] but their role in cold acclimation has not been previously described.

Within these networks, *RRTF1*, a redox-responsive transcription factor known to elevate ROS levels under stress[134] was targeting several core cold regulators, including multiple members of the *DREB* family. This supports a role as a signaling intermediate linking redox state with cold response[134,144]. This, to our knowledge, is the first evidence implicating this factor directly in cold acclimation in Brassicas but further studies on the impacts of loss of RRTF1 during cold acclimation is needed.

A defining feature of the accession-specific cold networks was diversification within core circadian oscillator components. These findings reinforce the role of the clock in gating abiotic stress responses[38] and regulating a substantial proportion of the plant transcriptome[21,145]. In sorghum, for example, researchers found a set of TFs including *LHY*, *REVILLE2* and other *CO-Like* genes that were responsive to chilling stress but only at night, consistent with previous work in Arabidopsis and bread wheat showing that the clock gates cold response to the evening[35,38,104]. Consistent with these studies, our tolerant and cabbage only networks also identified clock outputs such as *PRRs*, *LHY*, *GI*, and *CONSTANS/CONSTANS-Like* genes targeting a number of glucosinolate biosynthesis genes. These included *MYB28*, *MYB34*, and *MYB122*. Loss of function *MYB28* mutants show a drastic reduction in aliphatic (those that are primarily methionine derived) glucosinolates[146], although loss of *MYB29* is also required to fully abolish aliphatic levels in leaves[147]. Similarly, the loss of *MYB34* nearly alleviates leaf-derived indolics (derived from Tryptophan) alone, but a further reduction is seen in *MYB34/51/122* knock out lines[148]. Given evidence that sulfur from glucosinolate breakdown can be redistributed back to primary metabolism[68], clock or *ICE1* mediated regulation of glucosinolate biosynthesis could influence sulfur allocation and redox capacity during cold acclimation. The presence of *LHY* targeting *MYB28* was particularly intriguing as clock control of glucosinolates has been suggested through previous metabolomics work[20,129,149] but a direct regulatory role of the clock has not been shown. Considering *MYB28* also has multiple glucosinolate biosynthesis connections - in addition to *ICE1* targeting a glucosinolate catabolic enzyme - these networks provide intriguing evidence that sulfur and glucosinolate pools are differentially regulated in response to cold, and that that regulation may be partially under clock control.

Despite increasing recognition that the circadian clock gates abiotic stress responses[26,61], relatively few studies have profiled transcriptomes across the diel cycle under stress conditions across physiologically and genetically diverse accessions. As a result, our understanding of how temporal regulation shapes stress acclimation, and how much intraspecific regulatory variation exists within crops, remains relatively limited. Collectively, our results show that cold acclimation in *B. rapa* is driven by accession-specific regulation that is based on time-of-day specific dynamics that ultimately rewire photosynthesis, redox, and sulfur metabolic pathways. These findings further indicate that cold tolerance in polyploid crops cannot be fully understood based on gene identity alone, but also depends on which copy and the time of day it is expressed. This has strong implications for breeding programs that are aimed at optimizing tolerance to abiotic stress in polyploid crops.

Chapter 2: Tables

Table 1.) Genome assembly and annotation statistics. Genome names are listed at the top.

	O_302V	PCGlu	L58	WO-83	A03	VT123	R500
Genome Size	418.2Mb	421.8	428.4	436.3	429.7	432.3	398.2
Contig N50	14.9Mb	8.95	15.3	34.0	16.0	18.6	37.5
HiFi Cov.	62X	41.77	41.77	79.12	96.81	89.44	126.7
Illumina Cov.	65X	55	55	50	61	52	55
RNA-seq reads	1.2G	2.4	2.1	2.3	2.2	2.0	3.4
Iso-Seq reads	3.3M	7.1	5.9	5.1	5.9	4.4	9.0
Emb. BUSCO	99.0	98.7	98.8	98.8	98.8	98.8	99
N. Genes	38,746	39,473	39,146	38,886	39,122	39,062	39,581

Chapter 2: Figures

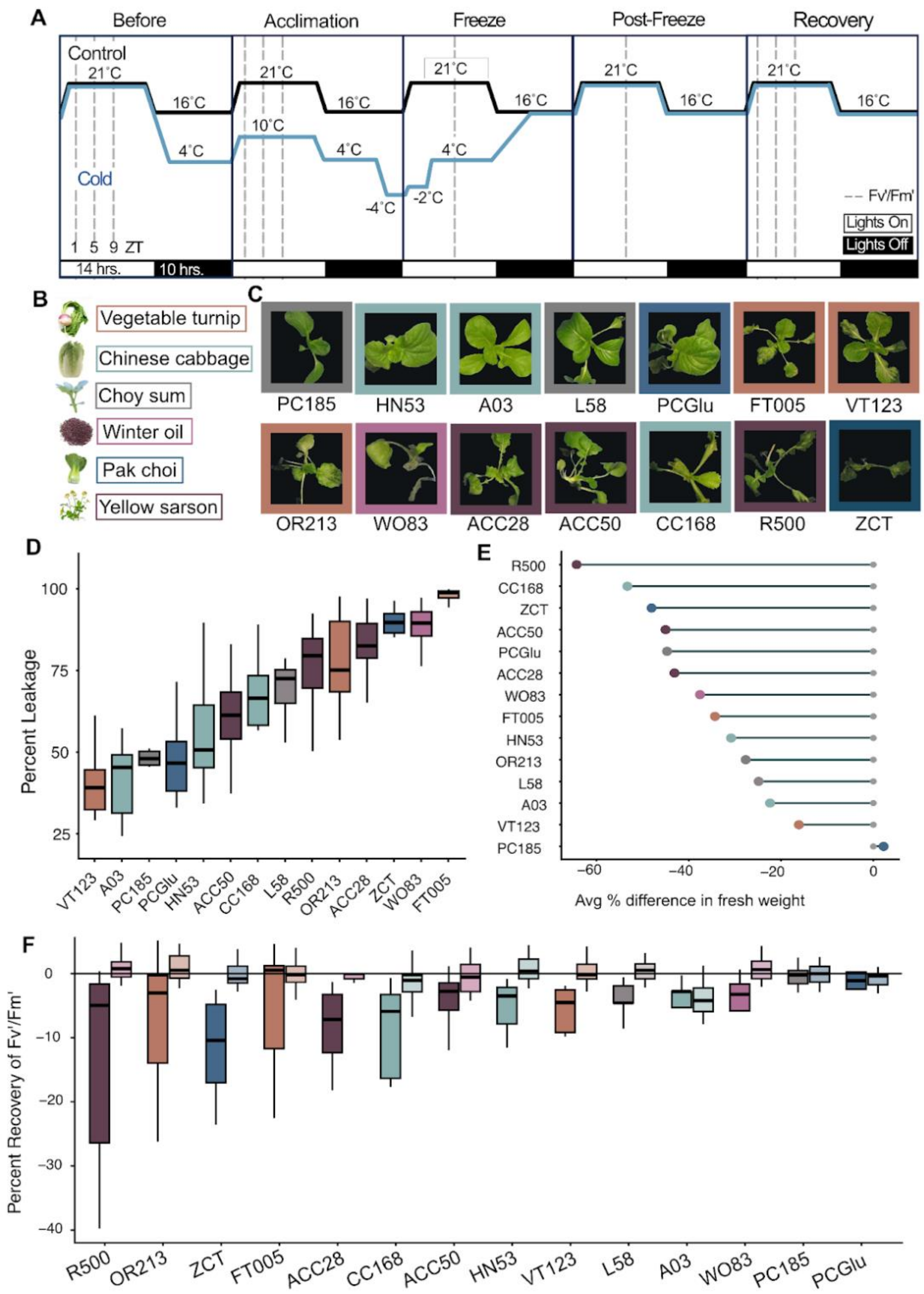


Figure 1.) Relative freezing tolerance among six Brassica rapa morphotypes

A) Experimental design for all freeze stress assays. Freeze plants were cold acclimated for 24 hours followed by two hours of -4℃ and two hours at -2℃ before slowly warming back to control conditions. Dashed lines indicate Fv'/Fm' collection at indicated ZTs (Zeitgeiber; hours after lights on). B) Six morphotypes were utilized in the freeze stress experiments which are color coded. C) Plants are ordered by damage score with the lowest score on the top left and the highest on the bottom right. Images of freeze plants taken the day following the freezing stress ('Post-freeze'). Boxes are colored by their respective morphotype as described in panel B. D) Electrolyte leakage assay quantifying the percent of ruptured cells following freezing stress. A lower percentage of leakage indicates a more tolerant plant. Boxes are colored by their respective morphotype. E) Average difference in fresh weight following freezing stress. Dots indicate the respective morphotype. F) Fv'/Fm' measurements at ZT9 on the 'Post Freeze' day (left boxes) and recovery (right boxes). Percent recovery in Fv'/Fm' was calculated as the difference between freeze stressed plants on day 4 ('Post-Freeze') and the difference on day 5 ('Recovery') from the mean of control responses taken on day 1 ('Before'). Boxes are colored by their respective morphotype where lighter shaded boxes represent recovery responses (right). The line at 0 indicates no difference between freeze or recovery and the mean Fv'/Fm' in control.

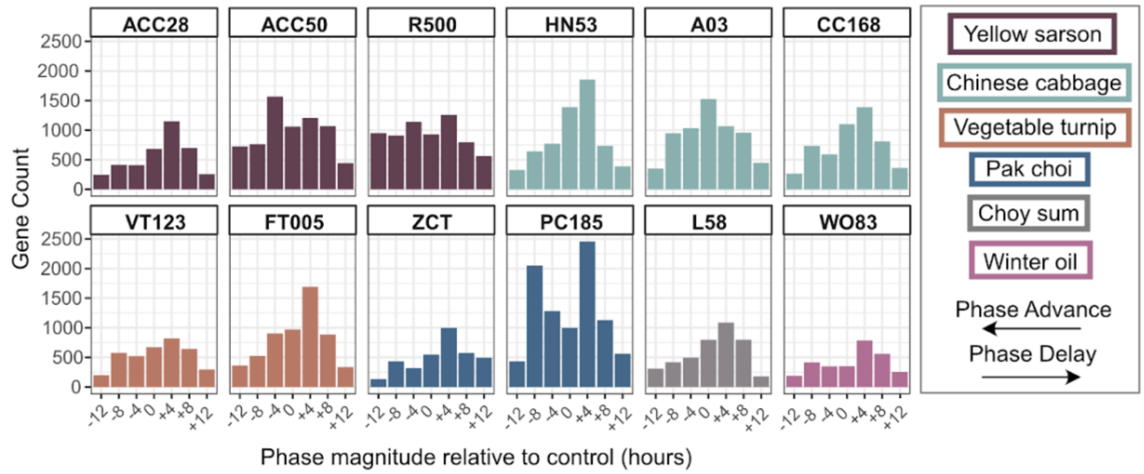


Figure 2.) Phase changes induced by cold are not morphotype specific.

Phase changes were calculated as the difference between the max expression in control (phase) from the max expression in cold. Changes are measured in hours. The “0” group contains genes that had a significantly different pattern but did not have a change in phase between the treatments. Morphotypes are colored based on the key on the right.

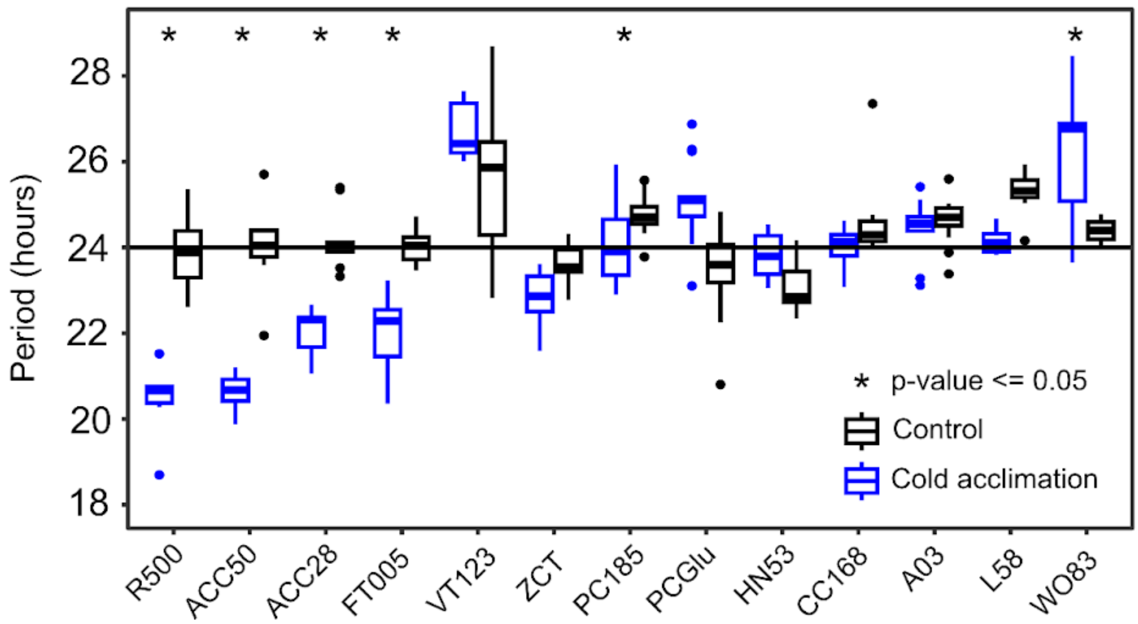


Figure 3.) Endogenous circadian period length varies between accessions in control and cold acclimation conditions.

Period length in hours as determined by leaf-movement in constant light and temperature for Cold (12ºC) and Control (22ºC) for each accession. A student's t-test was conducted between control (black) and cold (blue) acclimation groups The line represents period values at 24h as expected under diel conditions. Statistically significant responses are denoted with an asterisk (p-value <0.05). Accession identifiers are listed at the bottom.

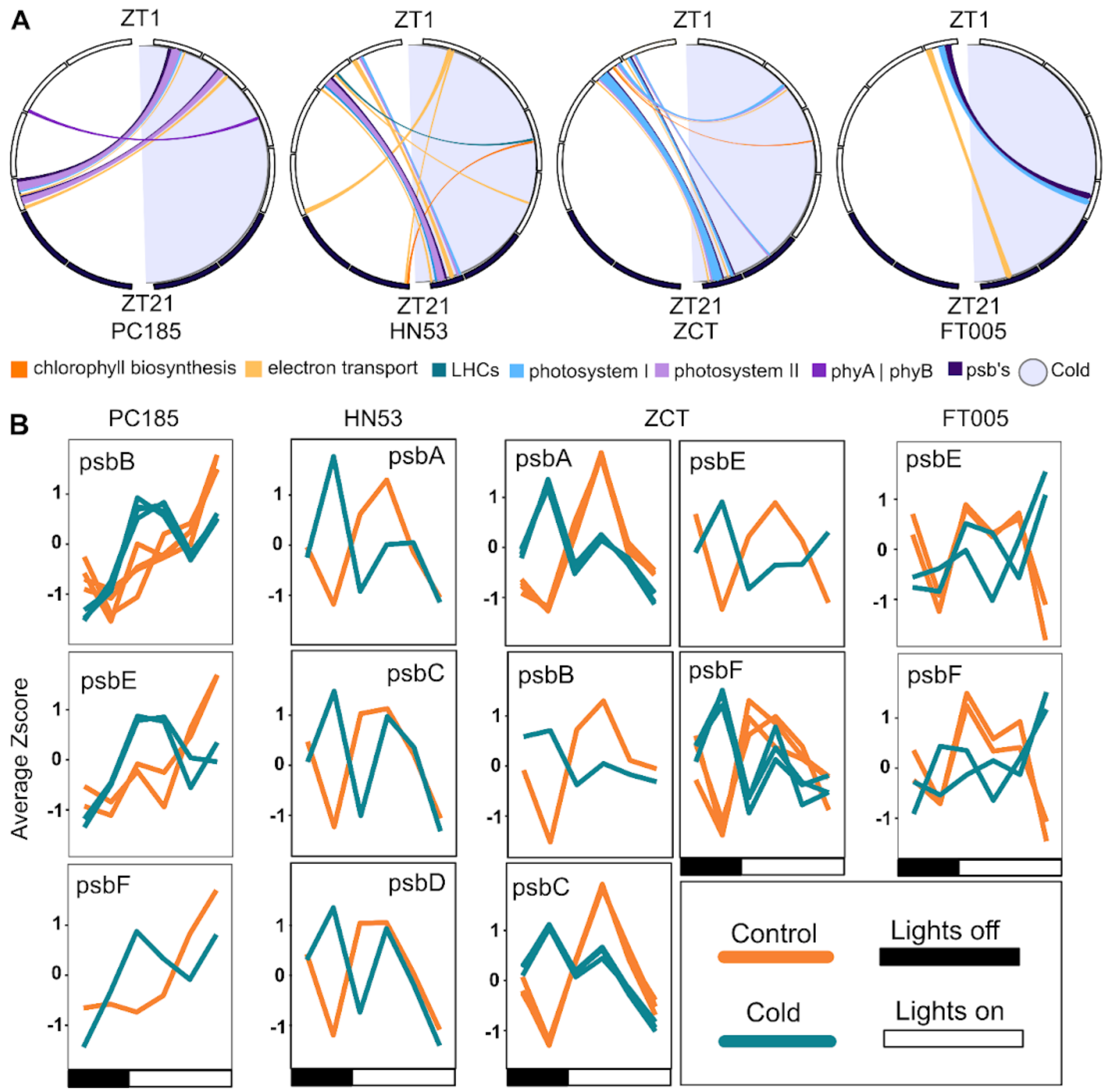


Figure 4.) Enrichment in photosynthesis related terms is time of day dependent

A) Visualization of temporal changes in phase between control and cold samples for genes involved in photosynthesis enriched GO terms. Ribbon sizes reflect the number of genes in each bar; bars reflect the number of genes going into the analysis for each PCG. Colors indicate specific parts of the pathway as indicated by the key. ZT is Zeitgeiber, or hours since lights on. Control phases are on the left (white background) while cold phases are on the right (blue). B) Expression of psb paralogs found in PCGs. Each line represents a copy of the ortholog. Gene symbols are included inside the boxes.

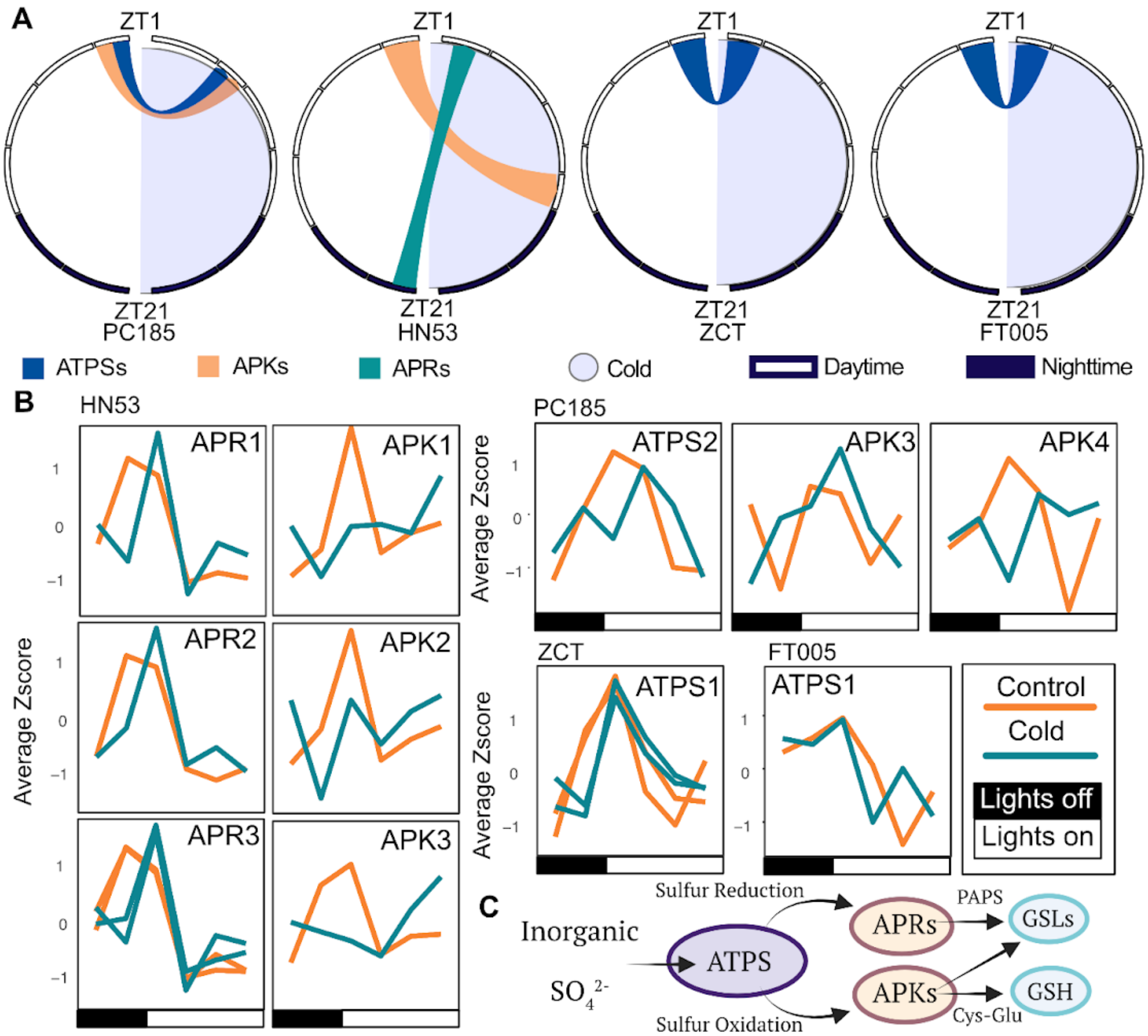


Figure 5.) Time of day differences in core sulfur assimilation genes under cold acclimation.

A) Visualization of temporal changes in phase between control and cold samples for genes involved in sulfur assimilation enriched GO terms. Ribbon sizes reflect the number of genes in each bar; bars reflect the number of genes going into the analysis for each PCG. Colors indicate specific parts of the pathway as indicated by the key. Control phases are on the left (white background) while cold phases are on the right (blue). B) Temporal expression patterns among B. rapa orthologs for key sulfur assimilation genes. C) Simplified schematic of the major branchpoints in sulfur assimilation. ATPS = ATP Sulfurylases, the first committed step in sulfur assimilation. APRs = Adenosine-5'-Phosphosulfate Reductases. APKs = Adenosine-5'-Phosphosulfate Kinases. PAPS = 3'-Phosphoadenosine-5'-Phosphosulfate. GSL = Glucosinolate. Cys-Glu = Cysteinylglutamic acid.

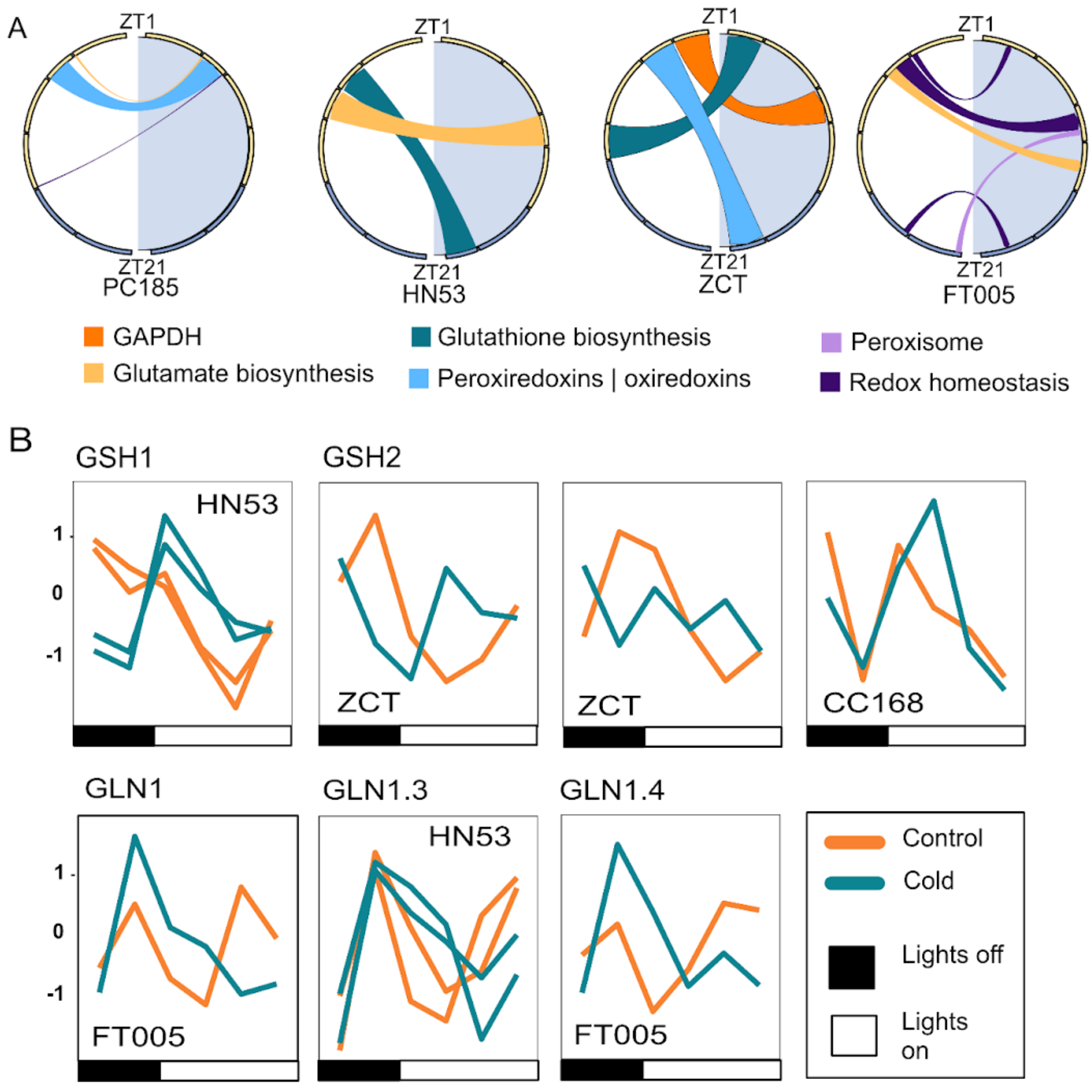


Figure 6.) Temporal alterations to redox homeostasis in freeze tolerant and intolerant lines.

A) Visualization of temporal changes in phase between control and cold samples for genes involved in redox metabolism enriched GO terms. Ribbon sizes reflect the number of genes in each bar; bars reflect the number of genes going into the analysis for each PCG. Colors indicate specific parts of the pathway as indicated by the key. Control phases are on the left (white background) while cold phases are on the right (blue). B) Diel expression patterns between cold and control among B. rapa orthologs for key redox associated genes.

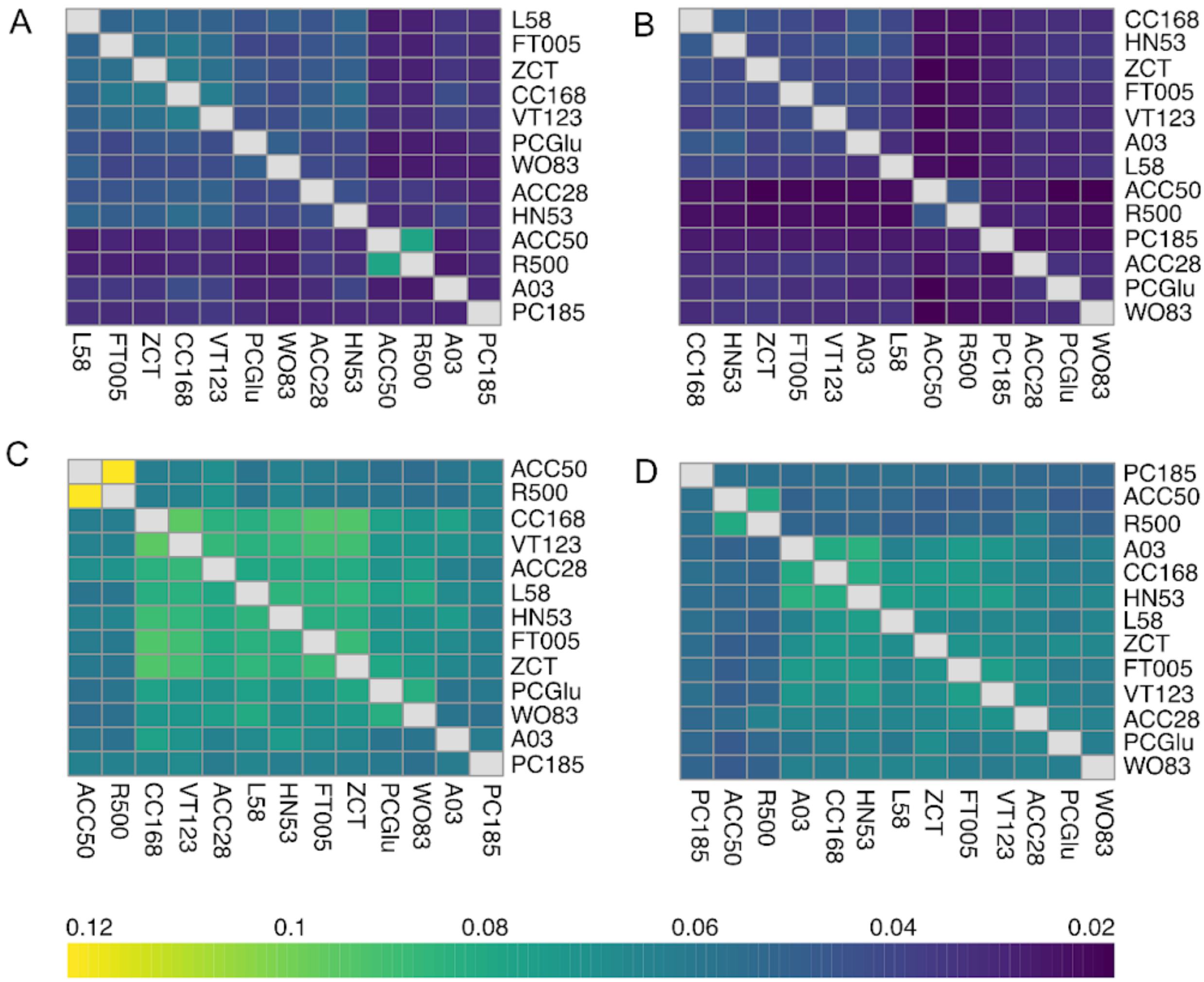
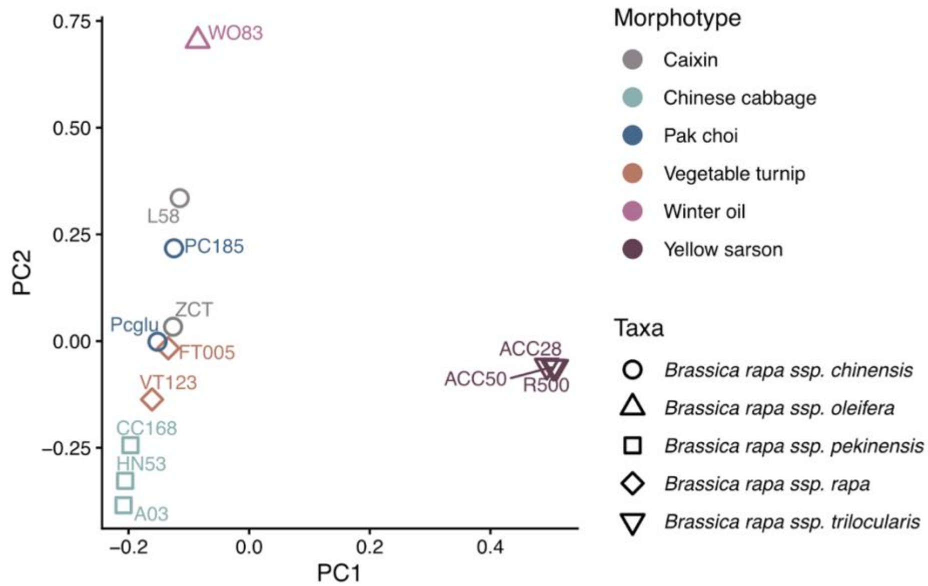


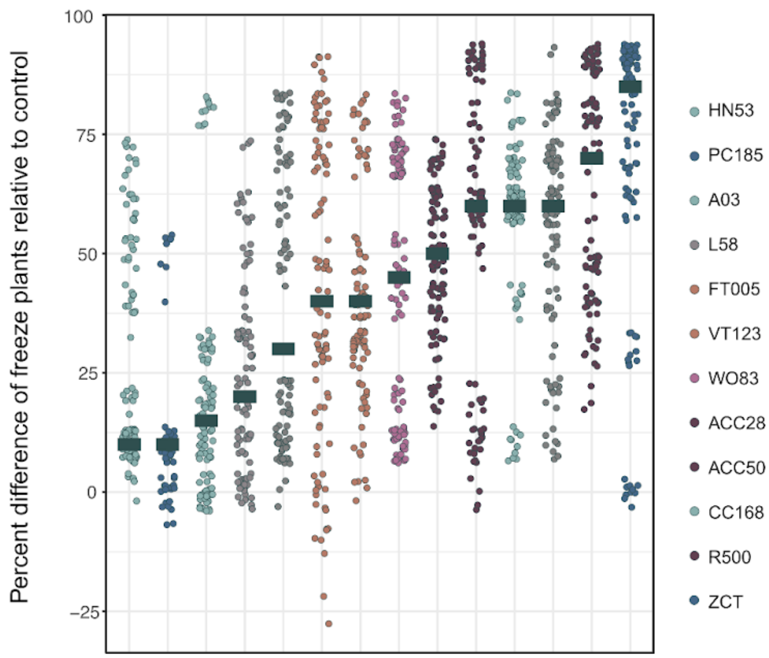
Figure 7.) Gene Regulatory Networks highlight paralog divergence among B. rapa accessions.

Heatmaps using the jaccard similarity index for each pairwise network comparison in control and cold networks individually. Scale below the heatmaps indicate the jaccard score for each comparison, where a higher number is more similar. A) Control jaccard scores using the B. rapa annotation. B) Cold jaccard scores using the B. rapa annotation. C) Control jaccard scores after converting to the Arabidopsis annotation. D) Cold jaccard scores after converting to the Arabidopsis annotation.



Supplemental Figure 1.) SNP-based Principal Component Analysis shows accessions cluster by morphotype.

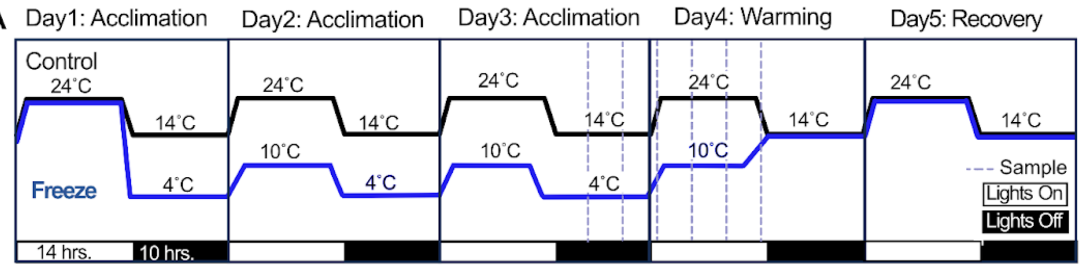
Shapes represent individual accessions as indicated next to the shape. Accessions are colored by their respective morphotype.



Supplemental Figure 2.) Quantitative damage score across B. rapa accessions under freezing stress.

Data points represent individual percentages of cold responses relative to controls and are colored by morphotype. Accession IDs are listed on the right.

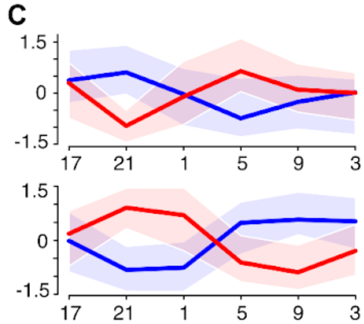
A



B

Accession	ExpressedGenes	kME_Only	kMed_Only	Prop_CR
WO83	26251	2904	10535	0.51
PC185	25842	8915	4268	0.51
CC168	25706	5254	7638	0.50
R500	27247	6542	6441	0.48
ACC28	26621	3850	8449	0.46
ZCT	26658	3506	8357	0.45
VT123	26049	3735	7839	0.44
HN53	25558	6111	5099	0.44
L58	25637	4085	7126	0.44
ACC50	26894	6832	4851	0.43
FT005	26288	5707	5693	0.43
A03	26579	6334	2701	0.34
PCGlu	27016	338	8053	0.31

C



Supplemental Figure 3.) Transcriptome rewiring following cold acclimation in 13 B. rapa accessions.

A) Experimental design for the five day cold acclimation RNA-sequencing time series experiment. Leaf tissue was collected every four hours for 24h as indicated by the dashed line. B) Numbers of genes with differential expression patterns (kME) or median expression (kMed) following cold acclimation. C) All cold responsive genes as called by DiPALM were clustered to visualize global expression patterns across the diel cycle.

Ch 3: Homoeolog expression divergence contributes to time of day changes in transcriptomic and glucosinolate responses to prolonged water limitation in B. napus

3.1 Summary

Water availability is a major determinant of crop production and rising temperatures from climate change is leading to more extreme droughts. To combat the effects of climate change on crop yields, we need to develop varieties that are more tolerant to water limited conditions. We aimed to determine how diverse crop types (winter/spring oilseed, tuberous, and leafy) of the allopolyploid *Brassica napus*, a species that contains the economically important rapeseed oilseed crop, respond to prolonged water limitation. We exposed plants to an 80% reduction in water and assessed growth and color on a high throughput phenotyping system over four weeks and ended the experiment with tissue collection for a time course transcriptomic study. We found an overall reduction in growth across cultivars but to varying degrees. Diel transcriptome analyses revealed significant accession-specific changes in time-of-day regulation of photosynthesis, carbohydrate metabolism, and sulfur metabolism. Interestingly, there was extensive variation in which homoeologs from the two parental subgenomes responded to water limitation across crop types that could be due to differences in regulatory regions in these allopolyploid lines. Follow up experiments on select cultivars confirmed that plants maintained photosynthetic health during the prolonged water limitation, while slowing growth. In two cultivars examined, we found significant time-of-day changes in levels of glucosinolates, sulfur- and nitrogen -rich specialized metabolites, consistent with the diel transcriptomic responses. These results suggest that these lines are adjusting their sulfur

and nitrogen stores under water limited conditions through distinct time-of-day regulation.

3.2 Introduction

Climate change is reshaping daily and seasonal environmental trends causing plants to experience developmental triggers that misalign with the optimal photoperiod[150,151]. Recent examples include early flowering of cherry blossoms in Japan[152] and early ripening of grapevines in wine growing regions[153] both triggered by early spring warming. In addition to temperature changes, water availability continues to be a limiting factor for crops leading to reduction in yield[154,155]. In 2017, it is estimated that over 58 million acres of cropland were irrigated (USDA, 2017 Census of Agriculture) thus making these crops vulnerable to crop loss. According to the UC Davis Center for Watershed Science, the 2015 drought in California accounted for $1.84 billion lost in direct costs[156]. Temperature and water availability is predicted to continue to limit the yield potential of our crops[157], challenging us to develop more resistant varieties to reduce the amount of irrigated water needed.

Plants must balance growth and tolerance to overcome stress[158]. Some plants maintain or speed up growth to try and 'escape' stress, while others slow growth to potentially survive beyond stress (avoidance strategy;[159]. This relationship is evident at the transcriptional level where the *GROWTH-REGULATING FACTOR7* (*GRF7*) transcription factor (TF) represses the cold and drought responsive *DREB2A* TF to inhibit stress response under normal conditions to maintain growth[160]. Overexpression of *DREB1A* increases stress tolerance but comes at a cost of reduction in growth and overall yield[33]. However, when expressing *DREB1A* under a promoter with time of day specific expression, a temporal increase in expression provides stress tolerance while growth is maintained[33]. Diel and circadian regulation of physiological and metabolic processes is an important component of abiotic stress response. The circadian gating of abiotic stress leads to time of day specific transcriptional responses to stress[38,102,161]. Limiting responses to certain times of day is thought to be critical for maximizing energy efficiency to maintain growth while responding to stress[26]. Several studies in diverse plant species have uncovered time of day specific responses to abiotic stress, and revealed that the largest variation in transcriptome responses is time of day[34,138,139]. Uncovering these time of day responses is critical for identifying transcriptional regulators controlling the physiological or metabolic responses associated with tolerance, yet most abiotic stress studies are done with single time point resolution. Additionally, there are limited time course studies in more complex polyploid crops where we have less understanding of the diel regulation of the transcriptome in response to stress and the amount of intraspecific variation.

Brassica napus (*B. napus*), a relative of Arabidopsis within Brasssicaeae, is an allopolyploid derived from the hybridization between *Brassica rapa* (*B. rapa*) and *Brassica oleracea* (*B. oleracea*) just ~7,500-12,000 years ago[39]. The Brassica genus underwent an ancient Brassica whole genome triplication event that occurred in the ancestor of the tribe Brassicaceae, resulting in three copies of each chromosome[162]. This event significantly contributed to the diversification and evolution of Brassica species, including important crops like cabbage, broccoli, and brussel sprouts[40]. When two distinct species hybridize, their genetic material merges, resulting in a combined genome. The genes that were once orthologous in their separate lineages, having evolved from a common ancestral gene, are now present together within this unified genome. Due to their unique origin and relationship, these co-existing genes are referred to as homoeologs. In this study, our analyses will focus on comparing the homoeologs between the two parental species of *B. napus*.

B. napus is commonly known as rapeseed and includes the second highest oil producing crop grown worldwide after soybean[163]. In addition to spring and winter oilseed varieties, *B. napus* contains other specialty crop varieties including rutabaga and swedish turnip. Like other Brassicacea, *B. napus* produces glucosinolates (GSLs), sulfur and nitrogen-rich specialized metabolites that are derived from amino acids such as methionine

(‘Aliphatics’), tryptophan (‘Indolics’), or phenylalanine (‘Aromatics’). These compounds are especially known for their production of the ‘mustard bomb’, or hydrolyzed byproducts, following herbivory[164]. GSLs play an important role in abiotic stress responses as well. In Arabidopsis, three auxin-sensitive *Aux/IAA* transcriptional repressor genes (IAA5, 6, and 19) induced by *DREBs* to maintain levels of aliphatic GSLs when grown in drought conditions[165]. Under the same drought conditions, the *iaa5iaa6iaa19* mutant line exhibited reduced plant biomass but exogenous application of a single GSL (4MSO) rescued growth[165]. Sinigrin, an aliphatic GSL found in several Brassicaceae, has been shown to induce stomatal closure indicating a direct role in stomatal regulation[166]. Under warming (25°C) and heat stress (32°C) Arabidopsis lines with altered GSL content, specifically lower indolic GSLs, exhibited reduced thermotolerance and lower auxin content following increased temperatures[167]. These results indicate an important link between GSL and thermotolerance, auxin metabolism, stomatal regulation, and drought response. Turnover of GSLs also contributes to replenishment of primary sulfur pools[68] stress responses through glutathione mediated reactive oxygen species (ROS) and electrophile scavenging[168]. Increased ROS levels will decrease photosynthetic rates thereby limiting available energy for growth[169]. Sulfur is also a critical component in chloroplasts, primarily in the form of sulfoquinovosyl diacylglycerols (SQDGs) which are localized in the thylakoid membrane, and Fe-S proteins that aid in photosynthesis and nitrogen assimilation[170]. Because of the interplay between sulfur and nitrogen metabolism, the ratio between sulfur and nitrogen pools is highly regulated to maintain a strict balance. For example, a study in *B. napus* showed that when a 1:4.5 or 1:5.75 ratio of Nitrogen:Sulfur was applied, seed yield increased by 42% to 267%, respectively[171,172]. However, when the same amount of nitrogen was applied without sulfur, there was no increase in seed yield or seed oil, and a reduction in yield when a lower rate of nitrogen was used without sulfur[172]. When soils are sulfur-deficient, there is less nitrogen available for uptake which can promote nitrogen leaching[171,173]. With up to 30% of sulfur allocated to GSLs in above ground tissues[174], and at least one molecule of nitrogen for each GSL, a more complete understanding of their regulation under stress conditions is critical for optimizing nutrient use efficiency in *B. napus* and other Brassicaceae crops.

Improved drought tolerance is a key target trait for the rapeseed industry[154,175,176] as number of siliques, number of seeds, and seed oil content are all negatively affected under water limitation[177] in addition to lower photosynthetic rates and lower vegetative biomass[154,176]. In the 2023 growing season, Canada - one of the largest global producers of rapeseed - experienced its fourth year of drought with 76% of agricultural land classified as either abnormally dry (D0; water limited), moderately dry (D1) or extremely dry[178]. In addition to temperate climates *B. napus* is grown in semi-arid and arid zones leading to diverse ecotypes with varying responses to water limitation[175].

To explore how the expanded genome in *B. napus* responds to water limitation and what transcriptome level regulation drives changes in growth and metabolic processes, we performed a prolonged water limitation experiment across 16 diverse accessions of *B. napus* covering winter/summer oilseed, tuberous, and leafy crop types. In this experiment, we collected image data across the 16 diverse accessions under well-watered (WW) and water limited (WL) conditions. At the end of the experiment, leaf tissue was collected at multiple time points for a diel transcriptome analysis to determine what changes to time of day regulation of biological processes were altered under water limitation. We found that all accessions reduced their growth to various degrees but estimate that they remain photosynthetically healthy. We found unique changes in transcriptome responses across accessions that were time of day dependent. Most striking was the extensive variation in which homoeolog(s) encoded on the two parental subgenomes was most responsive across accessions indicating genome-wide changes to regulatory regions in this relatively young species. In particular, two accessions had opposite regulation of GSLs suggesting unique temporal regulation of sulfur use under water limitation. A follow up study revealed that in fact these accessions altered their GSL stores under water limitation resulting in different time of day changes in GSL pools.

3.3 Results

3.3.1 Plants reduce growth but have higher 'greenness' in non-senescing tissue under water limitation

To identify *B. napus* accessions with improved responses to water limitation, we examined the phenotypic effects of prolonged water limitation on 16 *B. napus* accessions using the Bellwether Phenotyping System[179]. We exposed all accessions to an 80% reduction in water (20% full capacity) over four weeks and assessed growth and color through a high throughput automated system (Methods: Bellwether Experiments). We used the open source PlantCV software on 59,850 RGB and 59,848 NIR images to quantify treatment responses.

Overall, prolonged water limitation reduced the final size of all accessions (Figure 1). However, the difference in size between treatments varied among accessions with large (ex: Av; St) and small (ex: Mu; Qu) differences observed. We modeled the change in area over time (growth) using Bayesian Hierarchical models. Growth curves of all accessions in well-watered (WW) and water limited (WL) treatments were fit to a Gompertz model (see methods for the equation). Using these models, we tested the hypothesis that within each accession the WW group would have a final size, represented by the Gompertz model parameter A (asymptote), at least 50 percent larger than that of the WL group. This hypothesis was supported for each accession (posterior probabilities and effect sizes reported in Table S1). The hypothesis that the inflection point in the growth curve (B parameter) was at least 5% larger in the WW group than the WL group was also supported for each accession (Table S1). Lastly, the hypothesis that the growth rate (C parameter) of the WW group was 5% larger than the WL group was not significant for any accession (Table S1). These results show that despite the growth rate of WW and WL plants being similar, the WW plants grew larger by the end of the experiment, suggesting that water-limitation treatment restricts growth as opposed to slowing growth rate.

While we did not directly measure photosynthesis in this initial experiment we did quantify plant color. First, we used the multiclass Naive Bayes classifier in PlantCV[180] to categorize plant pixels into 'healthy' (green and purple plant pixels) and 'senescing' (yellow plant pixels) categories (Figure S1). We found that the percent of yellow tissue is similar or higher in the WW treatment (Figure S1). Accessions with a higher estimate of senescing tissue along with a larger overall area under WW treatment (Figure 1), likely suggests that plants are further along in development in comparison to WL plants with restricted growth.

We then examined the 'healthy' pixels. For only the pixels categorized as 'healthy' we extracted the hue circular mean. For reference, a hue value of 60 degrees is yellow and 120 degrees is green. Unexpectedly, we found that most WL plants were more green (values towards 120 degrees) under prolonged water limitation in comparison to well-watered (Figure 2) although the magnitude of the difference varied among accessions, with relatively extreme (Mu, St, Av) and mild (Ab) differences observed. In addition, the timing of when treatments began to separate for color among the accessions differed (Figure 2). For plant growth, all lines began to deviate around day 11; however, differences in plant color were more varied. For example, Ab doesn't appear more green until 8 days of water limitation while Mu occurs around day 4 (Figure 2).

3.3.2 NIR values are consistent with treatments

The NIR imaging system measures wavelengths of light absorbed by water. In general, lower NIR signal values suggest more water absorption because there is less signal reflected back to the NIR detector[179]. However, NIR signal can change with leaf thickness and tissue composition[179]. Our NIR imaging results match with treatment conditions (Figure S2). In nearly all accessions, average NIR signal is lower in the WW treatment compared to the WL treatment, suggesting higher water content in WW treatment plants. Differences in NIR signal between treatments typically matched differences in plant size. For example, Mu showed smaller differences in plant size between treatments and smaller differences in NIR signal. However, accessions like Av and St had larger differences in plant size and NIR signal between treatments (Table S2). Overall, results from plant size, growth, and color suggest that *B. napus* adjusted to prolonged water limitation by slowing above ground growth. The reduction in growth was expected but we were surprised to find that WL plants were more green. This led us to wonder what transcriptomic changes had occurred in these lines following prolonged WL.

3.3.3 Accessions exhibit variation in subgenome responsiveness that is time of day dependent

Gene expression is a dynamic process that is often under circadian or diel regulation, resulting in time of day control of physiology, metabolism, and abiotic stress response[34,138,181]. To capture these transcriptome dynamics we collected leaf tissue for all accessions every six hours over a twenty four hour period for RNA-sequencing after four weeks of prolonged WL.

We first examined whether the differences in growth response under WL could be explained by the genetic background of the accessions. We called Single Nucleotide Polymorphisms (SNPs) and clustered the accessions by PCA. Consistent with our documentation of these accessions, they grouped by crop type (Figure S3). When we mapped the difference in plant area to the PCA, we found that there is no relationship between crop type and growth response to WL conditions, indicating that the variation in growth is not explained simply by crop type. To identify significantly differentially patterned genes within each accession, we used the DiPALM R package[182], resulting in thousands of genes across accessions with differential patterns under WL (Table S3; 'Metadata' tab).

A comparison of the overlap of WL responsive genes across the 14 accessions revealed that the majority of WL responsive genes (~70%) are unique to one to two accessions (Figure 3a; light blue bars). However, an overlap with the Arabidopsis orthologs resulted in a higher proportion of shared genes (Figure 3a; dark blue bars) suggesting similar functional responses but variation in the homoeolog copy (encoded on either the BnA or BnC subgenome) that is responsive across accessions. We first looked into WL genes that were shared among 6 accessions since this was the relative halfway point. Within this group we found that the *NITRITE REDUCTASE 1* ortholog is responsive to water limitation in six accessions that differ in the respective BnC paralog (Figure 3b). We compared the averaged expression levels across time points for all the genes with significantly different expression patterns under WL and found that the BnC subgenome (*B. oleracea*) had slightly higher expression than the BnA subgenome (*B. rapa*) across accessions (Figure S4). To further examine the variation in which homoeolog is responding to WL conditions, we focused on a set of previously characterized sugar responsive genes[183]. Overall, the majority of accessions varied in the number of genes that were responsive under WL (Table S3; Metadata tab). Four orthologs of the *HOMEOBOX PROTEIN 6* gene are expressed under WW conditions across five accessions (Ab, Al, Mu, Ne, and St). While only a single BnC homoeolog responds to WL in Al, Mu, and Ne, two respond in Ab and three respond in St (Figure 3c).

A recent study in wheat observed time of day variation in subgenome contribution to expression level under circadian conditions[184]. To explore this in our dataset, we identified the time of peak expression (phase) for each gene by treatment and accession and calculated the change in phase between WW and WL. This resulted in five 'phase groups' that contained genes with an earlier phase under WL (negative), a later phase (positive), or no change in phase (0). Overall, the BnC subgenome contributes more for most phase change groups, but the proportion relative to the BnA subgenome varies across accessions (Figure S5). This analysis also revealed accession level differences in time of day changes of the WL transcriptome as shown by the direction of phase change under WL conditions (Figure 3d; Figure S5). For example, both St (large difference in

WL growth; Figure 1) and Mu (little difference in WL growth) have the greatest change in the +/- 6 h phase groups with overall 54% and 57% of WL responsive genes falling into these two groups, respectively. Yu (moderate difference in WL growth) is quite different having the largest proportion (36% of WL genes) in the +12 h group alone while other accessions had, on average, ~16% in this group. As a whole we did not find evidence that one subgenome dominates a single phase change group or growth response group, regardless of BnC having higher average expression overall.

Such large proportions of genes exhibiting similar phase changes (Figure S5) suggests coordinated regulation, and thus we would expect to find enrichment for biological pathways within a given phase change group. To explore this, we ran GO enrichment on the WL clusters from DiPALM and identified clusters that had a matching 6 h advance (St), 6 h lag (Mu) or a 12 h advance (Yu). Within these clusters (Figure 3e) we found enrichment for several biological processes including photosynthesis (St; large growth difference), sulfur metabolism (Mu; small growth difference), and gene silencing (Yu; moderate growth difference). See Table S3 'DiPALMClusters_GO_Terms' tab for a full list of enriched terms. We do not find evidence for subgenome bias within enriched biological processes suggesting that both the BnA and BnC subgenomes are contributing to these pathway level responses.

To more broadly identify the biological processes with time of day specific responses across all accessions, we generated a 'response score' as the difference between WW and WL expression at each time point for every gene and used these scores to identify global co-expression response modules. This allowed us to identify 11 time of day specific patterns that now describe the underlying WL responses of all 14 accessions (Figure 4a). For example, M2 contains genes with higher expression under WL in the morning, and lower expression under WL at night. M8 has the opposite pattern, while genes in M3 have higher expression at dawn (ZT1; 1h after dawn) and at the end of day (ZT13). Based on our analysis of the individual accession phase response groups, we hypothesized that although every accession was found in each module, the underlying function of the genes would be different.

3.3.4 B. napus alters time of day regulation of carbon fixation and photosynthesis under prolonged WL

We performed GO enrichment on each response score module and accession to determine what biological processes were associated with the time of day specific responses. This produced a list of approximately 2700 unique GO terms. We assigned each term into one of 34 'broad descriptors'; similar to a parent term (Table S3; ResponseScore_GO_Terms tab). Three out of the 14 accessions were enriched in abiotic stress suggesting that the majority of accessions had adjusted to the prolonged water limitation and were not showing signs of stress, consistent with the phenotyping data. Most accessions (12 out of 14) were enriched in carbohydrate metabolism, carbon fixation and/or sulfur related processes (Figure 4b and Table S3 'ResponseScore_GO_Terms' tab). Carbohydrate related terms involved in negative regulation or catabolism of carbohydrates were found in M3 (higher under WL at the beginning and end of day) and were enriched in Al and St (Figure 4b) while Ca was enriched in the same process in M8 (up at night). Genes involved in carbohydrate biosynthesis had more diverse patterns and were found in M1 (up before and after dawn; St), M2 (up during the day; Gr and Se), M3 (up at the beginning and end of day; Mu and St), M7 (down at night/predawn; Ab), M8 (up at night; Gr) and M9 (up at night; Da and Gr). Interestingly, Mu and St were enriched for both negative and positive regulation of carbohydrates in the same module (M3), suggesting that these accessions are turning over carbohydrates early morning and at the end of the day.

From our phenotyping color analysis, we didn't see any indications of reduced photosynthesis consistent with plants adjusting their growth to accommodate limited water availability. However, we did find that several accessions (Ab, Br, DH20, Gr, St, Yu) were enriched for changes to photosynthesis (Figure 4b) on the transcriptome level. For the full list of GO terms with their associated accession/module see Table S3 (ResponseScore_Go_Terms tab). While some accessions tended to be enriched for terms involved in light sensing (Br-M5; DH20-M8; St-M3 and M8), others were enriched in the light/dark reactions (Ab-M1), stomatal closure (Ab-M1 and M8; St-M1 and M3;Yu-M9), or had a few terms enriched in chlorophyll metabolism (Gr-M2; St-M1, M3 and M8; Yu-M6). We noticed that although some accessions were enriched in the same terms as another (Ab, Ca, St, Yu), the modules their genes were enriched in differed, implying unique regulation strategies in response to prolonged WL (Figure 4a; Figure 4b; Table S3). For example, while Ab and Yu were both enriched for stomatal closure (GO: 0090332), Ab's genes were enriched in M8 (higher WL expression in the evening and night) while Yu's were in M9 (little to no difference during the day but higher at night). Similarly, both Ab and St were enriched for the same term involved in light/dark reactions and carbon fixation but had opposing WL responses at the end of day, with Ab (M1) maintaining lower WL expression towards the end of the day and ramping back up at night while St (M3) had higher expression at the end of the day and came back down at night (Figure 4a).

As the time of day specific WL responses (response scores; Figure 4a) are the difference between WL expression and WW expression at a given time point, there are multiple expression patterns that could lead to a similar response score pattern. To examine the expression patterns themselves we chose two accessions (St; relatively large reduction in growth and Ab; intermediate reduction in growth, Figure 1) that differed in their temporal responses for photosynthesis (Figure 5a). We noticed that, in general, St was dramatically altering expression of carbon fixation related genes under WL, where the treatment responses were flipped (antiphase). In contrast, Ab showed a more subtle difference and appears to be maintaining a similar expression pattern overall only slightly up before/after dawn and ramping down expression midday/evening (Figure 5a). We wondered if other accessions had similar WL responses and if the responses differed by growth group (Metadata; GrowthPhenotypeGroups tab). To explore this, we examined the expression of the genes from the carbon fixation enriched process in Ab and St (Figure 5a) that were WL responsive in the other accessions (Figure 5b). Similar to St and Ab we see that WL expression was higher early in the morning in the large (Figure 5b; top row) and intermediate (Figure 5b; middle) growth groups and mixed for the low growth group (Figure 5b, bottom), with most of the genes falling into M2 (up in the morning and down at night). Interestingly, we found that the low growth group had few WL responsive genes while accessions with larger differences in growth had several WL responsive genes involved in carbon fixation, despite the lack of significant enrichment.

We were curious if the genes in Ab and St (GO:0015977; carbon fixation) were shared or if there was evidence for homoeolog divergence like we found with our list of sugar responsive genes (Figure 3c). While both accessions were enriched in the *PLASTID ISOFORM TRIOSE PHOSPHATE ISOMERASE* ortholog (At2g1170), Ab was enriched for a BnA copy (M8) while St was enriched for a BnC copy (M3). We also found divergence in the time of day that enriched homoeologs altered expression within a single accession (St), where a BnC copy of the ortholog *D-RIBULOSE-5-PHOSPHATE-3-EPIMERASE* (At5g61410), was enriched in M3 while the BnA copy was enriched in M1. This was not the case for other overlapping orthologs such as *TRANSKETOLASE1* (At3g60750) that have the same responsive homoeolog. Overall, based on the expression patterns of carbon fixation related genes (Figure 5a) and other photosynthesis related processes (Figure 4b), we find that Ab and St show unique diel regulation of photosynthesis under WW and WL conditions. This led us to wonder if there was a measurable difference in photosynthesis across the entire day or if the time of day specific shifts in expression resulted in the same net photosynthesis between treatments.

3.3.5 B. napus accessions maintain net photosynthesis and chlorophyll content under WL

To capture aspects of photosynthesis outside of transcript level responses we performed a follow up phenotyping experiment on a subset of accessions with differential growth responses. We grew these lines under the same 20% WL as done in the original experiment and captured photosynthetic related metrics including dark-adapted photosystem II efficiency (Fv/Fm), chlorophyll index (CI), total chlorophyll content, and stress related indices (NPQ: non-photochemical quenching; NDVI: Normalized Difference Vegetation Index). Images were taken weekly at ZT8 and were analyzed with PlantCV (https://github.com/danforthcenter/ktgreenham-brassica-drought-paper). Leaf tips were collected at the end of the experiment at ZT6 to quantify chlorophyll levels.

Consistent with the original experiment, plants exhibited few signs of stress and maintained high levels of photosystem II efficiency and CI (Figure 5c and Figure S6) under WL, particularly at day 21 (Figure S6). NPQ and NDVI also remained similar between WW and WL treatments throughout the experiment (Figure S7). Accession Ab (intermediate reduction in growth; Figure 1) had higher average chlorophyll index (CI) under WL (Figure 5c; $p < 0.0001$), but St (relatively large reduction in growth; Figure 1) showed no difference in average CI between WW and WL plants after 3 weeks (Figure 5c, $p = 0.05$). Surprisingly, we found no differences in extracted chlorophyll content in any accession (Figure S6c). This could be due to lower content in the leaf tips compared to the more basal tissue. It could also be due to components other than chlorophyll, such as nitrogen, reflecting in the same CI region ($CI_{Red\text{-}Edge}$;[185]. Regulation of nitrogen and sulfur pools are particularly critical for Brassica crops due to the production of the nitrogen and sulfur rich glucosinolates (GSLs). While GSLs are best known for their role in biotic defenses, more recent evidence indicates that they may also be important for abiotic stress responses including drought[165,186,187].

3.3.6 Accessions are altering sulfur and GSL pools in response to prolonged WL

Our transcriptomic analysis revealed four accessions (Ab, Av, Mu, St) that were enriched in sulfur and GSL metabolism (Figure 4) that include: fixation of inorganic sulfide (ex: *O-ACETYLSERINE (THIOL) LYASE (OAS-TL) ISOFORM A1*), synthesis of sulfur related amino acids (ex: *CYSTEINE SYNTHASE D1*) primary metabolites (ex. *GLUTAMINE SYNTHETASE 2*), GSL biosynthesis (*SUPERROOT1*), and negative regulation(*SDI1*) or catabolism (*BETA GLUCOSIDASE 26*) of GSLs. A full list of over enriched sulfur-related genes with their corresponding Arabidopsis ortholog can be found in Table S3 in the 'AllSulfurTerms'' tab. To explore modifications in diel regulation of WL genes involved in this pathway we first grouped genes by the parts of sulfur/GSL metabolism they were enriched in (Figure 6b; colored boxes). While most accessions were enriched in one or two steps of the pathway, St was enriched in all aspects that we looked at except for negative regulation of GSLs. In general, under WL Mu upregulated expression of sulfur related genes early in the morning (Figure 6c; purple, red, dark blue) while St was more variable. Ab and Av had little enrichment overall but reinforce our previous findings that accessions enriched in similar pathways (or in this case, a specific step of a pathway) tend to have differential time of day expression patterns under WL implying unique regulatory strategies. For example, Ab and Mu are both enriched in genes involved in GSL biosynthesis (peach) but the expression patterns are quite different. While Ab is ramping up throughout the day and peaking just before night in WL (M8) Mu expression is highest at night/dawn (M10) under WL.

We noticed that St had a large proportion of its enriched genes in M3. This module had enrichment in multiple steps of the pathway including aspects of primary sulfur metabolism and GSL catabolism. We were curious if any of these genes were found in multiple copies, and if so, what level of homoeolog expression variation was present. The majority of the sulfur metabolism and GSL catabolism genes were expressed as a single copy, with two exceptions. A TF well known for its' importance in aliphatic GSL biosynthesis (*MYB28*), and an atypical myrosinase (enzymes that catabolize GSLs; *BETA GLUCOSIDASE 33*; *BGLU33*), were expressed in multiple copies but only one homoeolog was WL responsive. We looked at all accessions that had at least one responsive copy of *MYB28* and/or *BLGU33 and* found variation in which accessions have at least one responsive copy, the number of homoeologs that are responsive, and which homoeolog is responsive among the accessions. This was most apparent in *BGLU33* which was responsive in seven accessions with one (St), two (Br, DH12, Mu, Se), or all three (Gr, Ne) expressed homoeologs called WL. Of those that had two responsive copies, most had one BnA and one BnC copy (Br, DH12, Se) while Mu had two BnA copies. This trend was consistent in the *MYB28* ortholog with accessions having one (Ab, St), two (Br) or four (Mu) of the six expressed homeologs called WL responsive. Overall, this transcriptome data suggests that these accessions have unique time of day and subgenome/homoeolog specific sulfur re-allocation strategies under WL.

3.3.7 Temporal variation in GSL content in a spring (St) and winter (Mu) oil

We found that St (rapeseed; Canola) had the most enrichment in primary sulfur metabolism and was also enriched in GSL degradation. We hypothesized that St may be allocating sulfur towards primary metabolites and away from GSLs, while other accessions like Mu (a winter variety) had relatively more enrichment in GSL metabolism including biosynthesis. Because GSLs are both nitrogen and sulfur rich, and because sulfur is a critical component of chloroplasts (i.e. sulfolipids), alterations to GSL levels may contribute to the ability of these accessions to maintain photosynthetic performance.

To determine whether the GSL metabolites were being altered under WL conditions, we performed a prolonged WL metabolomics experiment with St (relatively large reduction in growth; Figure 1) and Mu (relatively small reduction in growth) under similar conditions to the previous two experiments (Methods: Physiology and Metabolomics).

We measured photosynthetic rates throughout the experiment to ensure similar responses to the previous two experiments and observed similar photosynthetic rates of WW and WL plants throughout the 28 day experiment (Figure S8). After four weeks of WL we collected tissue punches at the tip of the leaves at ZT4, 12 and 20 for metabolomics and a single time point collection at ZT7 for chlorophyll content. We also collected whole leaves at the end of the experiment to quantify total nitrogen and sulfur content.

We found that St had significantly less total leaf sulfur content and is trending towards more nitrogen under WL (Figure 7a). This is consistent with what we would expect from this accession as it is a rapeseed cultivar[188] that has been bred for low levels of GSLs (less than 30 µmol/g dry, defatted meal;[189]. Interestingly, we found significant differences in sulfur-related compounds between Mu (winter oilseed) and St (spring oilseed) that were time of day dependent. Although both accessions had similar levels of sulfur in the leaf in WW (Figure 7a) St had significantly less sulfur in WL and was trending towards higher nitrogen content in WL. Sulfoquinovose, (SQ; Figure 7b) a critical component of sulfolipids (e.g. Sulfoquinovosyl diacylglycerols; SQDG's) and part of the thylakoid membrane, increases at night under WL in Mu while St increases in the morning. While both accessions have similar levels of total indolics (Figure 7c), Mu has approximately twice the amount of aliphatics. This is especially apparent at ZT20 under WL conditions where St has very low levels of aliphatics (Figure 7c and FigureS9a).

We then looked at individual GSLs and found that the largest differences were in the two indolics we detected (4OH and an isomer that is either 4ME or NEO; Figure 7c). Again, we see unique patterns among the accessions where Mu is increasing 4ME/NEO content in the morning under WL ($p<0.000$) while St is decreasing content at this time ($p=0.008$). Under WW, Mu appears to be slowly ramping up content throughout the day (4ME/NEO) and into the night while St has the opposite response. 4OH is somewhat more consistent for both accessions with a minor reduction under WL at night in Mu ($p=0.050$) while St shows no treatment effects for 4OH. We also inspected variation across the LC-MS run itself to account for any technical variation or matrix effects that might contribute to these differences. While there is some variation between injections within a given treatment/ZT (Figure 7d), peak areas are largely similar as seen in the overlap of peaks between biological/technical replicates.

Methionine derived aliphatics (NAP, PRO, SIN, GJV) were surprisingly inconsistent for St, with two of these (SIN; GJV) being entirely below our limit of detection (LOD; 1.0E4 Peak height or 13.3 Log Area; Figure S9a) while the other two (NAP; PRO) were rarely above our limit of quantification (LOQ; 5.0E4 Peak height or 15.6 Log Area; See methods). Although Mu does not appear to have treatment effects for any of the aliphatics, we did see that, similar to 4ME/NEO, the WW samples increase significantly at the end of the day (NAP: ZT4-ZT12, p=0.008; ZT4-ZT20, p=0.003; PRO: ZT4-ZT12, p=0.002). This trend appears similar for SIN but we were unable to capture significant differences. A single aromatic GSL (NAS), derived from phenylalanine, has significantly higher levels of ZT4 but only in Mu, although both accessions exhibited a treatment effect that was not specific to a particular ZT (Figure S9b).

3.4 Discussion

To assess how diverse crop types of *B. napus* alter transcriptome regulation of key biological processes after a prolonged period of water limitation, we imposed an 80% reduction in available water on 16 accessions over a four week period and captured phenotypic responses throughout the experiment using a high throughput phenotyping platform (Bellwether Facility, Saint Louis, MO). Our growth curve models showed that all accessions exhibited a reduction of above ground growth with differences in the magnitude of change between well-watered and water limited conditions. We found that all accessions appeared photosynthetically healthy and were not exhibiting signs of stress, indicating that accessions were adjusting to the lack of water early on. In Arabidopsis, water limitation has been shown to reduce leaf emergence, leaf number and biomass[190] with no significant effect on net photosynthesis or dark respiration[183]. Despite the lack of visible signs of stress, we found quite dynamic changes in the transcriptome.

3.4.1 Differential contribution of subgenomes to expression dynamics in response to WL

The 24 h time course RNA sequencing experiment on all accessions provided a unique perspective on the diel regulation of the transcriptome under well-watered (WW) and water limited (WL) conditions. We found that the majority of WL responsive genes were unique to few accessions that were not due to crop type. However, there was a much higher proportion of shared genes when converting the *B. napus* genes to their Arabidopsis ortholog suggesting overlap in functional responses. What was especially striking was the extensive variation in which homoeolog(s) among the two parental subgenomes was responding under prolonged water limitation. This was evident within and between subgenomes, with some accessions having a single significantly differentially expressed homoeolog from one subgenome, while both homoeologs were significant in other accessions. While the BnC subgenome (*B. oleracea*) had higher expression overall, our phase analysis showed that the BnA contribution (*B. rapa*) differed by time of day (phase) and/or by accession. The contribution of each subgenome to the phase groups varied across accessions. Some accessions, such as Ca, Av, and Br, showed a stronger bias toward representation from BnC, while others, including Ab, Gr, and Yu, exhibited a more balanced contribution. Notably, there were phase group-dependent differences in subgenome contributions within individual accessions. For example, in Mu, the +6 hour phase group displayed a greater contribution from BnC compared to BnA, unlike the other groups. Again, we did not identify any clear patterns correlating with crop type. However, a more time-resolved dataset may be necessary to obtain more accurate phase values and thoroughly assess these relationships. In general, these results indicate that both subgenomes are contributing to the overall response, but the magnitude of this contribution depends on the time of day and the accession.

Variation in the responsive subgenome and homoeolog copy was consistent across multiple pathways including sugar responsiveness, carbon fixation, sulfur allocation and GSL catabolism that likely resulted in differences in GSL content. These findings are consistent with previous Quantitative Trait Loci (QTL) and GWAS studies examining variation in GSL content. QTL and GWAS analysis combined with transcriptomics have associated variation in homoeolog expression of the Arabidopsis orthologs of MYB28, MYB29, and GLUCOSINOLATE TRANSPORTER-2 (GTR2) with altered levels of GSL content in B. napus seed[66]. Genomic deletions of MYB28 homoeologs is found in B. napus lines with low GSL levels[191] consistent with the role of this R2R3-MYB transcription factor in regulating aliphatic GSLs[146,147]. Multiple orthologs of the bidirectional GSL transporter GTR2 have been identified in B. napus that contribute differentially to GSL seed content. In addition to variation in protein sequence, altered tissue specific expression exists among the homoeologs[192]. These data highlight the inherent complexity of polyploid crops where multiple homoeologs contribute to a trait with differential regulation at the tissue level and following abiotic stress.[70].

The widespread variation in homoeolog responsiveness across *B.napus* accessions is surprising for a relatively young crop (approximately 7,800-12,000 years old;[39]. This variation might be partly explained by the ease with which its parental species, *B.rapa* and *B. oleracea*, hybridize and form new polyploids through spontaneous chromosome doubling[193]. Recent studies have revealed significant gene content variation among different morphotypes of these species[72], with some research indicating that nearly 20% of genes in *B.oleracea* are affected by presence/absence variation[54]. Given this inherent variability and potential to form new polyploids from different morphotype combinations, it is plausible that B. napus has multiple origins. While one study supports this hypothesis[194], more recent research has yielded conflicting results[195];[194]. Furthermore, introgression from diploid species into *B. napus,* which can yield meiotically stable tetraploid lines despite initially forming sterile triploids, may be another source of variation among cultivars[196]. Additional variation may arise from post-polyploidization processes, such as fractionation (loss) and homoeologous exchanges, which can shuffle and modify regulatory regions[197]. Rearrangements in the regulatory architecture, including promoter and enhancer sequences, could alter transcription factor binding sites and contribute to the divergence in homoeolog expression patterns and stress responsiveness. In *B.rapa*, divergence in circadian regulation of paralog expression has resulted in instances where one copy retains a more 'Arabidopsis-like' expression pattern that could be predicted based on the shared presence of conserved non-coding sequences[182]. We would expect a similar level of regulatory variation in *B. oleracea* that would contribute to additional variation in *B. napus*. The lack of information on the specific progenitors that gave rise to the accessions in this study makes it challenging to pinpoint the sources of variation.

3.4.2 Accession specific phase changes in photosynthesis related genes under WL

To compare the transcriptomic responses across accessions we generated a response score for every gene and clustered them based on their response patterns. This resulted in 11 distinct time of day dependent WL response patterns among the accessions. We found all accessions represented in each cluster but the biological processes were quite distinct. Even among accessions with similar growth responses to the WL condition (St and Av) we found very little overlap in the diel responses for similar pathways. We examined genes involved in photosynthesis and found dramatic phase changes across several accessions. Two accessions, St and Ab, with large growth differences tended to have high WL expression of genes involved in carbon fixation in the morning with decreasing expression throughout the day. This is in contrast to WW conditions where peak expression occurred from ZT7-ZT13 in Ab and at ZT7 and again at ZT21 in St. Across accessions the expression patterns for genes involved in carbon fixation varied dramatically with some showing minimal phase changes and others exhibiting an earlier phase of several hours. Although we observed these dynamic temporal changes in genes involved in photosynthesis, our CropReporter follow up experiment confirmed that these accessions were photosynthetically healthy under WL conditions. The variation in diel regulation of photosynthesis related genes in similarly responding genotypes suggests that these accessions adjusted to limited water using alternate regulatory strategies. Time of day dependent transcriptome responses to abiotic stress have been shown in other plant species[34,42,138,139], confirming the complexity of the regulatory network. In this study, we demonstrate additional complexity across genotypes that exhibit distinct expression pattern responses for photosynthetically healthy plants. This raises several questions about the timing of these shifts in expression during water limitation and whether they contribute to adaptation to the new conditions.

3.4.3 WL alters diel regulation of sulfur and glucosinolate metabolism

B. napus allocates sulfur and nitrogen supplies to glucosinolates (GSLs), specialized metabolites that are important for biotic (Barco and Clay 2019) and abiotic[165,166] defense. The balance between sulfur and nitrogen is critical for optimizing yield (McGrath and Zhao 1996; Jamal Arshad et al. 2010). The enrichment of GSL related terms for several accessions in our WL conditions led us to examine whether the metabolites were affected. We found evidence for unique diel regulation among the two accessions in both sulfoquinovose (SQ) and GSLs. Although both accessions had similar amounts of sulfur content in the leaves, Mu (low reduction in growth under WL) had significantly higher SQ under WL at night while St (high reduction in growth under WL) was significantly higher in the morning. We hypothesized that these temporal differences in sulfur pools would lead to time of day specific patterns of other sulfur intensive metabolites such as GSLs. We found opposing patterns of an indolic GSL between the accessions and as compared to SQ in WL conditions. While 4ME/NEO abundance increased significantly in Mu in the morning, in St 4ME/NEO abundance decreased in the morning suggesting distinct differences in sulfur allocation strategies under WL. In WW conditions Mu displayed an increase in GSL content (NAP, PRO, SIN, 4ME/NEO) throughout the day, while St showed little treatment response or decreased content (4ME/NEO), again highlighting alternate regulatory strategies among the accessions. Variation in GSLs is likely to be affecting available sulfur pools, thereby affecting primary metabolite pools (Cystine; Glutathione) and nitrogen use efficiency[171]. Further, as auxin and indolic GSLs such as 4ME/NEO are both derived from tryptophan, there is potential competition between these two pathways for sulfur reserves. Alternatively, there may be cross talk between these pathways where one pathway acts to mediate the other[165–167] to maintain plant growth under WL[167]. Further study into the relationship between diel regulation of sulfur allocation and abiotic stress response is needed to understand the tradeoffs in time of day specific changes to sulfur pools.

In the upcoming years, global crops will be subjected to more extreme temperatures including variation in water availability. To avoid reductions in crop yields, we must develop varieties more tolerant to these challenging environments. However, we found substantial variation in which homoeolog copy between parental subgenomes was most responsive and in the time of day they were altering expression under WL, further challenging our ability to identify good breeding targets. Regulatory regions may be more effective and conserved targets for improving crop tolerance under the effects of climate change.

3.5 Experimental Procedures

3.5.1 Plant materials and growth

Seeds from *B. napus* accessions were germinated in soil in a growth chamber for five days before transplanting sixteen seedlings per accession per treatment into four inch wide, 0.47 L pots (Pöppelman TEKU, Lohne, Germany). Plants were germinated and grown in MetroMix360 soil mixed with Osmocote Classic 14-14-14 fertilizer (Everris NA Inc., Dublin, Ohio, USA). Transplanted seedlings were randomly placed in the Conviron (Controlled Environments Limited, Winnipeg, Canada) growth house of the Bellwether Plant Phenotyping Facility (Donald Danforth Plant Science Center, St. Louis, MO). Growth house conditions were set to 22°C/18°C day/night temperatures, 14 hour day lengths, 40% relative humidity, and 400 µmol/m^2/s light intensity. Plants were equally separated into two treatments for the duration of the experiment: 100% watering (Well-Watered, WW) and 20% watering (Water Limited, WL). Target watering weights were determined as described in Fahlgren et al.[179]. All pots were watered once to 100% capacity on the first day of the experiment and then weighed twice per day to maintain a target water weight through day seven on the Bellwether phenotyping platform. Starting on day eight, WL plants were allowed to dry down to 20% capacity and maintained at 20% capacity until day 14 (seven days after water limitation treatment began). At this point all watering amounts were increased by 10% for an additional 14 days to compensate for plant biomass. Plants were automatically weighed and watered twice daily to maintain treatment water levels.

3.5.2 Image capture

Detailed descriptions of the Bellwether Phenotyping Facility and associated LemnaTec imaging systems (LemnaTec Scanalyzer) are available in[179]. On each day of the experiment, all plants moved from the controlled environment growth house to the imaging loop via a conveyor belt. The imaging loop contains two imaging chambers with visible light (Red Green Blue, RGB) and near infrared (NIR) cameras (Basler AG, Ahrensburg, Schleswig-Holstein, Germany). Two side view images from 0° and 90° angles were captured each day by the RGB and NIR cameras. Zoom levels were adjusted two times during the experiment as the plants grew. A total of 59,850 RGB and 59,848 NIR images were captured during the experiment. Images are available here: https://datacommons.cyverse.org/browse/iplant/home/shared/danforth_center/Ricono_et_al_Brassica.

3.5.3 Image analysis

Images captured by the Bellwether phenotyping platform were analyzed with PlantCV v3.14. Typical PlantCV image analysis workflows involve segmenting the plant from the image background by creating a mask and then using PlantCV analysis functions to extract data from the plant image. All PlantCV scripts used for image analysis are available in the public GitHub repository: https://github.com/danforthcenter/ktgreenham-brassica-drought-paper. Because some *B. napus* leaves can have a purple hue or become yellow as they age, simple segmentation methods on RGB images often fail to capture the whole plant. To segment plants in the RBG images, the PlantCV naïve bayes multiclass training module was used to classify pixels as "healthy," "senesced," or "background." To train the naive bayes classifier, the RGB values of pixels were sampled across multiple images to create a training dataset of 984 pixels belonging to each category. Pixels were chosen from green and purple leaves for the "healthy" category, from yellowing leaves for the "senesced" category, and from all non-plant aspects of the image for "background." The trained classifier was used to segment plants from all images and calculate the percentage of pixels in each plant that were classified as "senesced." After segmentation, a rectangular region of interest was positioned around the plant to further eliminate noise. The PlantCV "analyze_object" and "analyze_color" functions were used to measure the size of the whole plant and color characteristics of the healthy and senesced leaves in each plant image. For all RGB images, a separate PlantCV image analysis workflow was created for each zoom level. Resulting pixel values were converted to physical measurements (*i.e.* cm and cm^2) using a custom R script to combine the standardized data[179]. Plant area values from 0° and 90° angle images of each plant were summed.

For NIR images, only images from day 14 to day 26 were used so that the NIR signal data was measured from images at the same zoom level, since the zoom configuration impacts NIR values. Masks generated in the RGB workflow were resized and repositioned to segment the plant in the NIR image. NIR pixel intensities were extracted and the mean NIR signal was calculated using the "analyze_nir_intensity" function in PlantCV.

3.5.4 Statistical analyses

Data curation and statistical analyses were done in R[198]. All R scripts related to the phenotyping analysis are available at the public GitHub repository: https://github.com/danforthcenter/ktgreenham-brassica-drought-paper. Plant area was modeled per each accession and treatment group using a Bayesian Hierarchical growth model. We used a Bayesian model for the flexibility in accounting for heteroskedasticity and hypothesis testing. Our data variables were plant area (cm^2) and time which was expressed as days after start (days). We used a Gompertz growth model to estimate the model parameters where A is the asymptote, B is the inflection point, C is the growth rate of the growth curve, and t is time; and we used a logistic model to model the variance over time using the parameters *subA* (the asymptote on variance per treatment group), *subB* (the inflection point for the increase in variance), and *subC* (the rate at which variance increases). This was a distributional model using the Student T distribution and thus includes priors on the growth model parameters, the variance model parameters, and the degrees of freedom. We used lognormal priors for each estimated parameter in our growth model (A, B, C) and in the sub model for variance (*subA*, *subB*, *subC*). For the asymptote parameter we used a lognormal distribution with mu=130 and sigma=0.25. For the inflection point parameter we used a lognormal distribution with mu=15 and sigma=0.25. For the growth rate parameter we used a lognormal distribution with mu=0.25 and sigma=0.25. In the variance sub-model, we used lognormal priors with sigma of 0.25 centered at 20, 10, and 3 for *subA*, *subB*, and *subC* respectively. These thick-tailed lognormal distributions were chosen to make sure the data could pull away from the prior if necessary and because plant area must be positive. We used Student T prior distributions for smooth terms of variance with a mean at 0, standard deviation of 5, and 3 degrees of freedom and a gamma (2, 0.1) prior on the degrees of freedom. All priors and further details and the code to implement our models are on Github: https://github.com/danforthcenter/ktgreenham-brassica-drought-paper. Prior predictive checks were conducted using the plotPrior function from the pcvr R package[199] (Figure S10 and Figure S11). These models were fit in R using the brms (brms version 2.20.4, cmdstanr version 2.34.0) and pcvr packages (version 0.1.0). We used 2 chains, each with

1000 post-warmup draws. Our bulk and tail effective sample sizes based on MCMC range from 700 to 2200 with an average neff ratio of 0.57 and no neff ratios below 0.29. Finally, our Rhat values show good convergence with no Rhat values above 1.0106 for model parameters (Table S1). For details on our hypothesis testing see the results and discussion section and Table S1.

3.5.5 Tissue Collection and RNA-Sequencing

After 28 days (21 treatment days) on the Bellwether phenotyping system, leaf tissue was collected from the apex of the youngest leaf on each plant. Tissue was collected at ZT (Zeitgeber; hours after dawn) 1, 7, 13, and 19. For each accession, tissue was collected from three replicates of each treatment at each time point. RNA was isolated using the MagMAX Plant RNA Isolation Kit (Thermo Fisher Scientific, Waltham MA). RNA was submitted to the DOE Joint Genome Institute for library preparation and sequencing as part of a Community Science Program grant CSP504418. RNA sequencing was performed on a Novaseq S4 with PE150. Data is available through the NCBI GEO database (GSE261928, reviewer token: yrizcwcitfghbwl).

3.5.6 RNA-sequencing processing

Raw reads were mapped to a concatenated reference consisting of *B. rapa* R500 v1.2[200] and *B. oleracea* TO1000[201] genomes. This was accomplished by simply concatenating the source reference FASTA files together. The reads were trimmed and filtered by quality using Trimmomatic v0.33 (ILLUMINACLIP:TruSeq3-PE-2.fa:2:30:10 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:25 HEADCROP:14 MINLEN:50), and alignment was performed using HISAT2 2.1.0 (max_intron=12000, strandness=RF). Feature counts were created from bam files using the 'featureCounts' function from Rsubread R package (-F GTF -M --fraction -s 2 -p -B -C) and normalized using edgeR[198]. All counts from each sample are compiled into accession specific tables and provided on the NCBI GEO database (GSE261928). Genes with no variance in expression across the time series were removed; remaining genes are referred to as “processed genes" and are also available through the NCBI GEO database (GSE261928, reviewer token: yrizcwcitfghbwl). We further filtered out lowly expressed genes by retaining those with a mean log2 FPKM greater than zero in at least one time point. For the pattern detection in DiPALM, gene expression for any missing replicate (Table 1; ImputedReplicates tab) was imputed as the median of the two remaining replicates for that time point. All code for the RNA-seq analysis can be found in the associated Rmarkdown file on github https://github.com/GreenhamLab/Bnapus_WL_TranscriptomeTC.

3.5.7 Expression principal component analysis

Raw expression counts from FeatureCounts of all individual samples were merged with the script ‘combining_featCount_tables.py’ from the Bioinformatic Notebook[202], the combined counts table was loaded into DESeq2 v.1.26[203] under R v3.6.3[198]. Read counts were normalized with the variant stabilizing transformation and principal components were calculated with the plotPCA function and visualized with ggplot2[204].

3.5.8 SNP Calling and Principal Component Analysis

SNPs were called using GATK v4.1.4.1[205,206] on single replicates of RNAseq data for each accession, taken from control treatment plants at ZT1. Sequencing reads were first trimmed using Trimmomatic v.0.33[207] to remove Illumina TruSeq3 adapters. Trimmed reads were mapped to a concatenated *B. oleracea* acc.TO1000[201] and *B. rapa* acc. R500[200] reference genome using STAR v2.6.0[208] with settings “--outFilterMultimapNmax 1” and “--outFilterScoreMin 10”. We used a two-pass mode to improve mapping performance around splice junctions. Duplicates were marked and read groups were added with Picard v2.25.0[209]. Reads with Ns in the Cigar string, representing reads that span splice events, were split using the GATK v4.1.4.1 SplitNCigar tool. Each accession was processed with the pipeline independently, and genotyped independently with GATK HaplotypeCaller, using diploid settings. Individual gVCF files were combined with CombineGVCFs, and then jointly genotyped with GenotypeGVCFs, resulting in a single VCF file for the 16 samples containing 4,599,922 raw variants. After removing Indels with GATK SelectVariants, 3,439,690 SNPs remained.

A series of variant filters were used on this combined vcf to address possible technical issues related to the sequencing and alignment steps and produce a high confidence set of SNPs. These included filters for strand biases and mapping quality (e.g., removing sites with SOR $\geq$ 4; FS > 60, ReadPosRankSum > 10 and < -10, and MQ < 30), genotyping quality metrics (removing variants with QUAL < 0 and QD < 20) and site depth (removing variants with DP < 15). The distribution of values for all the filters used on the dataset was used to select thresholds. After filtering, 620,331 SNPs were retained and used for principal component analyses, performed with Plink v1.90b6.16[210] and visualized in R v 3.6.3[211] with ggplot2[204].

3.5.9 DiPALM Analysis

DiPALM[182] was used to identify differentially patterned genes for each accession. This pipeline identified sets of genes with either a change of phase (kME), or change in median expression level (Med), across the time course. Two accessions (Qu and Ze) were dropped from downstream analyses as they had strikingly low numbers of kME genes (17 and 16; respectively) and were unlikely to be informative. Due to the missing replicates and the need to impute for the DiPALM analysis, DiPALM calls for the remaining 14 accessions were validated by comparing the number of differentially patterned genes to the number of genes called by single time point analyses for the time points with three replicates. Linear models were run at each time point and accession and the final numbers were compared to the DiPALM calls. Results were largely comparable; for full details see the RMarkdown available at https://github.com/GreenhamLab/Bnapus_WL_TranscriptomeTC. For accessions with missing replicates, we selected the DiPALM calls that overlapped with the single time point analyses.

3.5.10 Subgenome comparison

To quantify the relative diel contribution of each subgenome (BnA, BnC) to the number of WL genes for each accession, we identified the average phase (peak timing of expression) of each gene and then calculated the phase difference between WW and WL. This resulted in five phase change groups (0 = no change; 6 = six hours earlier; 12 = twelve hours earlier; -6 = six hours delayed; -12 = twelve hours delayed). Within each phase change group we calculated the proportion of WL genes relative to the total number of WL genes for each accession. We then identified similar phase changes under WL in our DiPALM clusters and ran Gene Ontology (GO) analyses to identify differences in regulation of enriched biological processes for these clusters. Differentially patterned *B. napus* subgenome homoeologs (kME; Med) were compared to identify the proportion of shared genes that respond to WL among the accessions. For each WL gene, the number of accessions were summed and the proportion was calculated relative to the total number of shared genes from all accessions. Syntenic orthologs in Arabidopsis for each WL gene were also analyzed and the relative proportion calculated as described for subgenome-specific homoeologs.

3.5.11 Response score module assignment

To compare WL transcriptomic responses across accessions, we generated response scores by taking the difference between WL and WW expression for every gene at each time point for all accessions. The response scores for all accessions were used in a co-expression network analysis[119] to identify global response score modules across accessions. A Gene Ontology (GO) analysis was done for each accession per response module using previously compiled annotations[182] and their corresponding biological process. As the concatenated *B. napus* reference inherits gene annotations from the source R500 and TO1000 references, GO term mappings were generated by associating accessions from the source references to their orthologs in the TAIR10 reference (Arabidopsis) which has functional annotations. In two cases (M5; M9) some accessions had opposing responses within a given module/time point (*i.e.* some were up in WL and some down at a given ZT). In these cases the module was further split and GO was run separately. Results of each analysis were compiled and GO Terms were grouped and assigned to a "broad descriptor", analogous to a parent term, to reduce complexity and to aid in interpretation (Table 1; GO_Terms tabs).

3.5.12 Plant materials and growth for CropReporter Measurements

Six *B. napus* accessions (Ab, Ne, Mu, Av, St, and DH12) were grown in conditions mimicking the original experiment done on the Bellwether phenotyping platform. Seeds were germinated on wet whatman paper to improve germination rates. All but St were allowed to germinate for three days; St was given five days. Five replicates per treatment and accession were sown into four inch diameter, 0.47 L pots (Pöppelman TEKU, Lohne, Germany), filled with equal amounts of a custom Jolly Gardener soil mixture (made as closely as possible to MetroMix360, which was used in the previous phenotyping experiment, but was no longer available at the time of this experiment). The accessions were split into three groups of two accessions each, and germination and planting were staggered to allow subsequent imaging during the experiment to occur within a two hour window at the same time of day for all accessions. Plants were manually watered daily to a specific target weight as described above. Plants were watered to control levels with fertilized water for six days after planting, before the 20% water-limitation treatment began at seven days after planting. After this point, plants were only watered with reverse osmosis water. At 14 days after planting, seven days after the treatment began, watering amounts were increased by 10% to account for the growing plants, as was done in the previous phenotyping experiment.

3.5.13 CropReporter Imaging

Whole plant chlorophyll fluorescence and multispectral images were captured using the PhenoVation CropReporter (PhenoVation Life Sciences, Wageningen, The Netherlands). Images were captured from the top down, and the distance from the camera to the top of the plants was kept consistent at ~105 cm throughout the experiment. Plants were dark adapted for 30 minutes prior to imaging. We first captured the induction curve of the dark-adapted plants followed by a six minute light-adaptation with actinic light at a light intensity of 400 µmol/m^2/s. The induction curve of the light-adapted plants was captured as well. More detail is available in the methods described in Lazarević et al.[212,213]. Plants were imaged at ZT8 on day 0 (pre-treatment), and 7, 10, 14, and 21 days after the water limitation treatment began. Images are available on CyVerse: https://datacommons.cyverse.org/browse/iplant/home/shared/danforth_center/Ricono_et_al_Brassica.

3.5.14 CropReporter image and statistical analyses

Image analysis was done with the open-source software PlantCV[179,180] v3.14.1[214]. Data visualization and statistical analyses were conducted in R v4.3.1[198], largely using the pcvr package[199]. Values were normalized for plant size. Scripts are available in the public GitHub repository https://github.com/danforthcenter/ktgreenham-brassica-drought-paper.

3.5.15 Plant materials and growth for physiology and metabolomics

Seeds for *B. napus* accessions Ab, Mu, and St were imbibed on wet whatman paper and sown in a 50/50 mixture of Sungro Professional Growing Mix and ProMix BRK in 4.5 X 4 inch circular pots. Soil was dried down at 65˚C for four days and hand mixed intermittently. 15 plants per treatment were grown for St and Mu and 11 for Ab. All plants were grown in a Conviron chamber as described above, with slightly higher (~60-80%) relative humidity. Target weights were re-calculated to account for differences in soil types. All plants were fertilized with 100% FC liquid Hoaglands fertilizer upon sowing. All pots were maintained at 80% or 20% FC the first week of growth and target weights were increased by 10% thereafter. Water limitation was initiated after one week of 100% FC watering. Pots were manually weighed daily throughout the experiment to maintain water levels.

3.5.16 Photosynthetic rate

Photosynthetic rate (*A*) was measured using a Li-COR LI-6800F Portable Photosynthesis System (LI-COR, Lincoln NE, USA). Physiological responses were recorded at least once a week on five replicates throughout the experiment starting from one week after WL. Measurements were taken on the first fully unfurled true leaf for the first three weeks followed by leaf five (22-27 days after WL) and seven (28+ days after WL) to account for maturation and senescence. Chamber settings matched environmental conditions of the growth chamber. LiCOR measurements were as follows: flow rate, 500 mmol s^{-1}; CO_2 concentration, 400 mmol mol^{-1}; VPD, 0.8-1.1kPa; and leaf temperature, 22°C.

3.5.17 Chlorophyll content

Chlorophyll content (a; b) was measured for each Li-COR replicate (n = 5 per accession; N = 30) from the seventh leaf using a 9mm (~12-18mg) leaf punch taken at the top and left of the midvein. Each punch was immediately submerged in glass vials containing 2mL of 80% acetone and shaken overnight in the dark. The following morning, extracts were quantified on a spectrometer (Molecular Devices SpectraMax i3X) in a 2:1 ratio of extract: 80% acetone. Samples were run in triplicate.

3.5.18 Metabolomics

Metabolomics samples were collected at ZT4, ZT12 and ZT20 from the seventh leaf that was paired with chlorophyll leaves using the same 9mm core borer as described for chlorophyll content. To avoid rapid metabolite degradation associated with a wounding response, 50mL Falcon tubes were placed on dry ice and partially filled with liquid nitrogen to create ultra cold reservoirs. Leaf disks were immediately dropped into reservoirs and stored in liquid nitrogen until they could be transferred to individual screwtop tubes with a single 2mm tungsten carbide bead and stored at -80˚C until pickling. 'Pickling' involves fully submerging tissue in 40% w/v 100% HPLC grade methanol on dry ice prior to storage for at least two weeks at -80˚C to slowly denature myrosinases and other enzymes. Tissue was then homogenized using frozen adapters for a Qiagen TissueLyser II and homogenized at a rate of 30Hz for 45 seconds for one to two rounds, or until the tissue was fully homogenized. Samples were subjected to centrifugation at 15,000 rpm and 4˚C for 15 minutes to separate the supernatant from cellular debris. Supernatant was collected and spun once more at 15,000 rpm in 4˚C for five minutes before storing at -80˚C.

Samples were diluted (1:4) before injection onto a UHPLC (Ultimate 3000, Dionex) ZIC-cHILIC column (100 mm × 2.1 mm, 3 µm particle size, EMD Millipore Corporation, Billerica, MA) at a flow rate of 1 µL/min. Solvents A (0.1% v/v formic acid in water) and B (0.1% v/v formic acid in acetonitrile) were used as mobile phases for gradient separation. Gradient separation was as follows: 2 min from 98% B, 0.5 min to 88% B, 1 min to 85% B, 2.75 min to 72% B, 0.75 min to 70% B, 2 min to 65% B, 1 min to 55% B, 1 min to 45% B and 98% B maintained for 2 min. The column was equilibrated for 2 min with 98% B prior to the next injection. Samples were detected using a hybrid quadrupole Orbitrap mass spectrometer (Q Exactive, Thermo Fisher Scientific, San Jose CA) with an autosampler and sample vial block maintained at 4°C. Full scan MS (99–1500 *m/z*) were acquired with positive and negative mode using polarity switching. The sheath gas, aux gas, and sweep gas were set to 50, 20, and 1 psi respectively. The capillary temperature was 350˚C. The S-lens RF value was set to 55 (arbitrary units). The Aux gas heater temperature was 300 ˚C. Samples were processed in two batches, one for each accession. Biological samples were run in triplicate to account for technical variation between runs. Data were acquired using Xcalibur™ software version 2.1 (Thermo Scientific). Raw data was converted to .mzML files using MSConvert and targeted peak detection was performed in mzMine 2.53[215]. Visualization of chromatograms was performed using mzMine to assess potential drift in retention times (RTs) and to assign limits of detection (LODs, 2x background) and limits of quantification (LOQs, 5x background) for each GSL. For full details including mass ranges used for extracting features see Table S4. We normalized across technical replicates by summing the relative abundance over a representative negative mode region of each run (Mu: RT 3.20-6.90 and *m/z* 600-1200; St: RT 5.2 to 6.8 and *m/z* 640 - 1220) and then divided each relative abundance value by the sum of this region for that sample. This value was then multiplied by the average of the representative region across all samples by accession to account for variation across runs. Raw data is available on the Data Repository for U of M (DRUM; https://conservancy.umn.edu/collections/6c548d8b-0f3a-4f6b-b16e-4154c88136c0).

3.5.19 Soil analyses and leaf nitrogen content

The entirety of the seventh leaf was collected, weighed, and fully dried for sulfur and nitrogen content. Approximately 1g of dry tissue was sent to the Analytical Laboratory at the University of Minnesota's St. Paul campus (https://ral.cfans.umn.edu/).

3.6 Acknowledgements

We would like to acknowledge the loss of Dr. Todd C. Mockler, who passed away while we were preparing this paper. The work began in Dr. Mockler's lab with his mentorship guiding the initial project design. We wish to thank Mindy Darnell from the Bellwether Phenotyping Facility (RRID: SCR_019049), Kristina Haines and the Plant Growth Facility at the Danforth Center (RRID: SCR_024902), and Noah Fahlgren and the Data Science Facility at the Danforth Center. We would also like to thank Noah Fahlgren, Kerri Gilbert, Bob Vanburen, Miranda Haus, and Nadia Shakoor for help with time-course tissue sampling, Yao Cao for performing all of the RNA extractions, and Ananda Menon for the many hours he spent watering and maintaining the metabolomics plants. *Brassica napus* accessions were obtained from Rod Snowdon and Sarah Schiessl (Justus Liebig University) and Chris Pires (University of Colorado). The RNA-sequencing work (proposal: 10.46936/10.25585/60001202) conducted by the U.S. Department of Energy Joint Genome Institute (https://ror.org/04xm1d337), a DOE Office of Science User Facility, is supported by the Office of Science of the U.S. Department of Energy operated under Contract No. DE-AC02-05CH11231.

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3.8 Conflict of Interest Statement

The authors declare that they have no conflicts of interest.

3.9 Data Availability

All raw and processed data have been made publicly available. Images from the phenotyping experiments are available on Cyverse (https://datacommons.cyverse.org/browse/iplant/home/shared/danforth_center/Ricono_et_al_Brassica.). RNA sequencing data is available through the NCBI GEO database (GSE261928), reviewer token: **yrizcwcitfghbwl**. All scripts used to analyze the image data can be found on github (https://github.com/danforthcenter/ktgreenham-brassica-drought-paper). All code used to analyze the RNA sequencing data including the scripts used to generate the figures can be found on github (https://github.com/GreenhamLab/Bnapus_WL_TranscriptomeTC). Raw metabolomics data can be found at the Data Repository for the University of Minnesota (DRUM; https://conservancy.umn.edu/collections/6c548d8b-0f3a-4f6b-b16e-4154c88136c0) by searching the manuscript title. Software used in this study is open-source or available for free.

Chapter 3 Supplemental Tables

CH3 Table 1.) Table of Gompertz growth model parameters and hypotheses.

	b_A_trtWate	b_A_trtWellv	b_B_trtWater	b_B_trtWellv	b_C_trtWater	b_C_trtWellv	b_subA_trtW	b_subA_trtW	b_subB_trtW	b_subB_trtW	b_subC_trtW	b_subC_trtW	nu	lprior	lp__
Min.	0.3809343	0.38350663	0.44187396	0.39769967	0.35537574	0.32160015	0.499121	0.51802887	0.3101512	0.59842069	0.29425268	0.60884552	0.45980825	0.44151726	0.34907368
1st Qu.	0.46335235	0.43458908	0.50636577	0.44189556	0.39183355	0.38302629	0.577194	0.6028708	0.54695062	0.62140079	0.58860134	0.6385986	0.59146038	0.51061946	0.40172295
Median	0.49052612	0.47800844	0.58453469	0.48045545	0.41289285	0.40846953	0.66805559	0.67006276	0.63086867	0.6784434	0.64522144	0.66727995	0.6555727	0.54427853	0.41658377
Mean	0.49909193	0.49136496	0.56515652	0.4880041	0.42994641	0.41918034	0.64872392	0.65358808	0.60768305	0.674899	0.62531742	0.68151692	0.64790229	0.57142789	0.41579837
3rd Qu.	0.50847849	0.52019257	0.6033019	0.52011606	0.45233516	0.43710274	0.73559943	0.69447873	0.68724912	0.71237512	0.70458051	0.72074468	0.68972002	0.60122365	0.43533202
Max.	0.68151144	0.65485425	0.67947111	0.61376744	0.579556	0.54816625	0.74749293	0.76944145	0.83344284	0.78457544	0.80494839	0.79557164	0.87243209	0.89962905	0.47691521

CH3 Table 2.) Table of the Kruskal-Wallis chi-squared test results for the NIR phenotyping data for all accessions.

	b_A_trtWate	b_A_trtWellv	b_B_trtWater	b_B_trtWellv	b_C_trtWater	b_C_trtWellv	b_subA_trtW	b_subA_trtW	b_subB_trtW	b_subB_trtW	b_subC_trtW	b_subC_trtW	nu	lprior	lp__
Min.	0.9994745	1.00010823	0.99935723	0.99970047	0.99995549	0.99984784	0.99915454	0.99944721	0.9996525	0.99966745	0.99940112	0.99977119	0.99950374	0.99955169	0.99994039
1st Qu.	1.0001073	1.00074364	1.00033191	1.00144617	1.00080852	1.00100718	1.00045743	1.00011413	1.00041463	1.00029839	1.00027168	1.00028645	1.00023476	1.00000975	1.00100569
Median	1.00070563	1.00138515	1.00080395	1.00196864	1.00125343	1.00232097	1.00080659	1.00059114	1.00087525	1.00109088	1.00113517	1.00076019	1.00062443	1.00062176	1.00202135
Mean	1.00111393	1.00242136	1.00099728	1.00214051	1.00154733	1.00283636	1.00112544	1.00062505	1.00144974	1.00108191	1.0018517	1.000835	1.00093102	1.00101166	1.00244277
3rd Qu.	1.00154983	1.00435899	1.00161452	1.00250878	1.00205247	1.00409388	1.00144712	1.00081521	1.00135164	1.00145515	1.00267164	1.00108286	1.00135775	1.00209631	1.00375692
Max.	1.00449631	1.00609093	1.00327086	1.00532611	1.00514231	1.00745928	1.00703575	1.00226314	1.00829138	1.00415397	1.00749107	1.00255851	1.0028084	1.0034434	1.00568213

CH3 Table 3.) Metadata table for the transcriptome analysis.

Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star	par	fit
(A_trtWellwa	84.8519571	8.16928685	71.2149575	98.352115	Inf	1	*	A	Ab_fit
(B_trtWellwa	1.92523337	0.18698311	1.62134288	2.23777405	Inf	1	*	B	Ab_fit
(C_trtWellwa	-0.0021567	0.00789654	-0.0148924	0.01147004	0.6286645	0.386		C	Ab_fit
(A_trtWellwa	29.3534346	6.12782356	19.1717175	39.5347475	Inf	1	*	A	Al_fit
(B_trtWellwa	1.63959524	0.26722986	1.21781478	2.07729163	Inf	1	*	B	Al_fit
(C_trtWellwa	0.0229128	0.00943621	0.0075593	0.03823471	104.263158	0.9905	*	C	Al_fit
(A_trtWellwa	106.23503	9.5304557	90.95447	122.46879	Inf	1	*	A	Av_fit
(B_trtWellwa	1.80767399	0.26707018	1.38044988	2.2576872	Inf	1	*	B	Av_fit
(C_trtWellwa	-0.0094509	0.01060736	-0.0265887	0.00709903	0.24223602	0.195		C	Av_fit
(A_trtWellwa	51.2812607	7.55864738	39.2063125	63.6095025	Inf	1	*	A	Br_fit
(B_trtWellwa	2.16716655	0.30988483	1.69686785	2.7008463	Inf	1	*	B	Br_fit
(C_trtWellwa	0.00557366	0.00922518	-0.0092458	0.02062753	2.6036036	0.7225		C	Br_fit
(A_trtWellwa	64.6683385	7.23418509	53.0498625	76.892385	Inf	1	*	A	Ca_fit
(B_trtWellwa	1.50977749	0.19190764	1.204559	1.82738063	Inf	1	*	B	Ca_fit
(C_trtWellwa	-0.0154789	0.00793787	-0.0283563	-0.0029204	0.02722137	0.0265		C	Ca_fit
(A_trtWellwa	72.7649993	5.68557887	63.5675425	82.106305	Inf	1	*	A	Da_fit
(B_trtWellwa	2.55827245	0.26400097	2.15197098	3.00111183	Inf	1	*	B	Da_fit
(C_trtWellwa	-0.0002548	0.01035097	-0.0167005	0.01715835	0.94552529	0.486		C	Da_fit
(A_trtWellwa	118.569467	12.7229284	97.572765	139.715923	Inf	1	*	A	DH12_fit
(B_trtWellwa	2.58445856	0.25065859	2.1818805	3.00728863	Inf	1	*	B	DH12_fit
(C_trtWellwa	-0.0038224	0.0078539	-0.0168514	0.00923351	0.46735143	0.3185		C	DH12_fit
(A_trtWellwa	73.2961365	13.217832	51.897575	95.66595	Inf	1	*	A	DH20_fit
(B_trtWellwa	2.44364641	0.26421813	2.02498893	2.88628453	Inf	1	*	B	DH20_fit
(C_trtWellwa	0.01448402	0.00730572	0.00224752	0.02665548	42.4782609	0.977	*	C	DH20_fit
(A_trtWellwa	54.188878	6.72978875	43.3745925	65.4283925	Inf	1	*	A	Gr_fit
(B_trtWellwa	1.71011004	0.22837354	1.33716378	2.09170538	Inf	1	*	B	Gr_fit
(C_trtWellwa	0.00534442	0.00835089	-0.0089842	0.01853977	2.90625	0.744		C	Gr_fit
(A_trtWellwa	42.2667154	5.04929661	34.253205	50.46867	Inf	1	*	A	Mu_fit
(B_trtWellwa	1.88692717	0.44862285	1.17557973	2.66522765	Inf	1	*	B	Mu_fit
(C_trtWellwa	-0.0188205	0.01458375	-0.0437172	0.00459407	0.11111111	0.1		C	Mu_fit
(A_trtWellwa	62.9417684	8.00792412	50.66419	76.4742075	Inf	1	*	A	Ne_fit
(B_trtWellwa	1.9251672	0.28835362	1.47479385	2.45115275	Inf	1	*	B	Ne_fit
(C_trtWellwa	-0.006804	0.01051895	-0.0238028	0.01083039	0.36518771	0.2675		C	Ne_fit
(A_trtWellwa	45.6374884	4.4067635	38.38083	53.040565	Inf	1	*	A	Qu_fit
(B_trtWellwa	2.2084496	0.4034027	1.56264065	2.8733155	Inf	1	*	B	Qu_fit
(C_trtWellwa	-0.0133154	0.0128806	-0.0336161	0.00878691	0.18063754	0.153		C	Qu_fit
(A_trtWellwa	44.8100334	6.73051557	33.82376	56.05885	Inf	1	*	A	Se_fit
(B_trtWellwa	3.59739548	0.70781466	2.52244363	4.82693835	Inf	1	*	B	Se_fit
(C_trtWellwa	-0.0058757	0.01622575	-0.0318641	0.02042112	0.55400155	0.3565		C	Se_fit
(A_trtWellwa	110.546313	14.2995567	87.84552	134.93232	Inf	1	*	A	St_fit
(B_trtWellwa	2.43793552	0.45607475	1.70213405	3.20466598	Inf	1	*	B	St_fit
(C_trtWellwa	0.00296346	0.00926518	-0.0128753	0.01732882	1.69179004	0.6285		C	St_fit
(A_trtWellwa	78.4626502	8.24746242	65.91787	92.037205	Inf	1	*	A	Yu_fit
(B_trtWellwa	1.96250902	0.35382412	1.4191898	2.55322655	Inf	1	*	B	Yu_fit
(C_trtWellwa	-0.0015514	0.00954932	-0.0177273	0.01381944	0.80668473	0.4465		C	Yu_fit
(A_trtWellwa	53.9546451	4.83234445	46.0647075	61.77131	Inf	1	*	A	Ze_fit
(B_trtWellwa	2.21646792	0.37457615	1.62930348	2.839235	Inf	1	*	B	Ze_fit
(C_trtWellwa	-0.0124636	0.01194232	-0.0321634	0.00705098	0.16482236	0.1415		C	Ze_fit

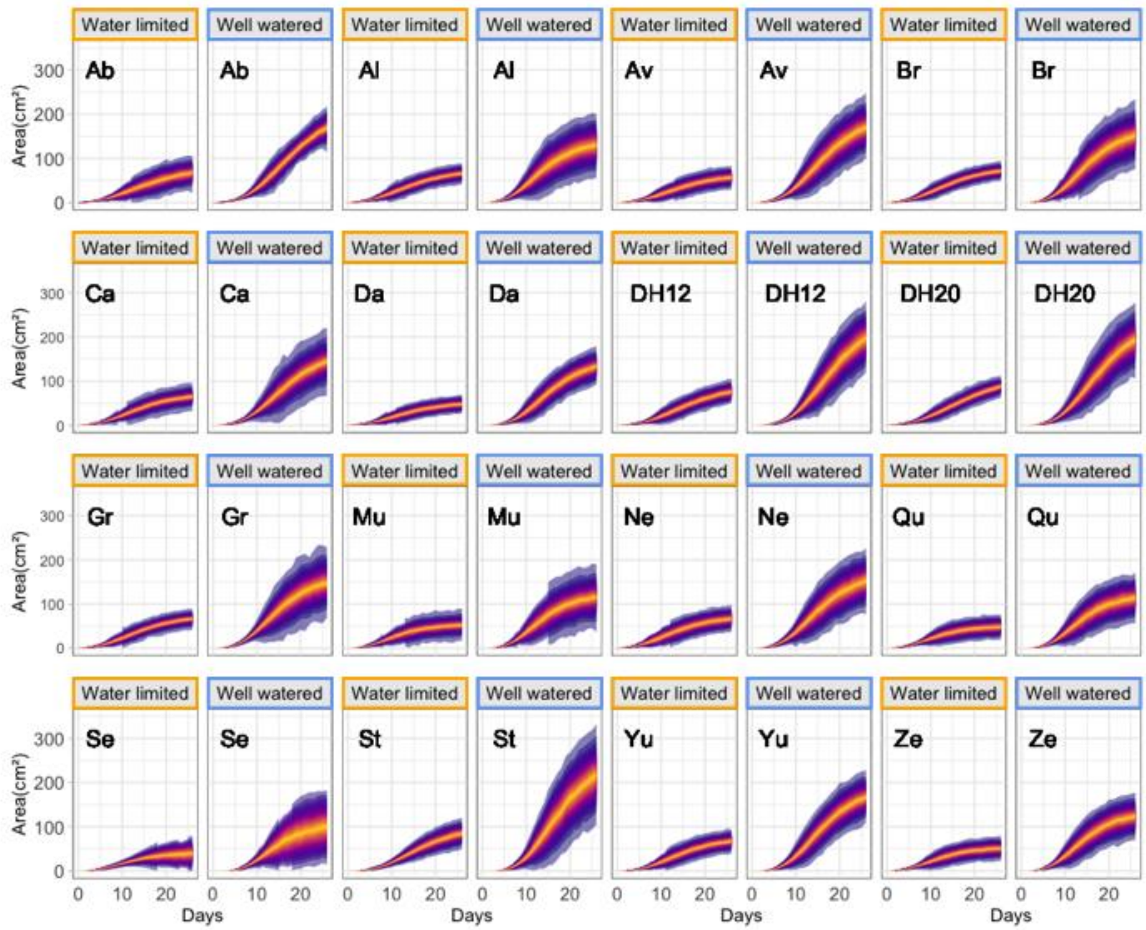
CH3 Table 4.) Table of the glucosinolates analyzed in the metabolomics analysis. Mass ranges used for extracting the glucosinolate features for St and Mu including the retention start and end times for detection.

Class	Abbreviation	Feature	exact monoisotopic *m/z* [M-H]-	RT start	RT end	Centroid RT	Polarity
aliphatic	SIN	sinigrin	358.0271935	8.8	9.14	8.97	Negative
aliphatic	NAP	gluconapin	372.0428475	8.18	8.69	8.4	Negative
aliphatic	PRO	progoitrin	388.0377615	9.3	9.65	9.55	Negative
aromatic	NAS	gluconasturtiin	422.0584975	7.5	7.93	7.7	Negative
indole	4OH	4-hydroxyglucobrassicin	463.0486605	9.9	10.55	10.2	Negative
indole	4ME	4-methoxyglucobrassicin (or Neog	477.0643105	8.12	8.54	8.4	Negative
Sulfinated su	SQ	Sulfoquinovose	244.025284	11.1	11.28	11.19	Negative

CH3 Table 5.) NIR values and statistics used to quantify treatment responses.

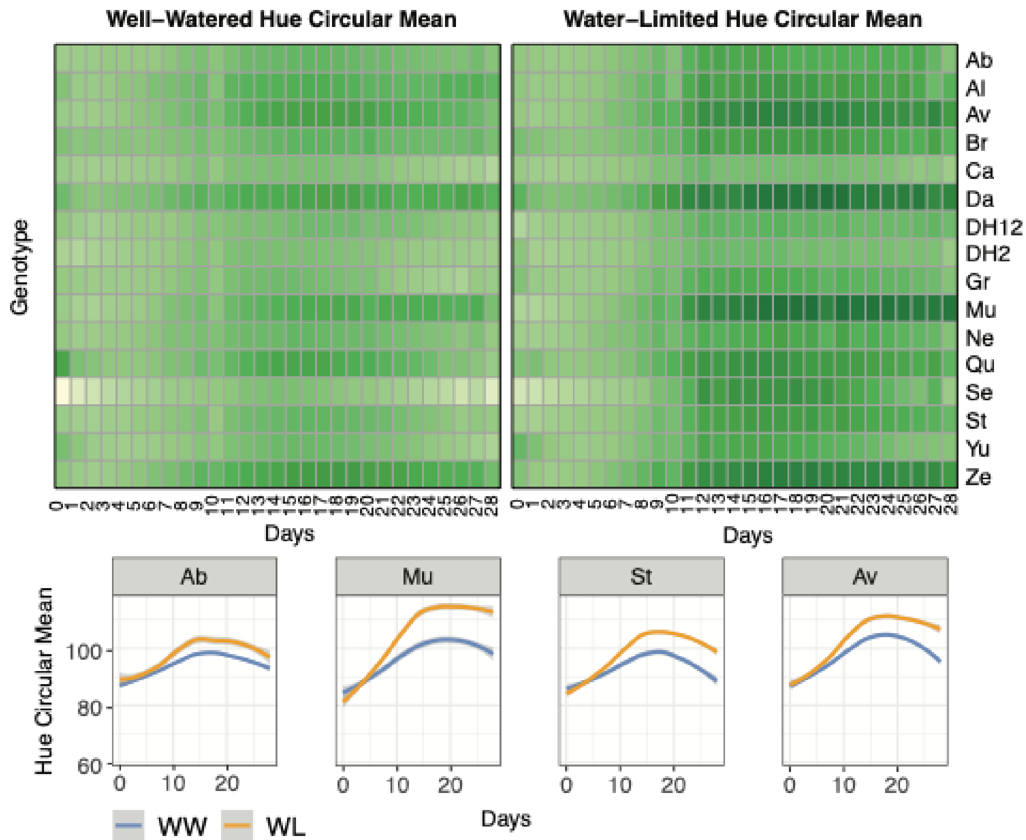
Accession	Experiment d	Kruskal-Wall	df	p-value
Ab	Day 15	8.2045	1	0.004179
Ab	Day 20	4.3393	1	0.03724
Ab	Day 26	15.364	1	8.87E-05
Al	Day 15	12.82	1	0.000343
Al	Day 20	0.53333	1	0.4652
Al	Day 26	17.188	1	3.39E-05
Av	Day 15	9.5511	1	0.001998
Av	Day 20	7.0848	1	0.007774
Av	Day 26	13.642	1	0.0002212
Br	Day 15	8.8651	1	0.002907
Br	Day 20	5.6333	1	0.01762
Br	Day 26	14.49	1	0.0001409
Ca	Day 15	10.023	1	0.001546
Ca	Day 20	3.1527	1	0.0758
Ca	Day 26	17.188	1	3.39E-05
Da	Day 15	3.9901	1	0.04577
Da	Day 20	0.5102	1	0.4751
Da	Day 26	1.0355	1	0.3089
DH12	Day 15	7.5696	1	0.005936
DH12	Day 20	6.0167	1	0.01417
DH12	Day 26	17.501	1	2.87E-05
DH20	Day 15	9.3196	1	0.002267
DH20	Day 20	2.8167	1	0.09329
DH20	Day 26	8.642	1	0.003285
Gr	Day 15	1.2784	1	0.2582
Gr	Day 20	4.1209	1	0.04236
Gr	Day 26	8.2045	1	0.004179
Mu	Day 15	0.75142	1	0.386
Mu	Day 20	0.24	1	0.6242
Mu	Day 26	0.41051	1	0.5217
Ne	Day 15	5.8182	1	0.01586
Ne	Day 20	0.64103	1	0.4233
Ne	Day 26	5.4602	1	0.01945
Qu	Day 15	7.3636	1	0.006656
Qu	Day 20	2.1618	1	0.1415
Qu	Day 26	5.2855	1	0.0215
Se	Day 15	3.0057	1	0.08297
Se	Day 20	1.641	1	0.2002
Se	Day 26	6.9602	1	0.008334
St	Day 15	10.023	1	0.001546
St	Day 20	7.35	1	0.006706
St	Day 26	0.96023	1	0.3271
Yu	Day 15	10.506	1	0.00119
Yu	Day 20	2.625	1	0.1052
Yu	Day 26	11.251	1	0.0007956
Ze	Day 15	4.2969	1	0.03818
Ze	Day 20	1.8	1	0.1797
Ze	Day 26	9.5511	1	0.001998

Chapter 3 Figures



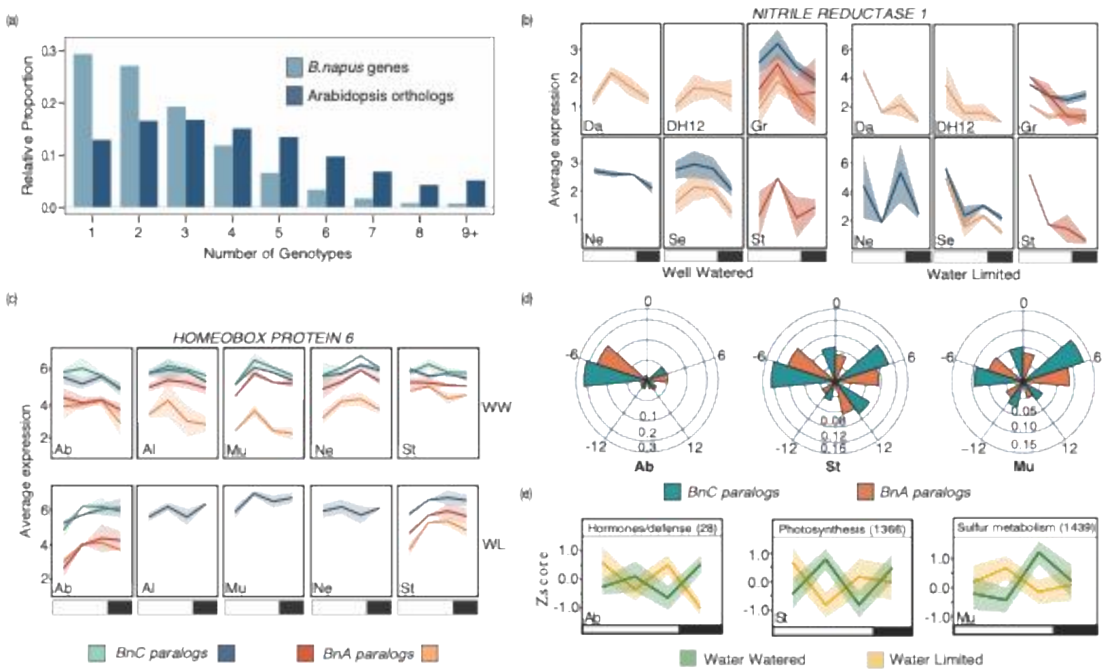
CH3 Figure 1.) All accessions reduced plant area under four weeks of water limitation

Growth curves for well-watered and water-limited groups of each accession with Bayesian posterior predictive intervals. Bayesian Hierarchical growth modeling was used to fit plant area data to a Gompertz growth model. Posterior predictive intervals are visualized for each accession and treatment (orange border, water-limited; blue border, well-watered). The ribbon color gradient represents the probability that the population mean falls within that interval, with the yellow area having the highest probability. Growth models of all accessions showed reduced final size in water-limited plants. The asymptote parameter (*A*) was at least 50% larger in the well-watered group for all accessions, suggesting that water limitation reduced final plant size in all accessions.



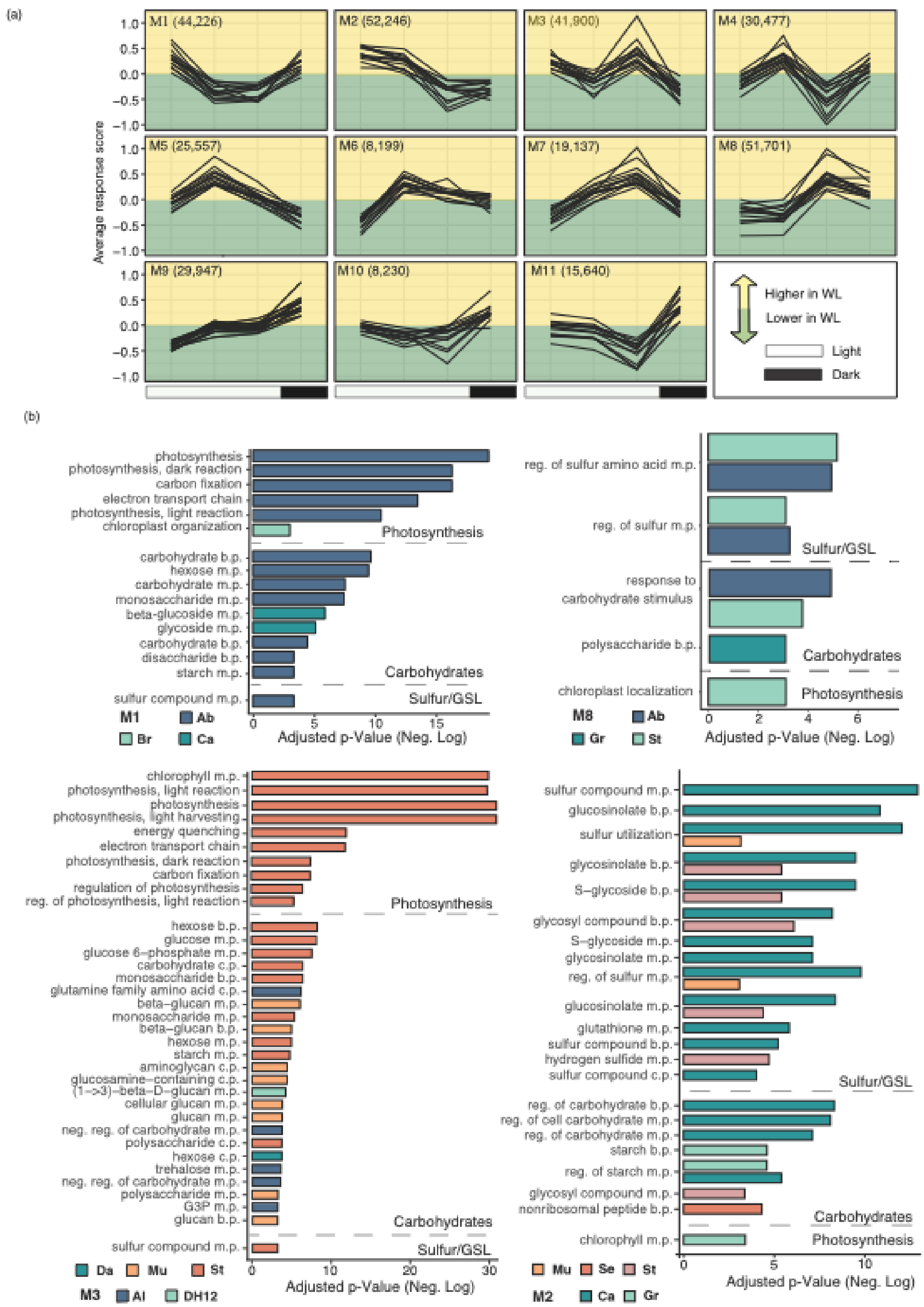
CH3 Figure 2.) Variation in hue circular mean across accessions under water limitation

Heatmaps of average hue circular mean values of plant pixels classified as green (Top). Examples of difference in hue circular mean under water-limited and well-watered conditions varies across *B. rapa* accessions (bottom).



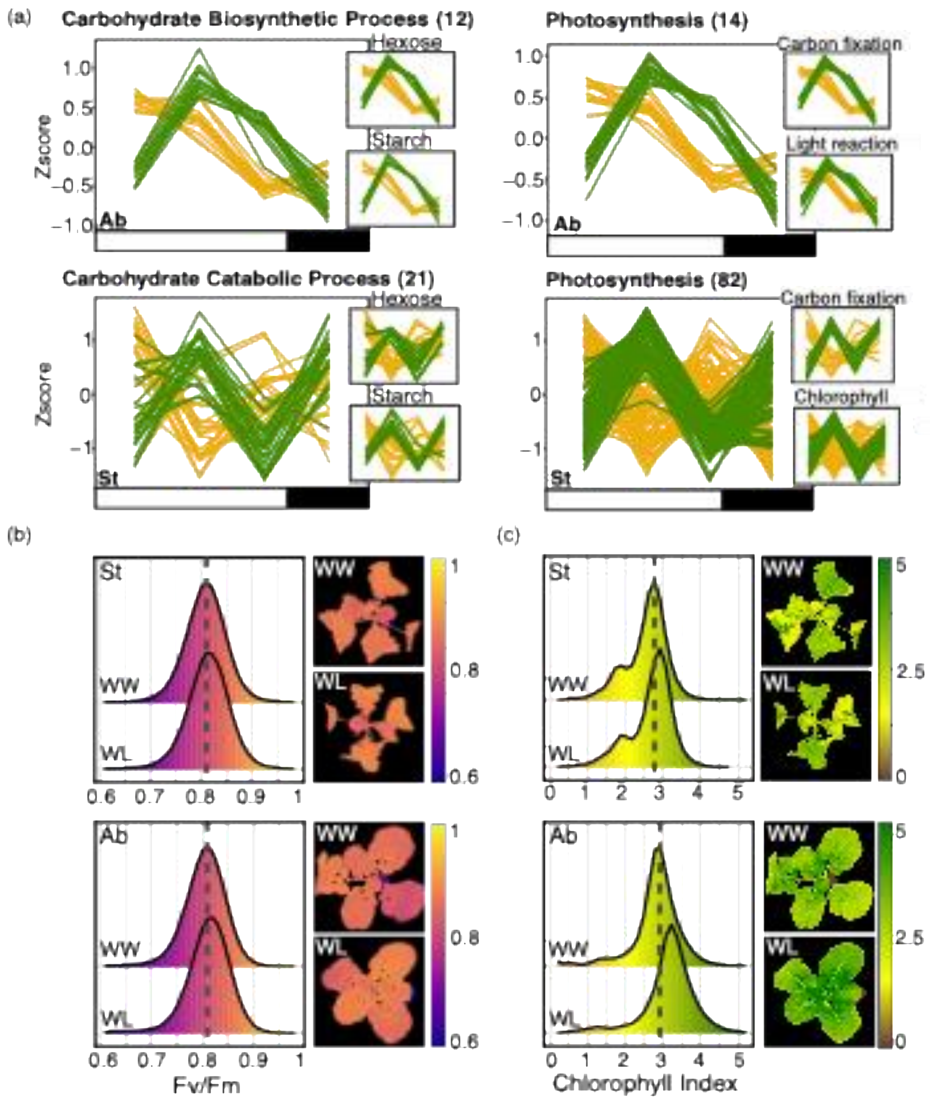
CH3 Figure 3.) Extensive divergence in the responsive paralog of conserved and sugar responsive genes.

(a) Overlap of all shared WL *B. napus* genes (light blue bars) compared to the overlap with the Arabidopsis orthologs (dark blue bars) across accessions. (b) Expression of the *NITRATE REDUCTASE 1* ortholog that was WL in four accessions (from Panel a) with extensive variation among accessions for which paralog was responsive. Bars under the panels indicate lights on (white) and lights off (black). (c) Average expression for orthologs of *HOMEOBOX PROTEIN 6* that exhibit differences in the number of WL responsive paralogs across accessions. (d) Distribution of phase change groups for all WL genes by subgenome in St (large difference in growth), Yu (moderate difference in growth), and Mu (small difference in growth). Numbers outside the radial plots indicate the phase group (difference in the timing of peak expression in hours between WW and WL) and numbers inside the circles represent the relative proportion of WL genes in a phase shift group relative to all WL genes for each accession; proportions are associated with the ring directly above the number. (e) Enriched Gene Ontology process for the DiPALM clusters representing the phase groups -6 for St, 12+ for Yu and +6 for Mu as shown in Panel d. Yellow lines indicate WL patterns and green are WW patterns. Shading is one standard deviation of the Z-score of gene expression. Numbers inside the parentheses indicate the number of enriched genes in that cluster. Bars under the panels indicate lights on (white) and lights off (black).



CH3 Figure 4.) Accessions exhibit similar diel transcriptional responses under WL but the genes underlying these patterns are enriched in different biological processes.

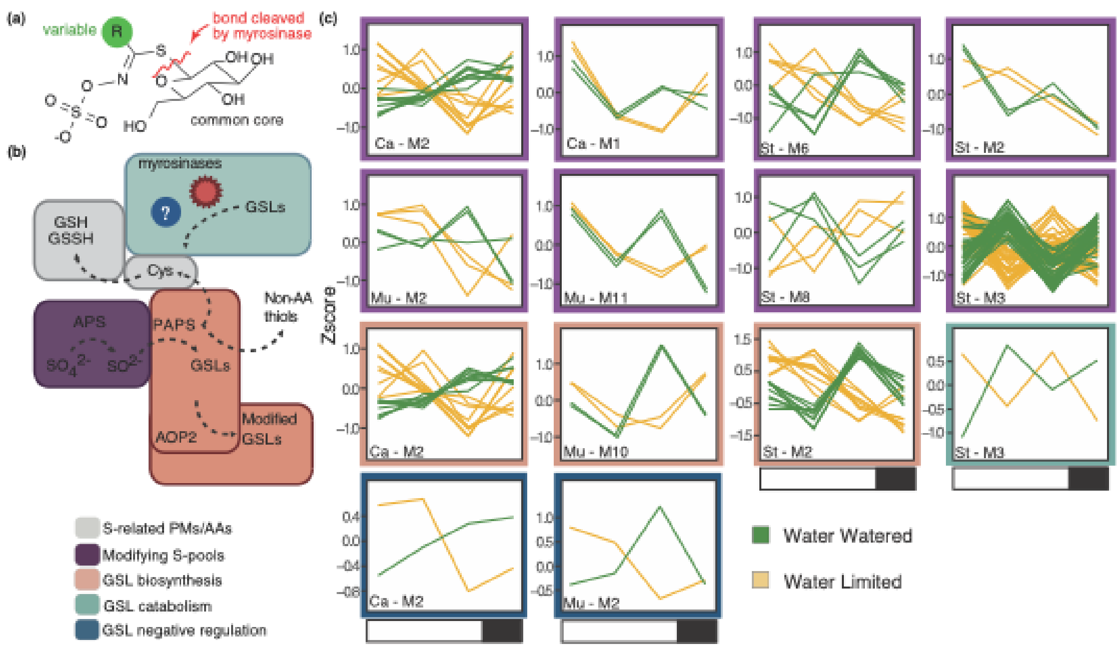
(a) Global temporal responses to WL were determined by calculating a response score and used to build a global co-expression network. Representative patterns (eigengenes) display variation in the time of day that accessions are responding to WL. Each line is the average response score for each accession in a given module. The number of genes within a module are provided in parentheses. (b) Enriched Gene Ontology (GO) terms for response modules M1, M2, M3, and M8. GO terms are listed on the y-axis and the negative log of the associated p-value is along the x-axis. Each bar represents the significance of a given GO term by accession (colors). Terms are grouped by broad descriptor ('parent term'). Biosynthetic process = b.p.; catabolic process = c.p.; metabolic process = m.p.



CH3 Figure 5.) Temporal modifications in transcript levels of enriched photosynthesis and carbohydrate genes and no change in net photosynthetic efficiency under prolonged WL in St and Ab.

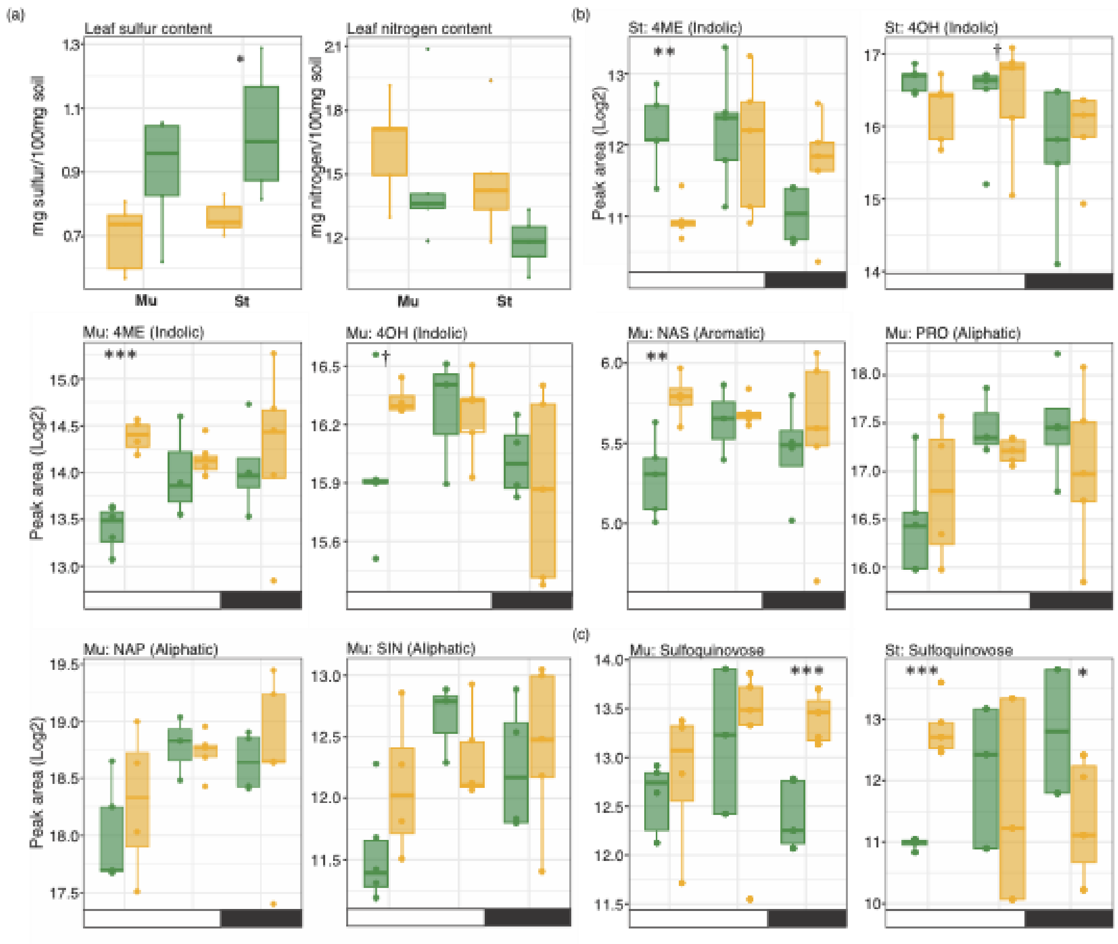
(a) Expression profiles of WL (yellow) and WW (green) genes in carbohydrate and photosynthesis related pathways for Ab and St. Bars under the panels indicate lights on (white) and lights off (black). Each line represents a gene with the total number of enriched genes indicated in parentheses. Note that while the polysaccharide insert represents GO terms involved in catabolism of polysaccharides, the hexose insert is related to biosynthetic terms. All other panels are described broadly (ie: 'metabolism'/'biological process') unless noted otherwise. (b) Efficiency of photosystem II (Fv/Fm) in St (top) and Ab (bottom) under WL or WW for 3 weeks. No significant difference in mean Fv/Fm from WW is seen under WL in either St or Ab ($p = 0.23$ and $p = 0.19$, respectively). 3) Chlorophyll Index (CI) in St (top) and Ab (bottom) under WW or

WL for 3 weeks. Significant change in mean CI is seen under WL in Ab ($p < 0.0001$) but not in St ($p = 0.05$).



CH3 Figure 6.) B. napus accessions exhibit altered regulation of sulfur metabolism and glucosinolate (GSL) genes at different times of day in response to WL.

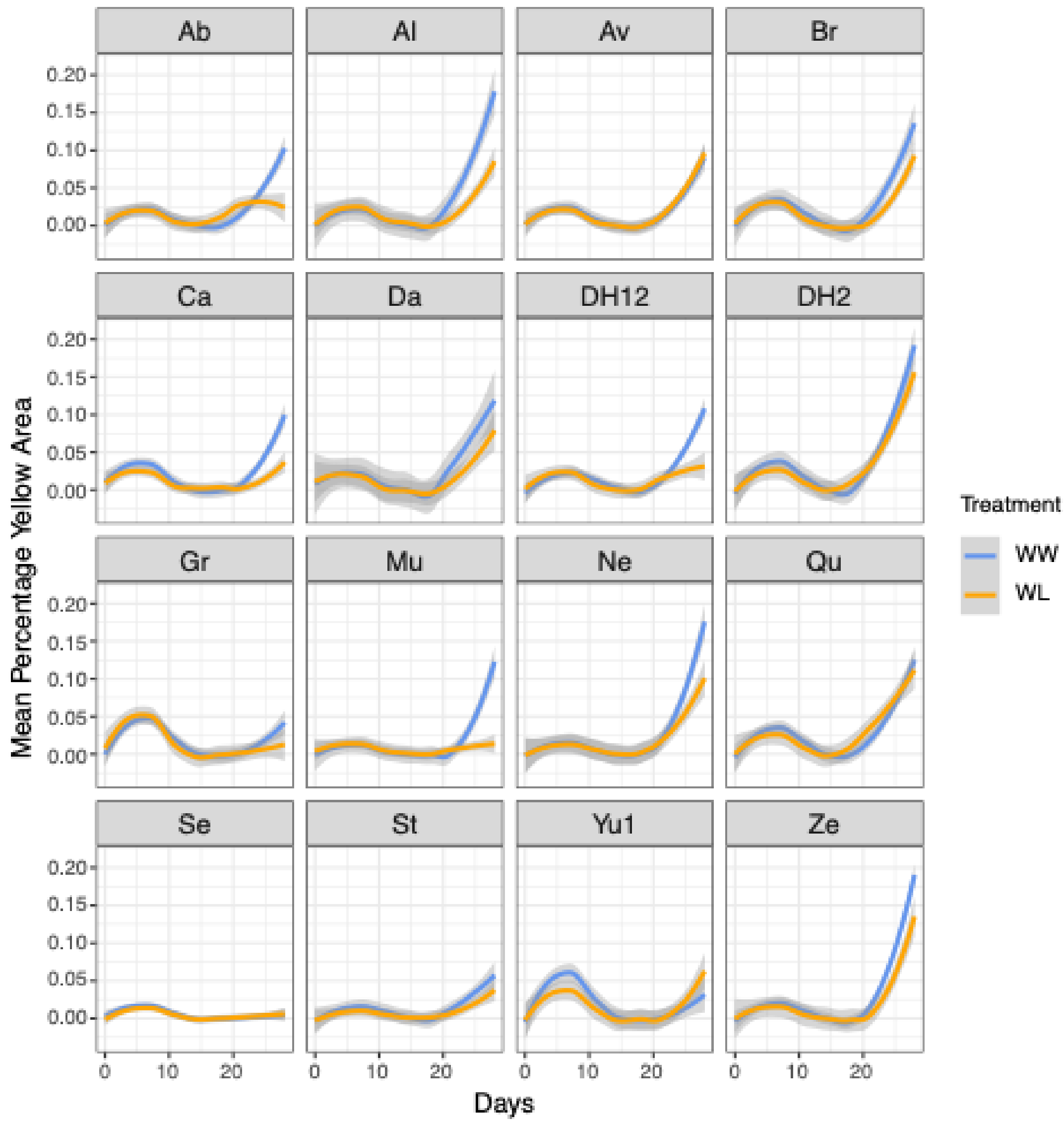
(a) A model of a generalized glucosinolate which is composed of three parts; the thio-glucose moiety, a sulfated thio-hydroximate, and a R (variable) group derived from an amino acid precursor. The conical view of GSL degradation requires GSL specific enzymes known as myrosinases. (b) Simplified schematic of the interaction between sulfur and glucosinolate metabolism. APS (Adenosine 5'-phosphosulfate) reduces sulfate (SO_4^{2-}) to sulfite (SO_3). PAPS (3`-phosphoadenosine 5`-phosphosulfate) sulfonates precursor GSLs before they are further modified by 2-oxoglutarate-dependent proteins such as AOP2. Sulfur is also allocated towards primary metabolites (PMs) such as glutamate/glutamine that can combine with cysteine to produce glutathione (oxidized: GSSH; reduced: GSH); amino acids (AA) such as Cysteine (Cys), and sulfolipids (ex: sulfoquinovose-diacylglycerols; SQDGs). (c) Expression patterns of genes enriched in modifying sulfur pools (purple), GSL biosynthesis (peach), and reduction of GSLs (aqua; dark blue), glutamate/glutamine (light blue) and sulfolipids (gray) as indicated by colored boxes. Each line is a single enriched gene. Bars indicate lights on (white) and lights off (black).



CH3 Figure 7.) Differential time of day accumulation of glucosinolates in a spring (St) and winter (Mu) cultivar.

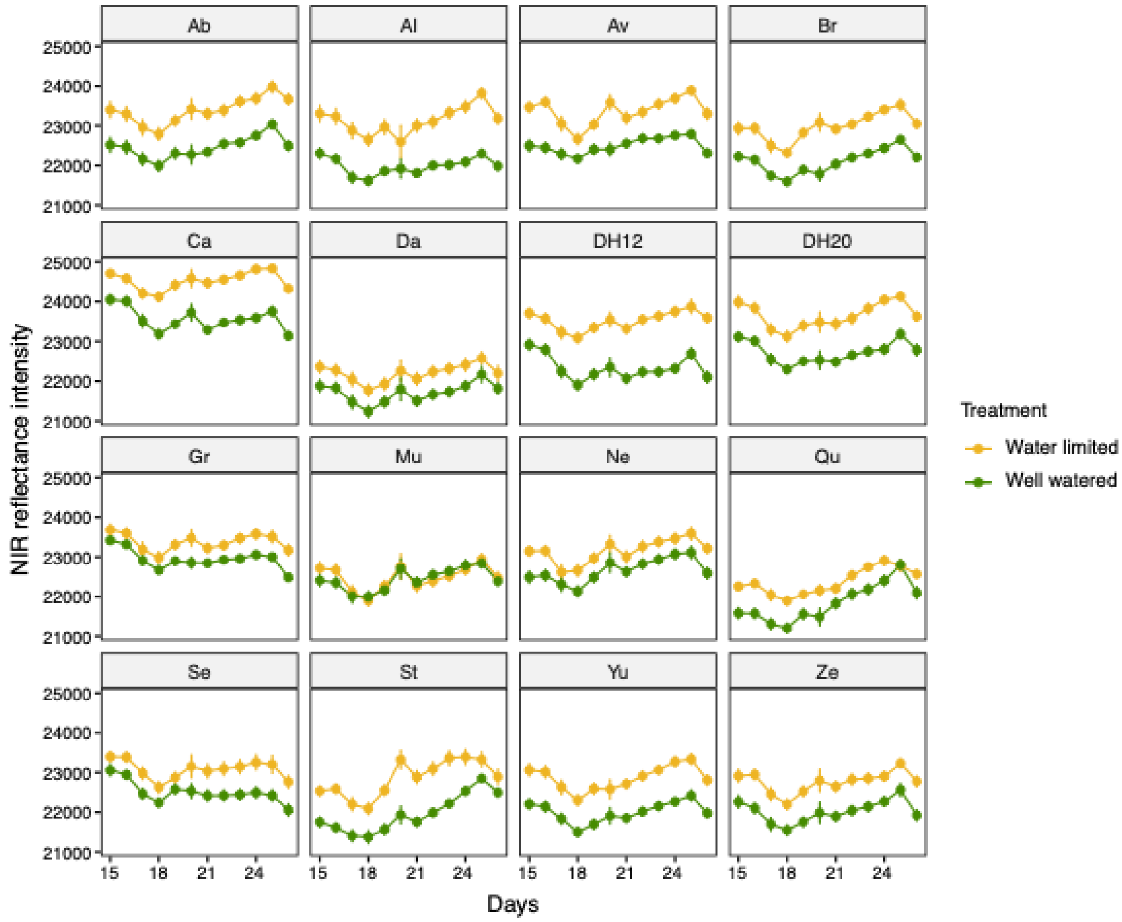
(a) Glucosinolate content (log Peak area) under well-watered (green) and water-limited (yellow) conditions at each ZT (Zeitgeber; hours after lights on). Bars indicate lights on (white) or lights off (dark). Derivative abbreviations are listed above each panel with the glucosinolate class in parentheses. 4ME = 4-methoxyglucobrassicin/neoglucobrassicin; 4OH = 4-hydroxyglucobrassicin; NAS = gluconasturtiin; PRO = progoitrin; NAP = gluconapin; SIN = sinigrin. (b) Sulfoquinovose content (log Peak area) in Mu and St. (c) Total sulfur and nitrogen content in the sixth leaf of St and Mu collected at ZT7.

Chapter 3 Supplemental Figures



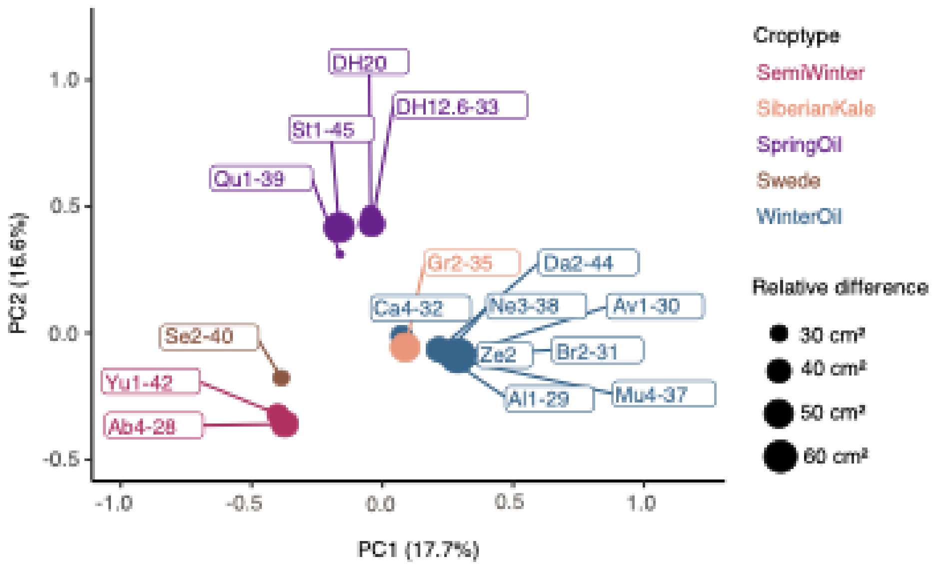
CH3 Supplemental Figure 2.) Percent of plant pixels classified as 'yellow' by a naive bayes classifier.

B. napus accessions exhibit reduced growth with water limitation (Figure 1). Several accessions on average have a lower percentage of plant pixels classified as 'yellow' at the end of the water limitation treatment compared to the well-watered treatment.



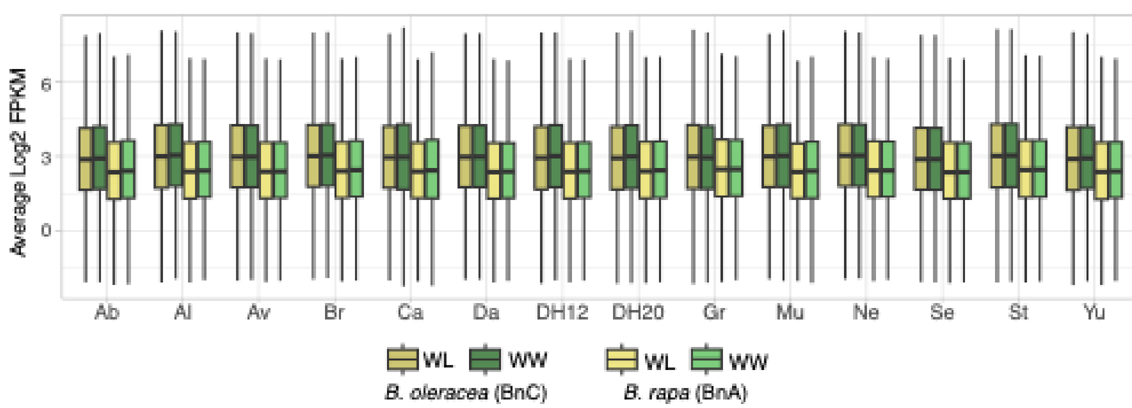
CH3 Supplemental Figure 3.) NIR signal is higher in most plants in water-limited treatment, suggesting lower water content than in well-watered plants.

Only NIR reflectance from day 14 to day 26 are shown, when images were taken at a consistent zoom level. Values were averaged across replicates and colored by treatment.



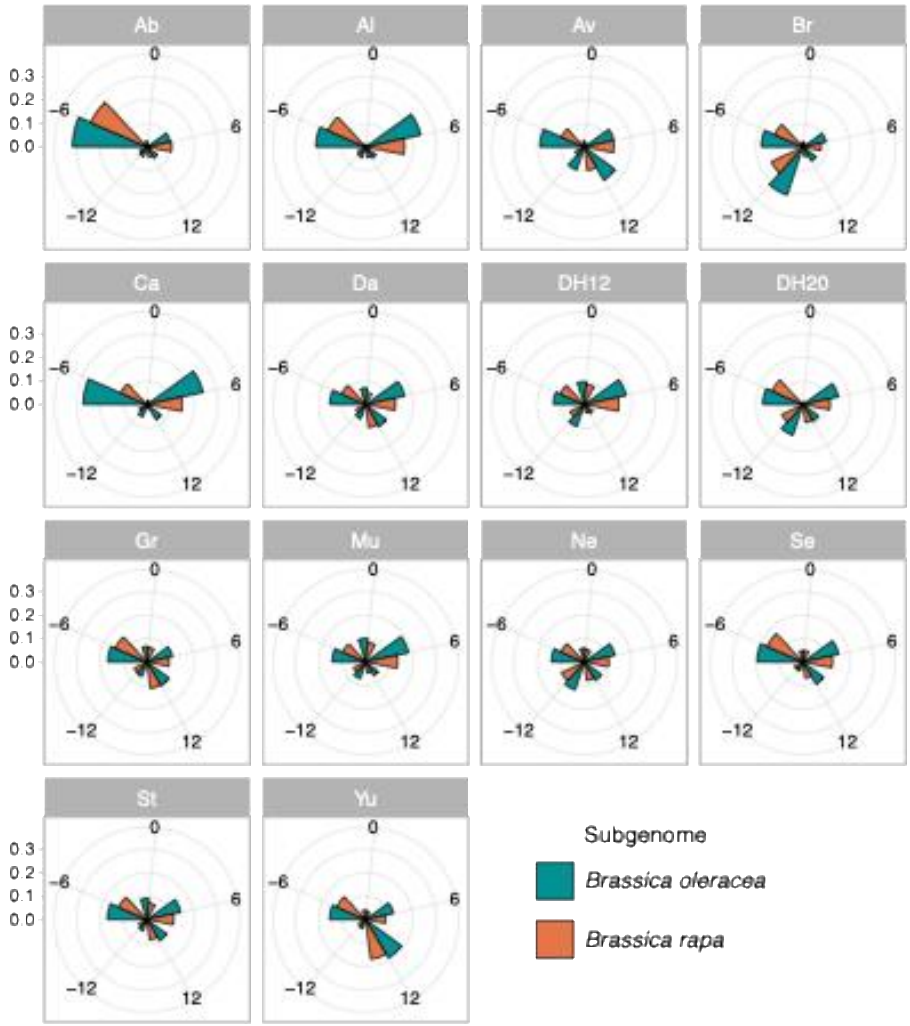
CH3 Supplemental Figure 4.) Accessions cluster by crop type, not differences in plant area.

Reads were mapped to the R500 genome and Single Nucleotide Polymorphisms (SNPs) were called using GATK. Plant area was quantified as the number of pixels using the plantCV software. Pixels were averaged by treatment for each accession. Relative difference in plant area was calculated as the absolute value of water-limited area minus well-watered. PCAs were made using the ggplot2 package in RStudio 4.3.1.



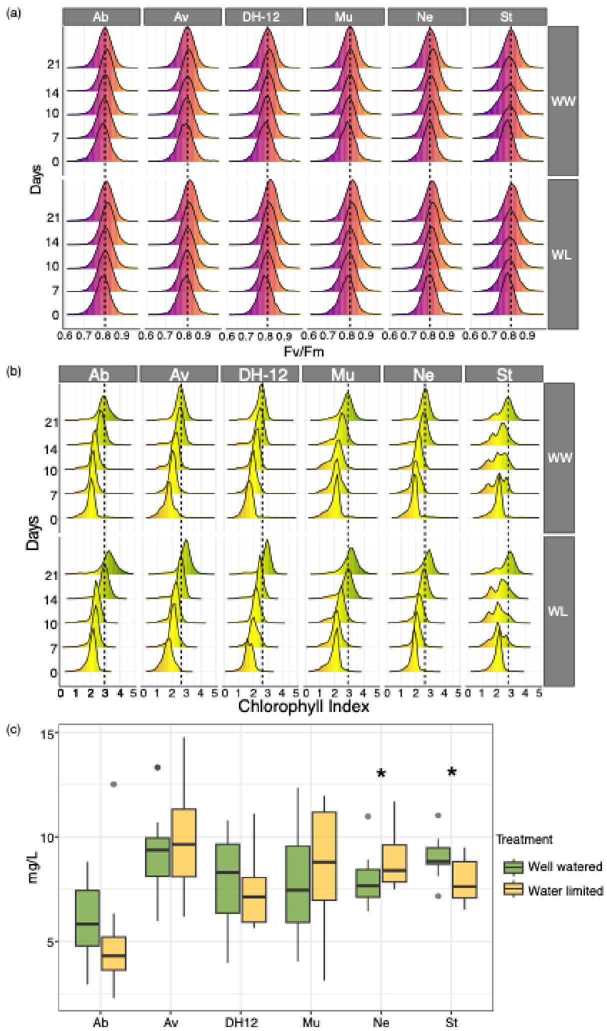
CH3 Supplemental Figure 5.) *B. napus* subgenomes exhibit slight bias in transcript levels but treatment levels are the same.

Transcript counts were normalized and the log sum was used to compare expression levels of all expressed genes by accession. Accessions are listed along the x axis.



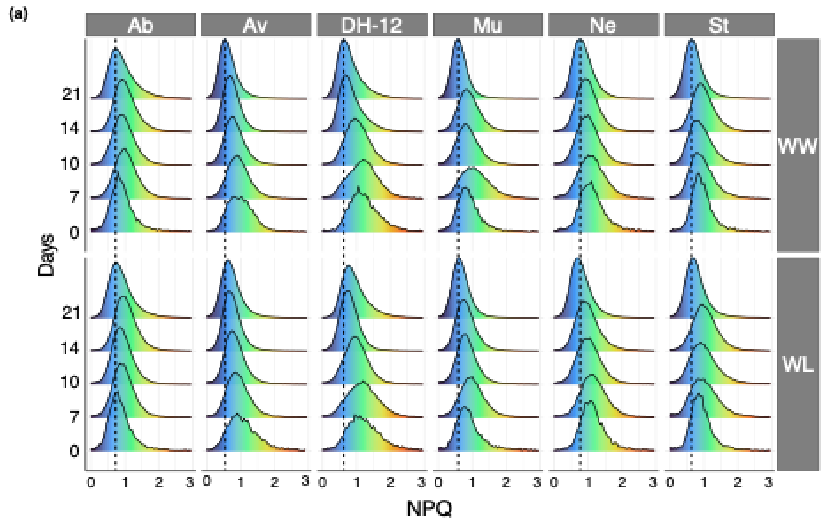
CH3 Supplemental Figure 6.) Subgenome contribution varies by time of day and accession.

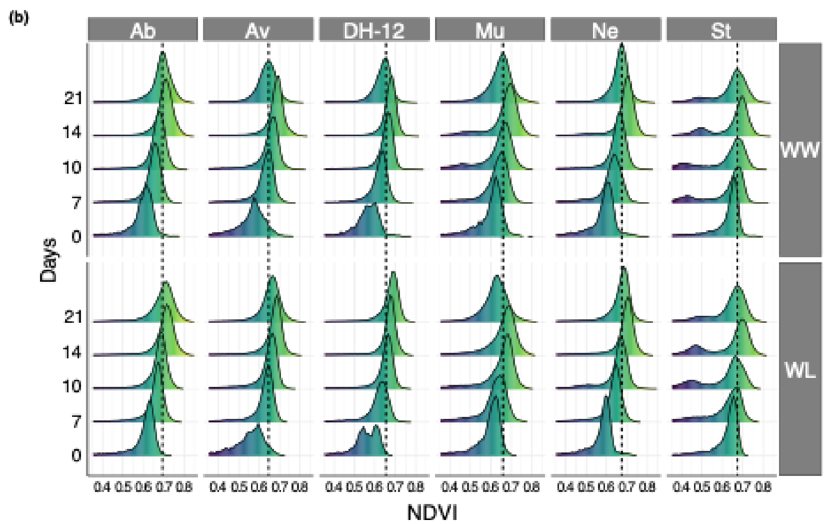
Phase groups were calculated by subtracting the timing of peak expression (phase) of WL from the phase of WW for every gene and accession. Proportions were calculated as the number of genes by subgenome in a given phase group relative to the total number of genes for that accession/subgenome.



CH3 Supplemental Figure 7.) Some accessions have higher chlorophyll fluorescence under prolonged water limitation but have little differences in net photosynthetic efficiency or chlorophyll content

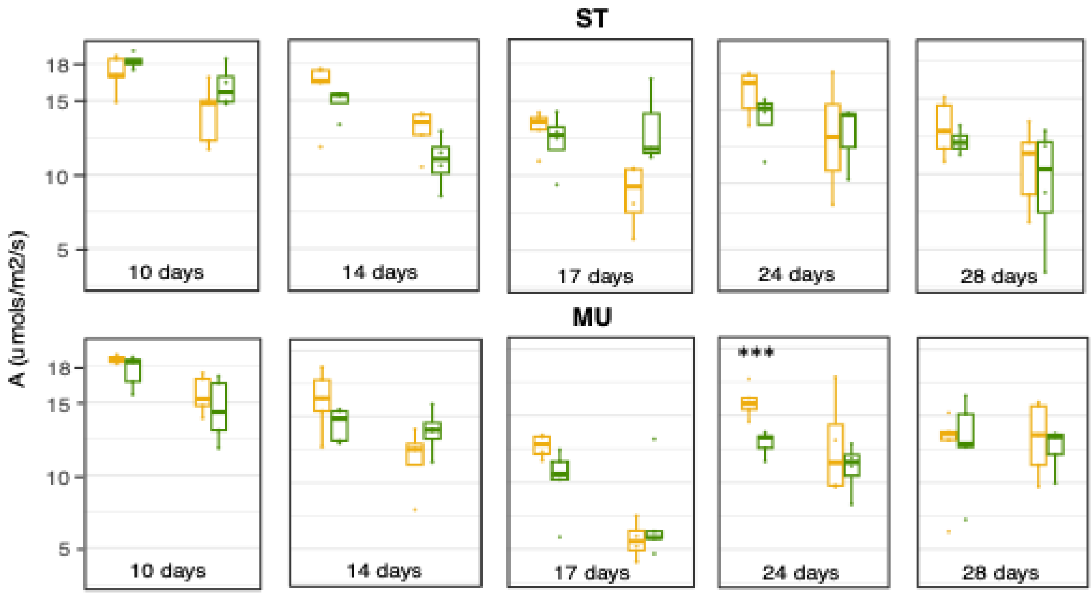
Fv/Fm reflectance (a) and chlorophyll fluorescence (b) images were captured using the CropReporter (PhenoVation Life Sciences, Wageningen, The Netherlands), and were analyzed using PlantCV v3.14.1. Joyplots were created using the "pcv.joyplot" function of the pcvr package (Sumner et al. 2023) in R v4.3.1 (R Core Team 2020). Values were normalized for plant size and averaged across replicates. Histograms are colored by value. Vertical dashed lines indicate peak (mode) of the well-watered treatment on day 21. (c) Total chlorophyll content was measured from leaf tissue in well-watered (WW) and water-limited (WL) plants at ZT6. Accessions are grouped along the x-axis. Ne and St had significantly higher ($p = 0.038$) and lower ($p = p = 0.003$) chlorophyll content after three weeks of water limitation.





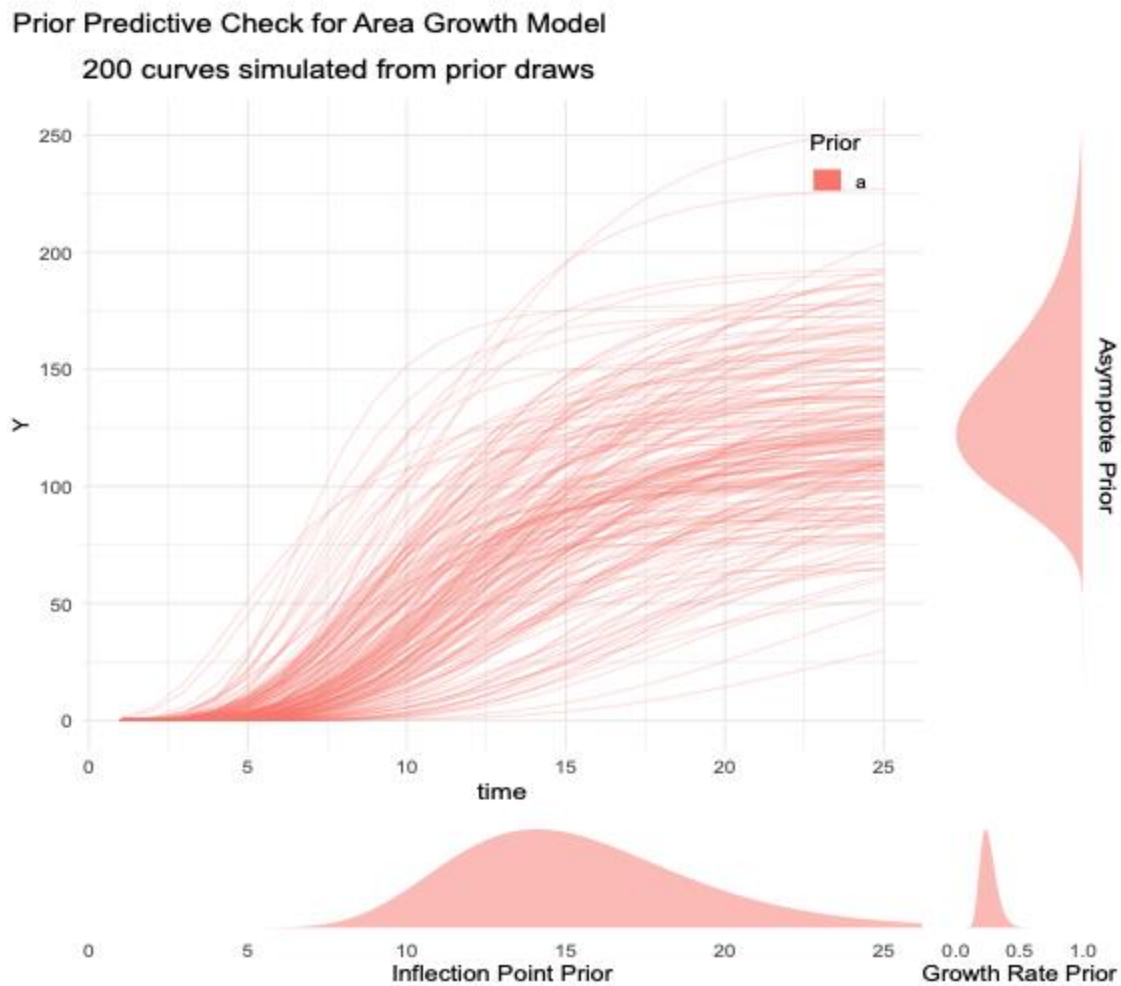
CH3 Supplemental Figure 8.) Accessions do not show signs of stress under prolonged water limitation.

Non-photochemical quenching of chlorophyll fluorescence (NPQ; a) and normalized difference vegetation index (NDVI; b) reflectance images were captured using the CropReporter (PhenoVation Life Sciences, Wageningen, The Netherlands) and were analyzed using PlantCV v3.14.1. Joyplots were created using the "pcv.joyplot" function of the pcvr package (Sumner et al. 2023) in R v4.3.1 (R Core Team 2020). Values were normalized for plant size and averaged across replicates. Histograms are colored by value. Higher NPQ values indicate more light energy being dissipated in the form of heat and therefore not being used by the plants for photosynthesis, while higher NDVI values indicate greener plants. Vertical dashed lines indicate peak (mode) of the well-watered treatment on day 21.



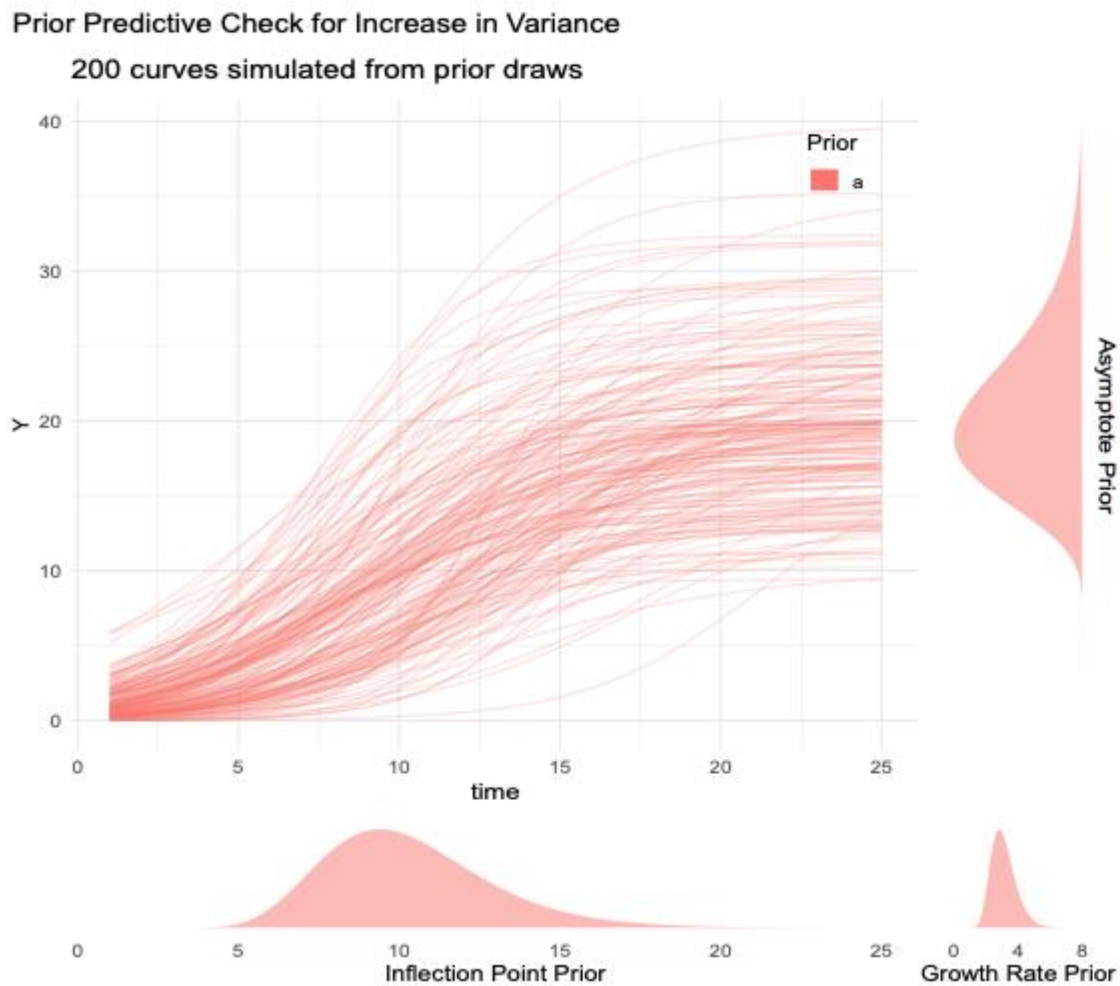
CH3 Supplemental Figure 9.) St and Mu maintain similar carbon assimilation rates throughout four weeks of water limitation.

With a single exception (Mu; 24 days) there was no significant difference between treatments in carbon assimilation (A = µmols of assimilated carbon per square meter per second) indicating accessions had adjusted to the treatment. Measurement days are indicated at the bottom of each panel. Asterisks indicate a significant difference between treatments (*** is p =< 0.001). Yellow boxes indicate water-limited; green are well-watered.



CH3 Supplemental Figure 10.) Prior predictive check for area growth model.

We used lognormal priors for each estimated parameter in our growth model (*A*, *B*, *C*) and in the sub model for variance (*subA*, *subB*, *subC*). For the asymptote parameter we used a lognormal distribution with mu=130 and sigma=0.25. For the inflection point parameter we used a lognormal distribution with mu=15 and sigma=0.25. For the growth rate parameter we used a lognormal distribution with mu=0.25 and sigma=0.25. Prior predictive checks were conducted using the plotPrior function from the pcvr R package (Sumner et al. 2023)



CH3 Supplemental Figure 11.) Prior predictive check for increase in variance submodel.

We used lognormal priors for each estimated parameter in our growth model (A, B, C) and in the sub model for variance (subA, subB, subC). For the asymptote parameter we used a lognormal distribution with mu=130 and sigma=0.25. For the inflection point parameter we used a lognormal distribution with mu=15 and sigma=0.25. For the growth rate parameter we used a lognormal distribution with mu=0.25 and sigma=0.25. Prior predictive checks were conducted using the plotPrior function from the pcvr R package *(Sumner et al. 2023)*.

Chapter 4: Conclusions

The rotation of the Earth generates 24h cycles of light creating predictable fluctuations in environmental conditions. Across all kingdoms of life, organisms have evolved an endogenous timekeeping system, known as the circadian clock, to anticipate and adapt to these recurring changes[43]. This endogenous time-keeping mechanism aligns physiological, metabolic, and stress responsive processes with the external day-night cycle, enabling organisms to respond to environmental stimuli at specific times of day[26]. The intrinsic ability to determine 'when' you are within a day confers a benefit through the ability to anticipate and therefore prepare for upcoming changes such as sunrise, colder nighttime temperatures, or fluctuations in resource availability[216,217]. Such anticipation is governed by 'zeitgeibers', or 'time givers' which are external cues that synchronize the clock with the surrounding environment[218]. In mammals, circadian control of the sleep-wake cycle is governed by light perceived by a central oscillator (suprachiasmatic nucleus; SCN) in the hypothalamus, and disruption of this cycle has known detrimental impacts on human health[219]. Avian migration can also be attributed to daily and seasonal circadian regulation, as well as monarch butterflies[220,221]. In other kingdoms such as microbes, particularly those in rhizomes or ruminant intestinal systems, light and temperature zeitgeibers are lost but host feeding patterns can still entrain their clocks[222]. Circadian regulation is conserved across diverse taxa and contributes to how organisms adapt to their local environments[23]. In addition to circadian regulation, diel, or 24h, regulation also coordinates physiological and metabolic processes with environmental change. However, diel rhythms are directly driven by external light:dark and temperature transitions and therefore depend on continued environmental cues. In the absence of these cues, only circadian rhythms persist under constant conditions due to the activity of an endogenous oscillator[26]. These regulatory systems are synergistic and often modulate overlapping components of the same pathway. Photosynthesis, for example, is governed both by direct light availability (diel control) and by circadian gating of photosynthetic gene expression, chlorophyll biosynthesis, and carbon assimilation[181]. Because these regulatory systems are tightly intertwined, parsing endogenous circadian control from light-driven diel regulation is difficult and requires experiments to be performed under circadian conditions in the absence of external cues. The analyses presented in this dissertation were performed under diel conditions and therefore reflect the integrated outcome of both circadian and diel regulation.

The complexity of the circadian clock has diversified and expanded across diverse taxa, with relatively simple oscillators in cyanobacteria and algae, to multi-component networks present in higher plants[14,23]. As sessile organisms, plants are restricted to their environment and must precisely allocate available energy stores between primary (growth) and specialized (stress responses) metabolism[223]. These processes are inherently dynamic throughout the day as they respond to variations in light availability and temperature that directly influence carbon assimilation and resource optimization. It is no surprise, then, that a major regulator of plant physiology and metabolism is circadian and diel regulation[26]. However, despite extensive evidence that circadian regulation underpins plant growth[224], the role of circadian and diel regulation in structuring stress responses across genetically diverse crops remains relatively underexplored. This gap in knowledge is particularly relevant in the context of climate change which is amplifying both the magnitude and the predictability of global temperatures and weather patterns, thereby intensifying environmental pressures faced by crops worldwide[45]. Extreme temperature events, including late spring frosts as well as prolonged periods of elevated heat are exposing crops to multiple abiotic stressors within the same growing season[225]. In parallel, altered precipitation patterns are exacerbating water limitation. Although drought is often associated with high temperatures, water scarcity represents a compounding constraint on agricultural productivity[225,226]. Modern agriculture has partially buffered the effects of water limitation through irrigation; however, approximately 60–70% of global freshwater withdrawals are already allocated to agriculture, and much of this supply ultimately depends on rainfall-driven hydrological cycles which are not always replenished to the same degree from year to year[227]. Climate change disrupts these cycles and increases competition for freshwater resources, inevitably exposing our crops to prolonged periods of water limitation[228,229].

Despite rising global temperatures, late spring frosts account for a substantial proportion of annual yield losses[230]. To reduce frost risk farmers may delay planting; however, planting later in the season also runs the risk of encountering extreme high temperatures later in the season which can be equally detrimental to yields[231]. Another way to mitigate this is to breed cold 'hardy' plants which are better able to withstand freezing temperatures. The frequency of extreme temperature events is projected to increase, posing a significant challenge to farmers and breeders alike. To overcome these challenges we need more stress tolerant crops that can withstand various osmotic stressors including extreme temperatures and reduced water availability. Although cold and drought are distinct stressors they both impose cellular dehydration and osmotic imbalance resulting in similar changes to physiology, metabolism and gene expression, and therefore can be jointly referred to as 'osmotic stressors'[232].

Much of our current understanding of the circadian clock and its role in mitigating stress responses comes from the diploid model plant Arabidopsis. However, most of our crops are polyploid and have undergone whole genome duplications or triplications[5], resulting in an increased number of gene copies (paralogs; homoeologs). Recent work in non-model crops such as *Brassica napus* (canola) and *Triticum aestivus* (bread wheat) have revealed extensive time of day diversification in paralog responsiveness under stress[20,38]. However, a more comprehensive understanding of how these paralog specific temporal patterns translate into physiological and metabolic outcomes is warranted.

To resolve the temporal dynamics underlying osmotic stress in polyploid crops, this dissertation integrates time-of-day resolved transcriptomic, physiological, and metabolic analyses to identify regulatory and metabolic mechanisms underlying osmotic stress tolerance in *Brassica* species. In the first chapter, 14 *B. napus* accessions were exposed to four weeks of prolonged water limitation with leaf tissue collected for RNA-sequencing every six hours over a 24h period. In the second chapter, I examined diel regulation of cold acclimation across a *B. rapa* diversity panel. Thirteen accessions representing six distinct morphotypes were cold acclimated for three days before performing a time-series RNA-sequencing experiment to profile cold acclimated transcriptomes. A complementary freeze stress experiment was performed in parallel to assess if accessions exhibiting more similar acclimation responses also displayed similar freeze tolerances. Across both studies, osmotic stress induced widespread temporal reprogramming of gene expression, with thousands of genes exhibiting altered responsiveness across the day in either shifts in phase or amplitude across the diel cycle. This divergence occurred predominantly on the paralog level, revealing an additional layer of regulatory complexity in which duplicated genes exhibited distinct temporal patterns across accessions, contributing to highly diverse and accession specific diel stress responses.

In *B. napus*, prolonged water limitation across 14 accessions uncovered extensive variation in both the number and identity of homoeologs exhibiting stress-responsive expression. Nearly 70% of responsive genes were unique to a single accession, indicating highly accession specific regulation. However, when gene models were converted to their syntenic *Arabidopsis* orthologs, overlap increased substantially, demonstrating that divergence was driven largely by differential paralog usage. This divergence was particularly evident in pathways related to carbon fixation and sulfur metabolism. For example, glucosinolate biosynthesis genes such as *MYB DOMAIN PROTEIN 28* (*MYB28*) and *BETA-GLUCOSIDASE 33* (*BGLU33*) differed in the number and identity of responsive homoeologs among accessions, corresponding with accession specific differences in diel accumulation of glucosinolates and sulfoquinovose. That such extensive temporal diversification is present in canola - a species domesticated ~8,000 years ago[39] - suggests the rapid diversification of regulatory partitioning following polyploidization.

Similarly, cold acclimation in *B. rapa* revealed pronounced paralog divergence, most notably within photosynthesis and sulfur assimilation pathways. Multiple *PHOTOSYSTEM II REACTION CENTER PROTEINs* (*psb*) paralogs varied in both copy number and temporal responsiveness across accessions, including both tolerant and intolerant lines. Through Gene Regulatory Network analyses I uncovered accession-specific regulatory connections that identified a link between core cold regulators (*ICE1*; *DREB* family members), sulfur and glucosinolate metabolism (*MYB28*; *BGLU33*), and circadian clock components (*LHY*; *GI*; *PRR7; TOC1*). The identification of *LHY* targeting *MYB28* is particularly intriguing as it provides evidence for a potential mechanism for circadian regulation of glucosinolate biosynthesis during cold acclimation, extending previous metabolomic observations of diel glucosinolate control[20,129,149].

Across both chapters, the amount of intraspecific variation in stress response among the accessions was striking, both in the variation of the time of day the response occurred as well as which ortholog was responsive. Although SNP-based analyses clustered accessions by morphotype, their diel physiological and transcriptional profiles frequently diverged - even within a morphotype - demonstrating that temporal regulation of stress responses is more variable than previously expected. Collectively, these findings establish that genome duplication enables rapid temporal diversification of paralogs across core metabolic pathways, particularly photosynthesis and sulfur metabolism. By revealing how duplicated genes are differentially utilized across the day under stress, this work highlights paralog-specific temporal regulation as a key mechanism underlying stress resilience in polyploid crops and underscores the importance of incorporating diverse accession panels to capture the full breadth of variation that is clearly present within Brassica species. The combination of genetic diversity, polyploid history, and pronounced temporal divergence positions Brassica as a powerful model system for dissecting the mechanisms underlying stress adaptation in crops.

Beyond illustrating paralog divergence, this work underscores the central role of sulfur metabolism in osmotic stress adaptation. Sulfur is an essential macronutrient that supports primary metabolism, redox homeostasis, and the synthesis of specialized defense compounds[233], yet its temporal regulation under stress isn’t fully understood. Given its integration with photosynthesis, redox homeostasis, and specialized metabolism, resolving how sulfur pathways are coordinated across the day may be critical for understanding - and ultimately improving - stress resilience in polyploid crops.

A central hypothesis emerging from this work is that the rapid divergence observed among paralogs may be driven by variation in cis-regulatory regions, including conserved noncoding sequences (CNSs) and transcription factor binding motifs that confer temporal specificity[19]. For example, gain of clock associated elements (*i.e* Evening Element[145]) could underlie the time of day specificity in cold responsive gene expression we observed. Future work looking at enrichment of binding motifs in CNSs in the paralogs followed by promoter-reporter assays will reveal sequence divergence in cis-regulatory elements and if that contributes to paralog-specific temporal expression patterns. Dissecting these regulatory differences among the paralogs/accessions will provide mechanistic insight into how temporal partitioning evolves following polyploidization.

Collectively, this dissertation research demonstrates that adaptation to osmotic stress in *Brassica* crops is fundamentally structured by time of day and paralog diversification. Temporal partitioning of duplicated genes emerges as a key mechanism underlying both inter- and intraspecific variation in stress responses, highlighting the importance of integrating time-of-day regulation into research aimed at improving complex agronomic traits. By accounting for which paralog is responsive, and the underlying temporal dynamics, polyploid crops become rich reservoirs of regulatory diversity with substantial potential for advancing stress-resilient breeding strategies.

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